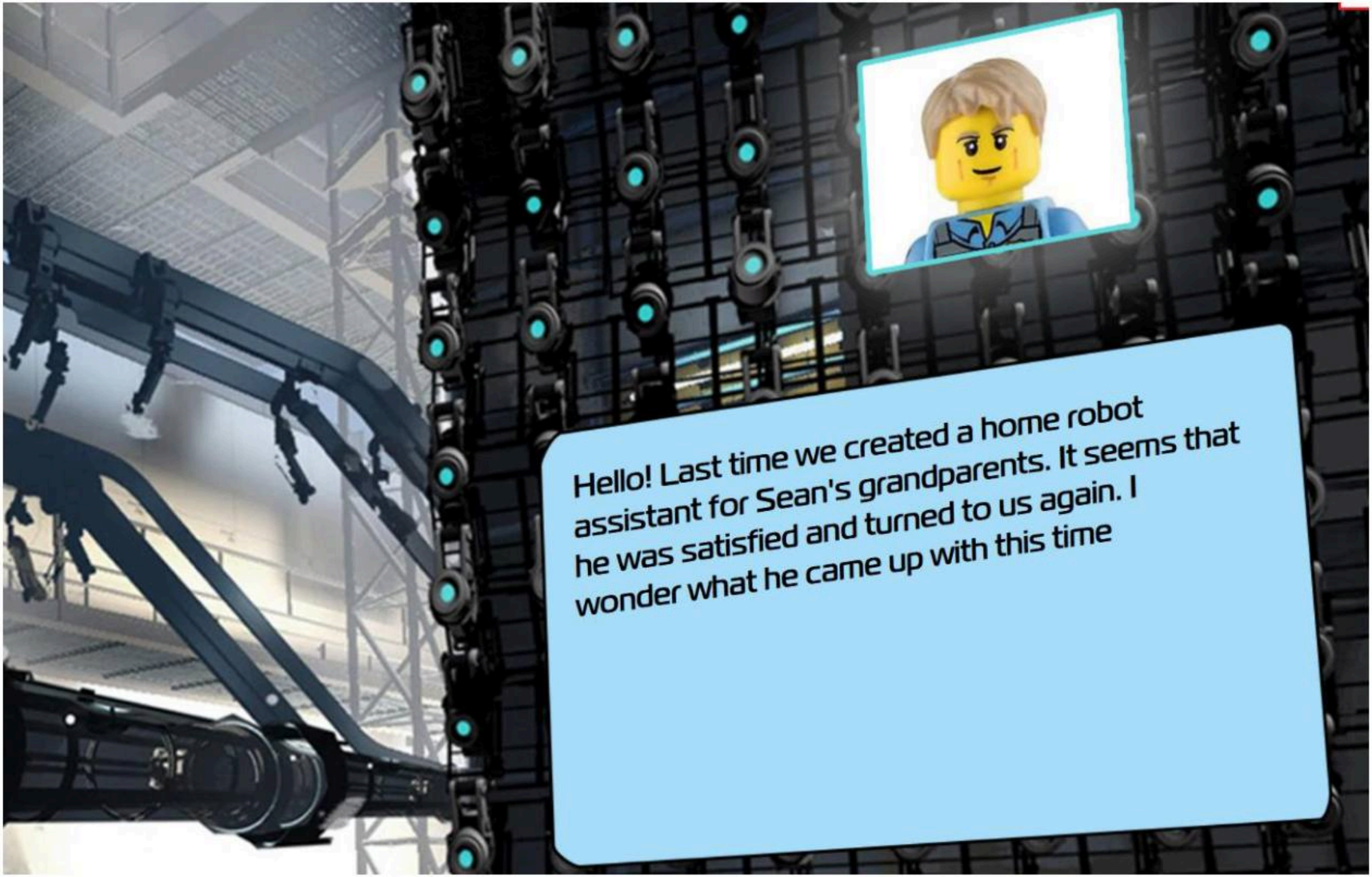
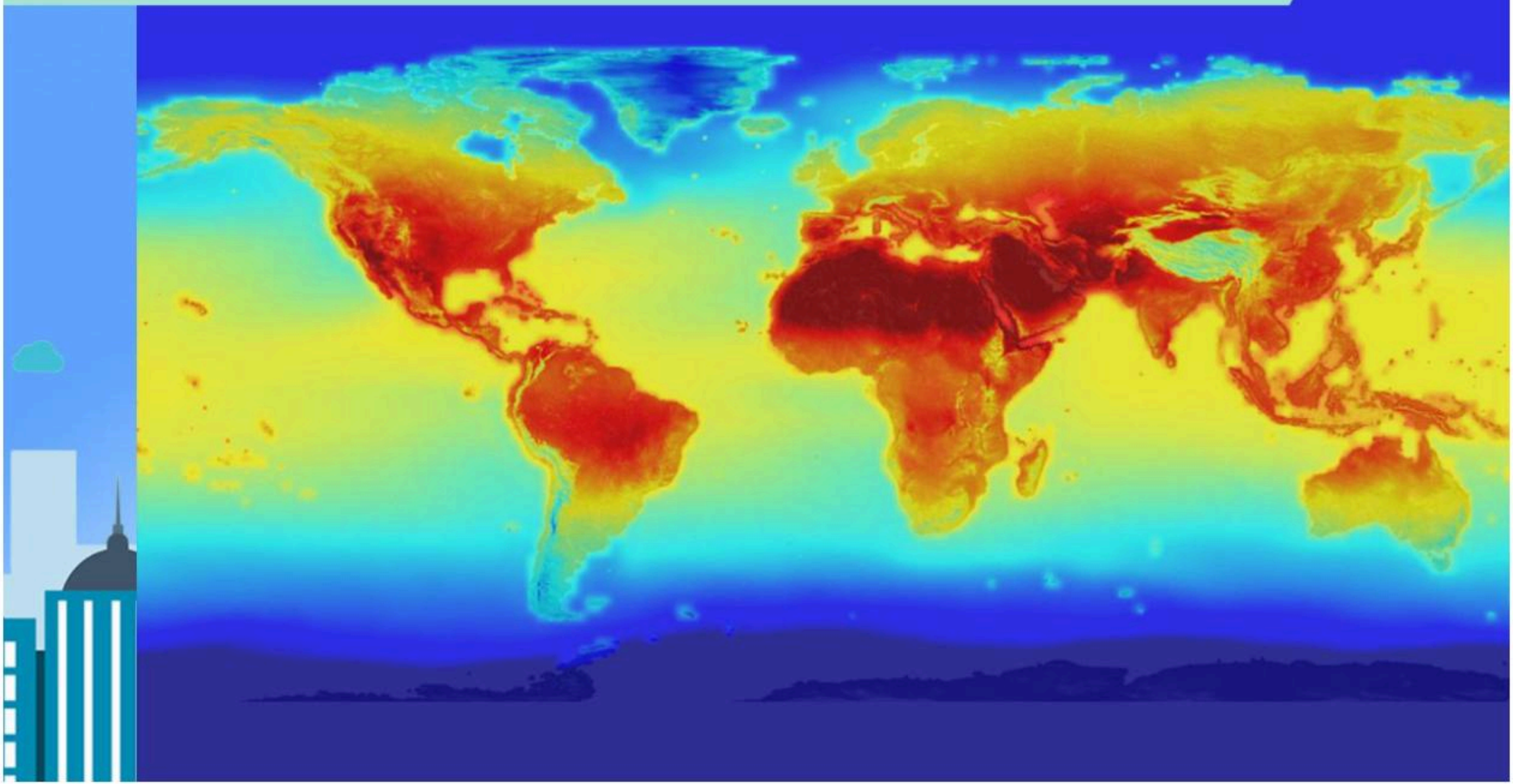



Hello! Last time we created a home robot assistant for Sean's grandparents. It seems that he was satisfied and turned to us again. I wonder what he came up with this time

I have friends in different countries, and they all write from time to time about some deviations from the norm. Someone is very hot, and someone, on the contrary, is very cold. Hurricanes became more and more frequent.



Can we somehow investigate what is happening on the planet? We might even be able to figure out how to make things better!



Hello! I think we need to start with the most general view of the state of affairs. That is, you need to visit different parts of the Earth and check what is happening there

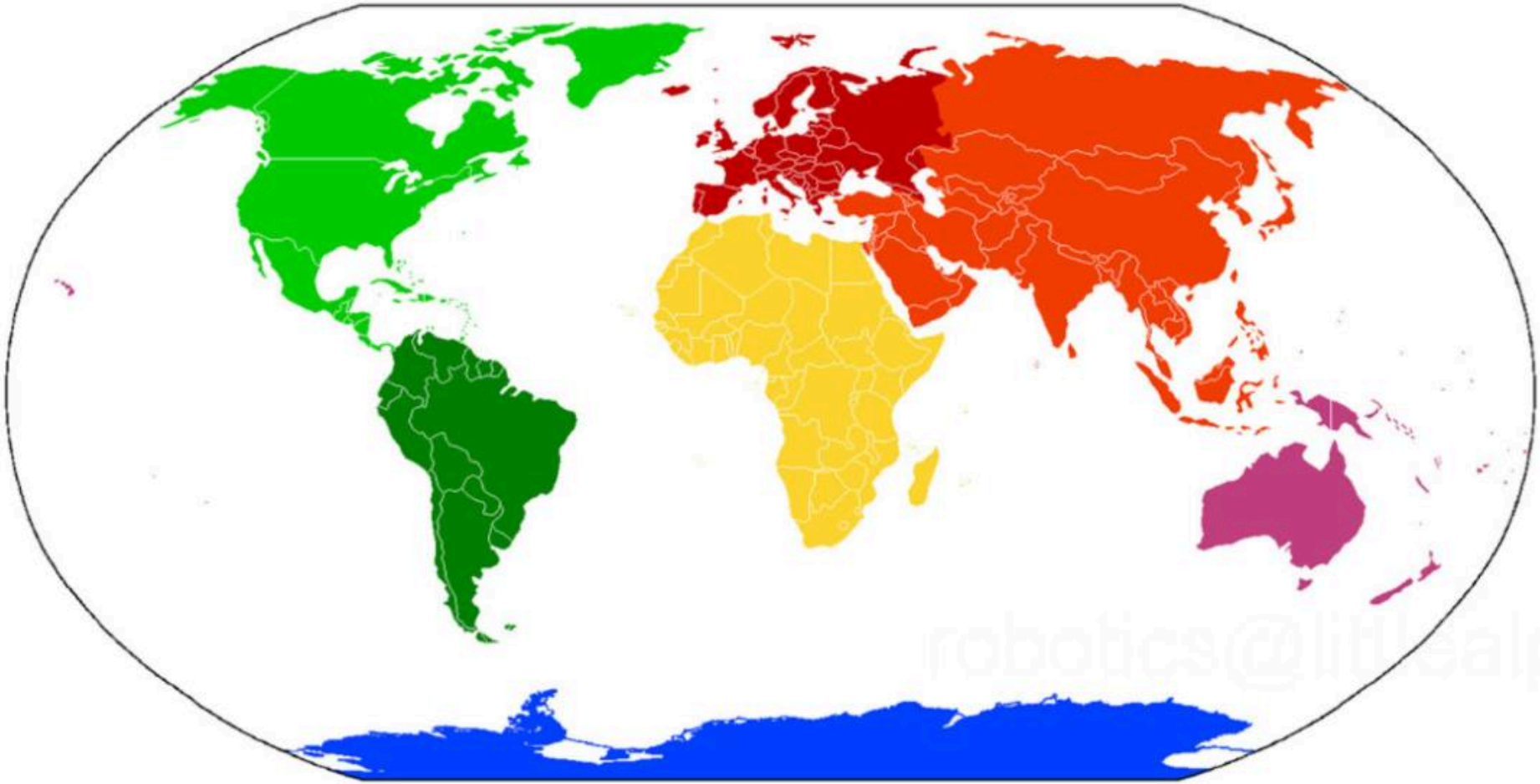


Earth

Do you know that our planet, the Earth, has the shape of a sphere, and most of its surface is covered with water? People live on land, parts of which are called continents.



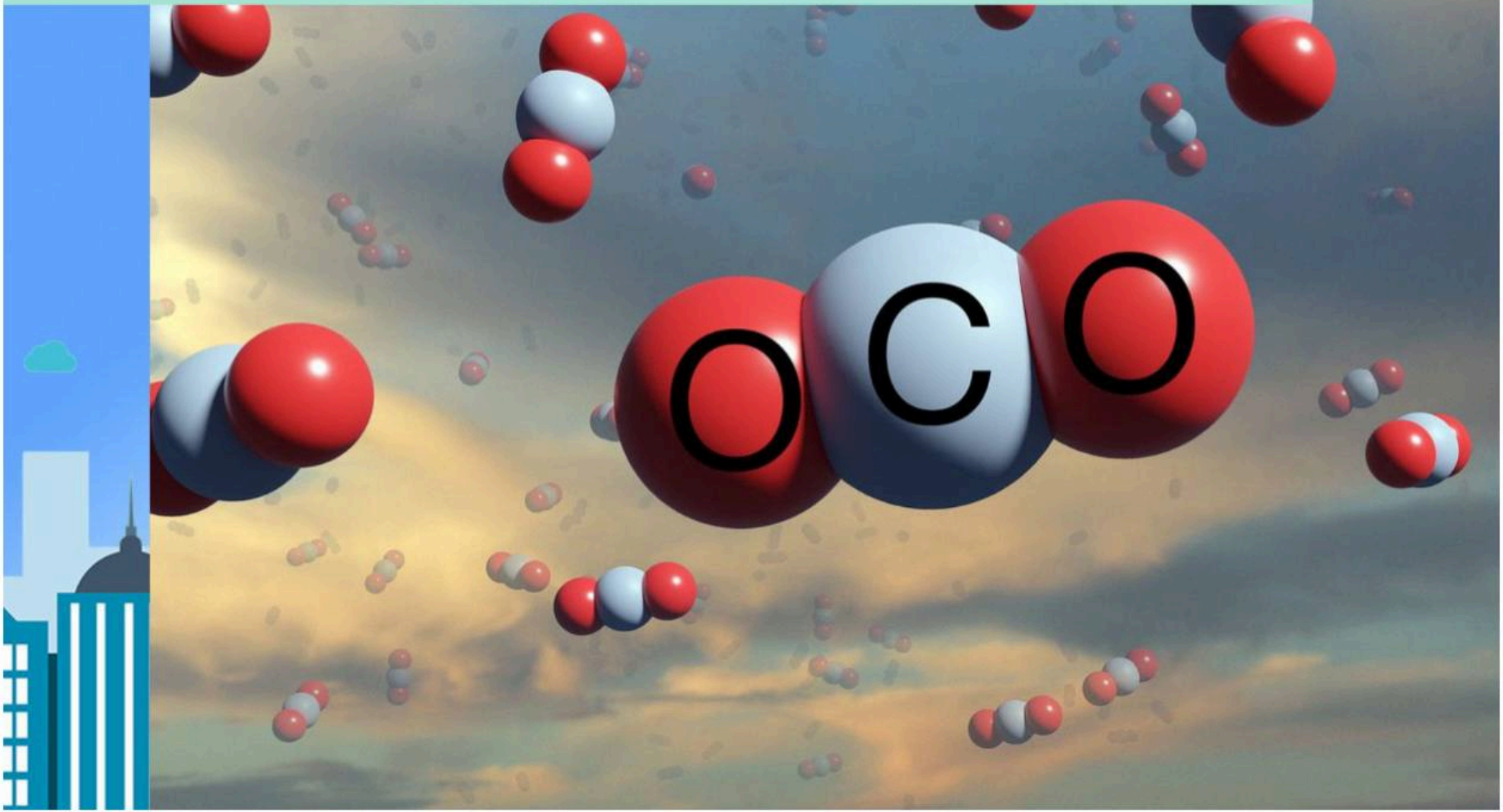
There are only seven continents: Europe, Asia, North and South America, Antarctica, Africa and Australia. Often Europe and Asia are combined into Eurasia, then there are six continents :)



The earth is quite large, and to overcome the distances between continents, it is best to use an airplane. It can take you to almost any point on Earth in a few tens of hours.



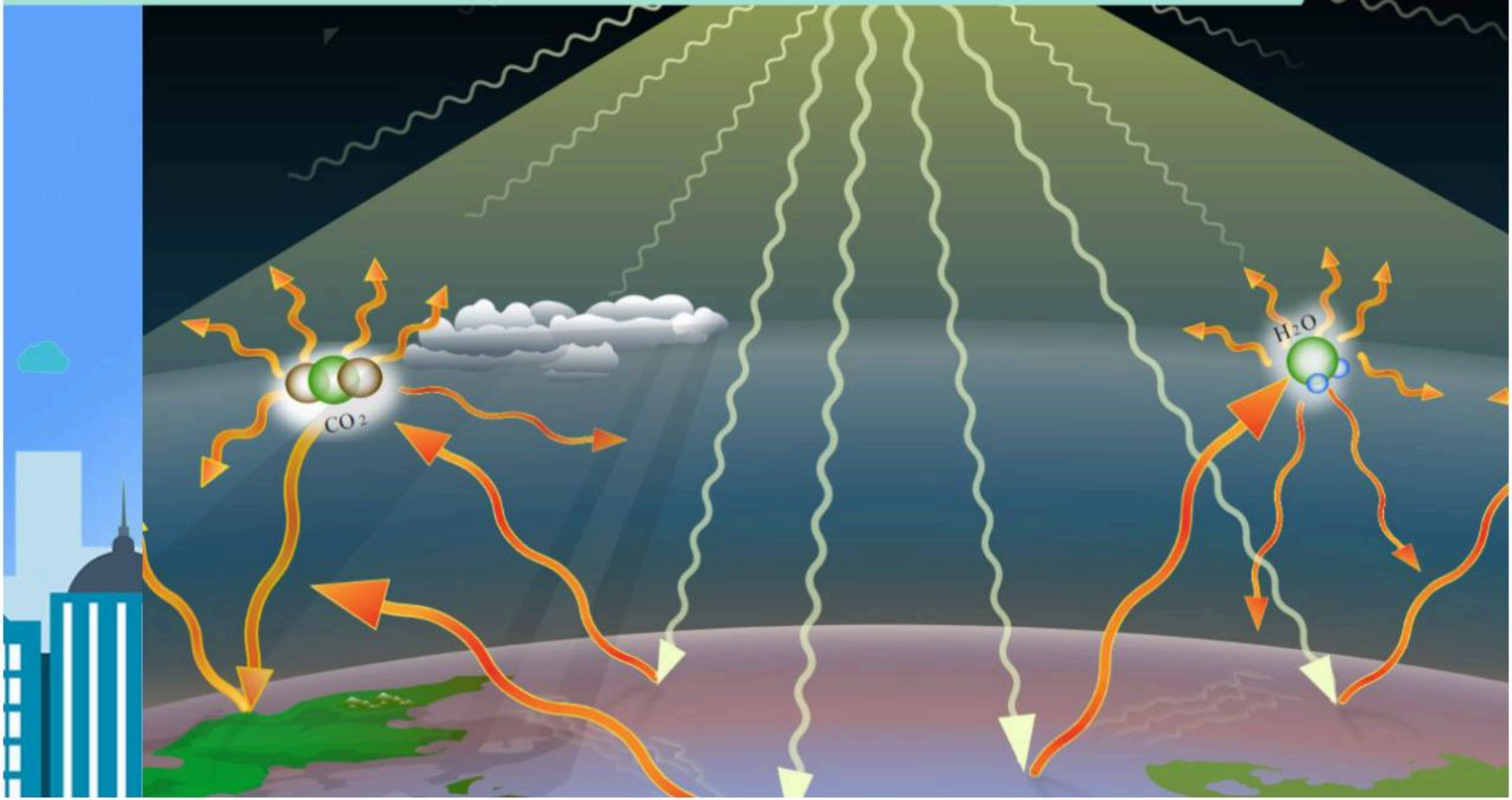
Scientists suggest that this abnormal heat may be caused by human activity, namely, greenhouse gases.



The fact is that we burn a lot of coal and oil to produce energy, which releases carbon dioxide, which does not release the sun's rays back into space.



Because of this, the Earth heats up more and more, and these processes are called the greenhouse effect.



To reduce this impact, scientists are looking for ways to produce energy that do not result in greenhouse gas emissions. For example, nuclear power plants are very environmentally friendly in this regard.



We propose to build an electric plane that will have solar panels on the wings. Such an aircraft will be environmentally friendly and will be able to stay in the air for a very long time and even collect data on the state of the atmosphere for a long time!



And since you are still quite a young pilot, we will equip the plane with systems for autonomous exit from the hangar and take-off, as well as a landing assistant.

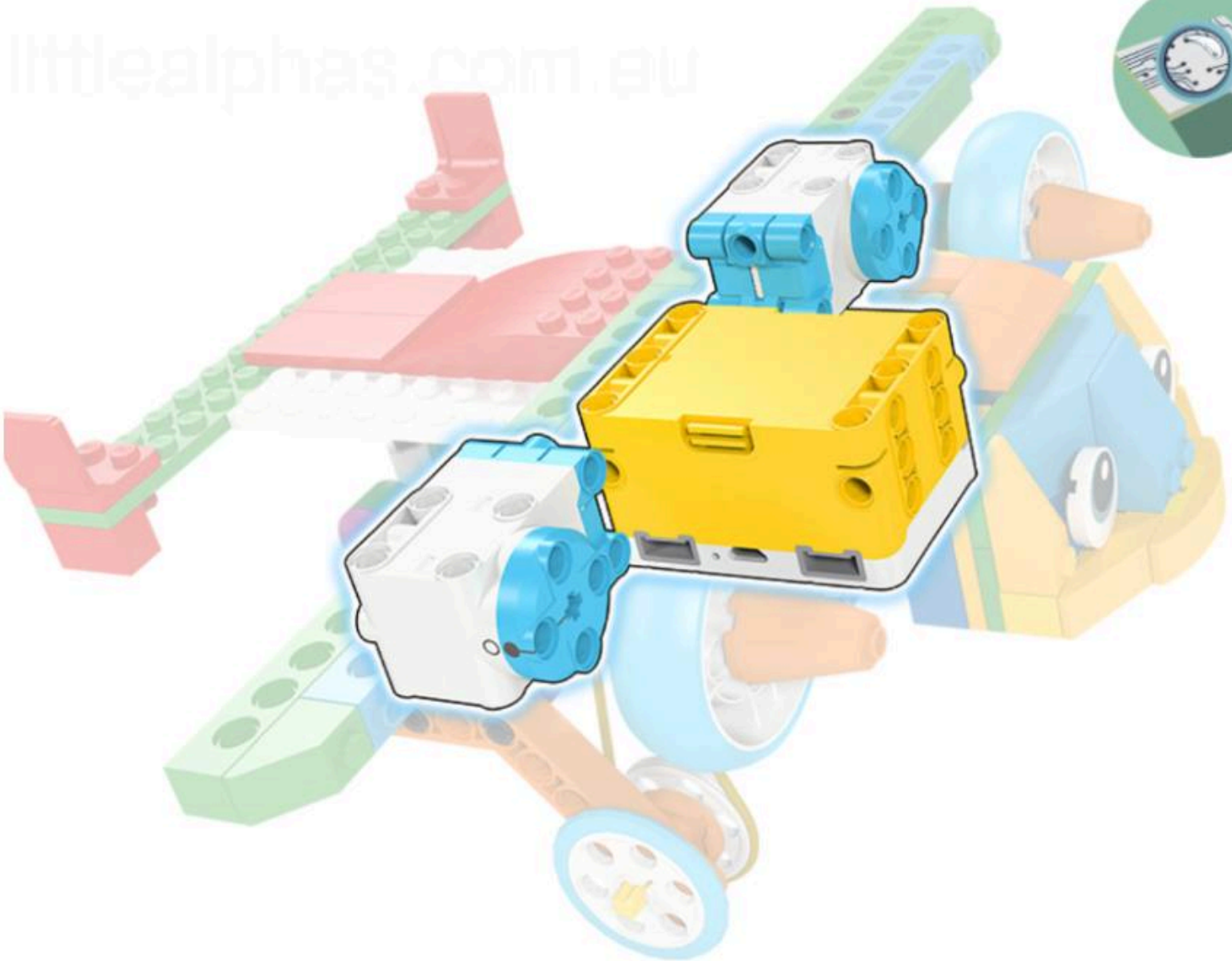


We must create a robotic airplane that:

- will allow you to quickly inspect large areas
- will have robotic features to facilitate control
- will use electricity to drive and solar panels to generate it



Two motors located on the wings are used to rotate the screws and drive the wheels.



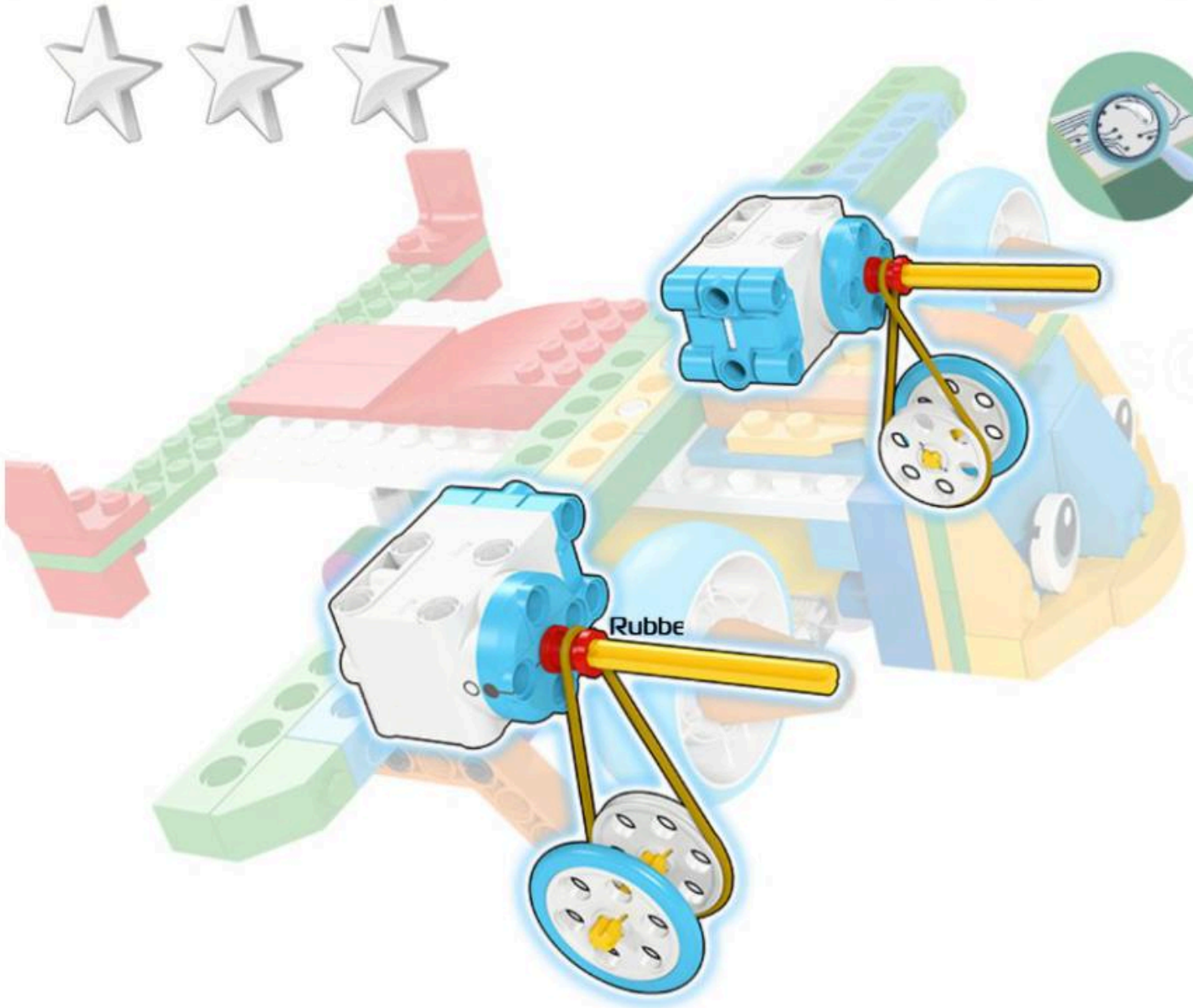


Find:

- Smarthub
- Left motor
- Right motor



The rubber belts allow rotation transfer from the motors to the wheels and turn the axis of the rotation for 90 degrees.



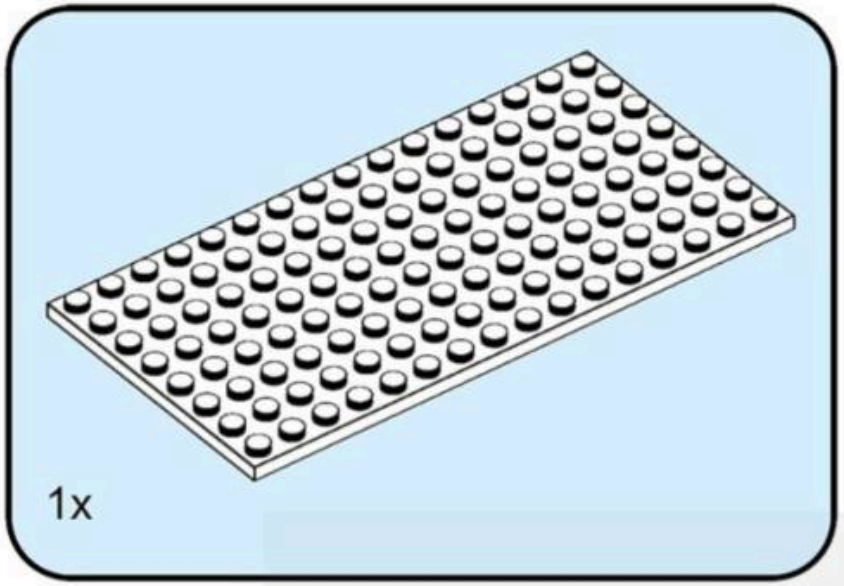


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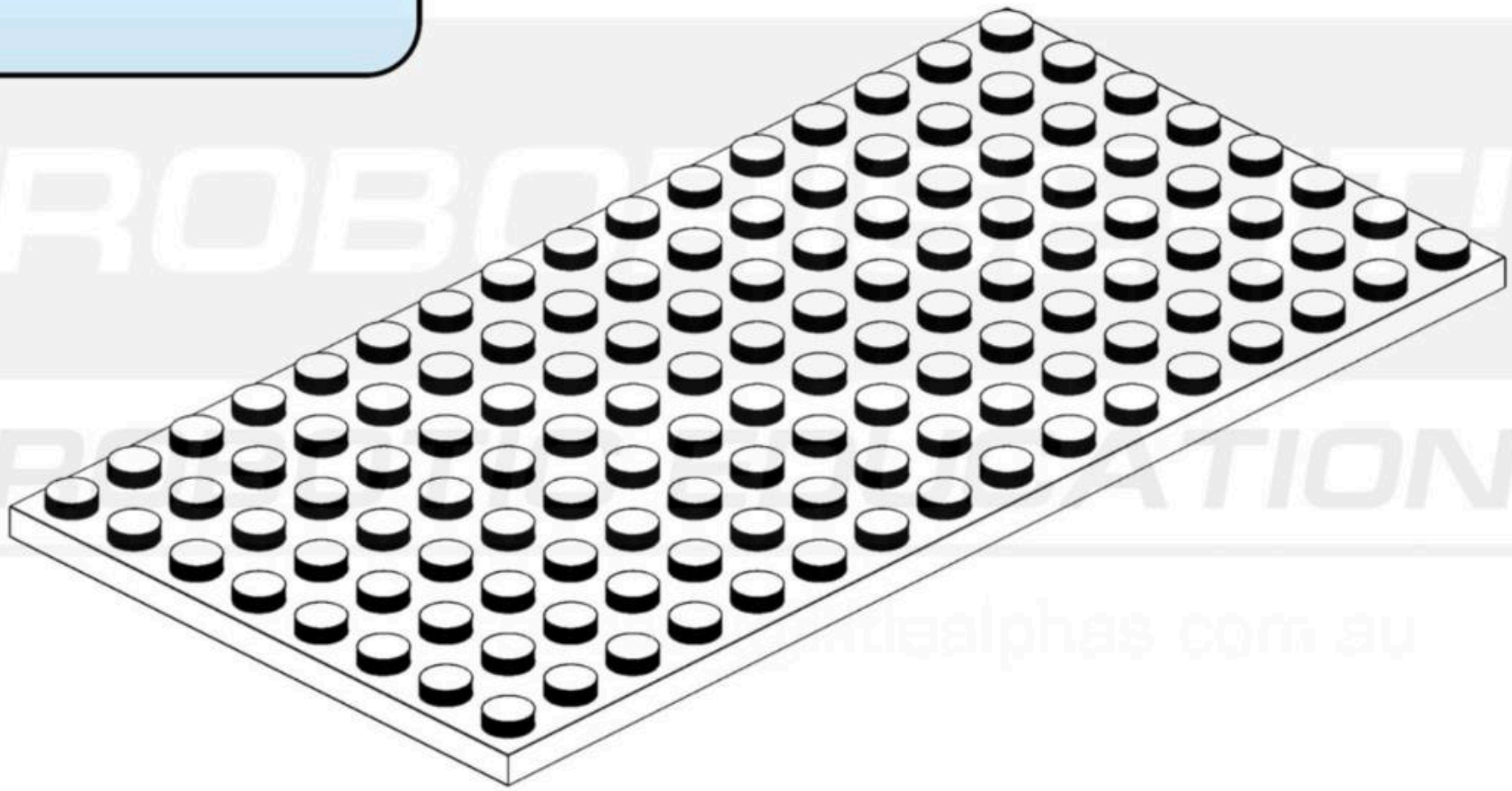
- motors
- pulleys
- rubber belts

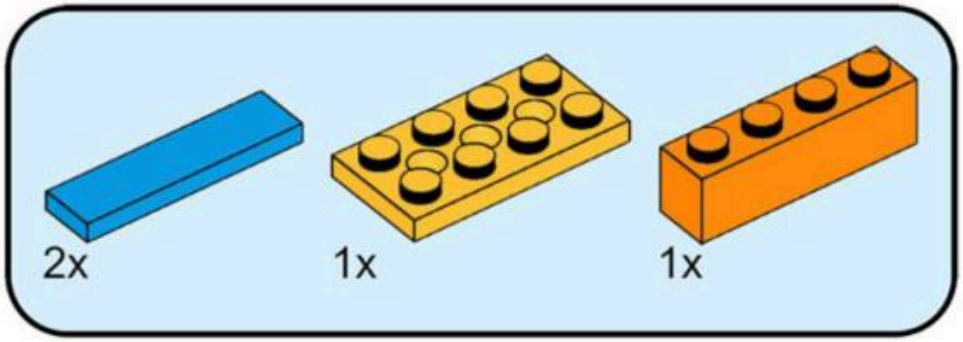
Design features

The rotation from the motors to the wheels is transmitted thanks to belt gears. They not only reduce the speed, but also turn the axis of rotation by 90 degrees

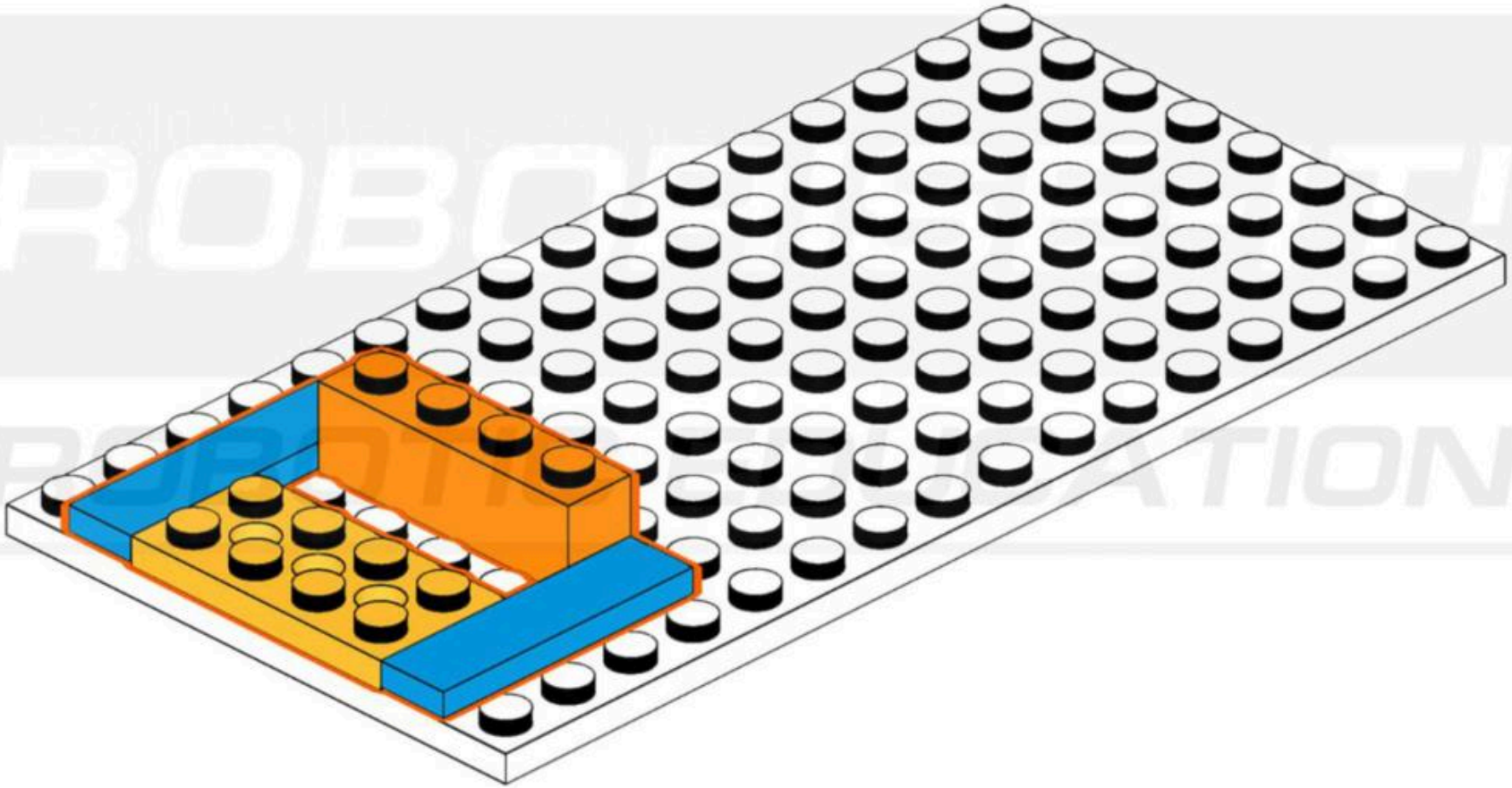


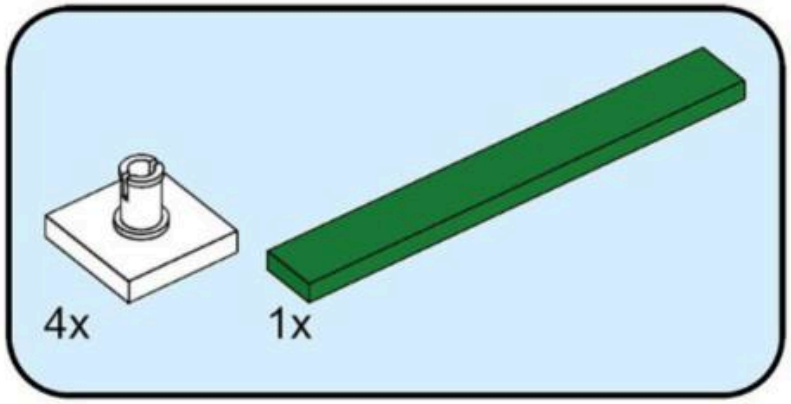
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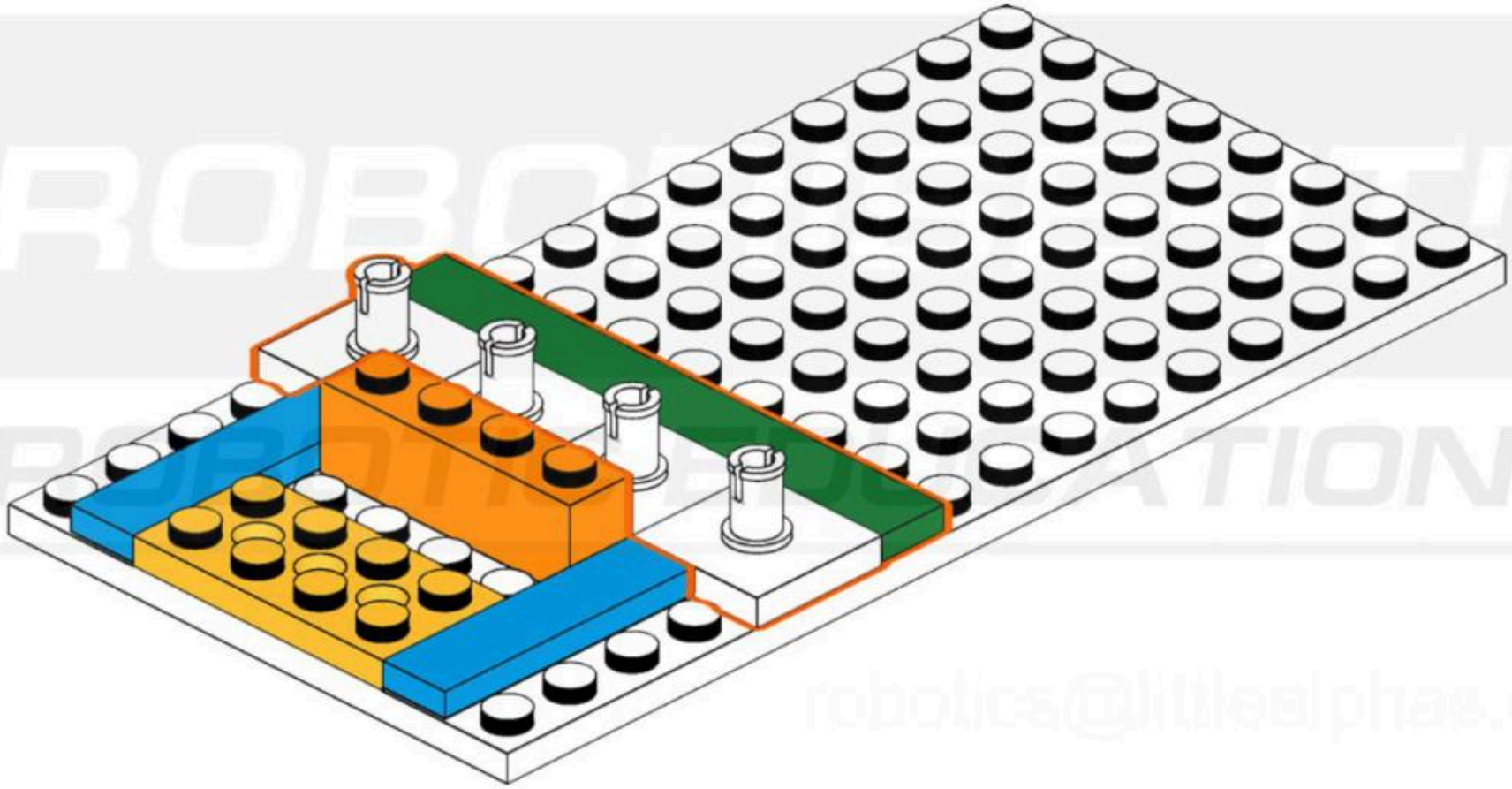


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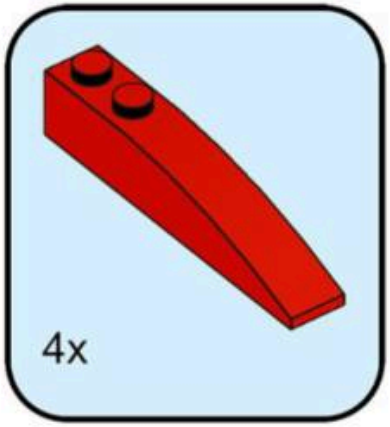




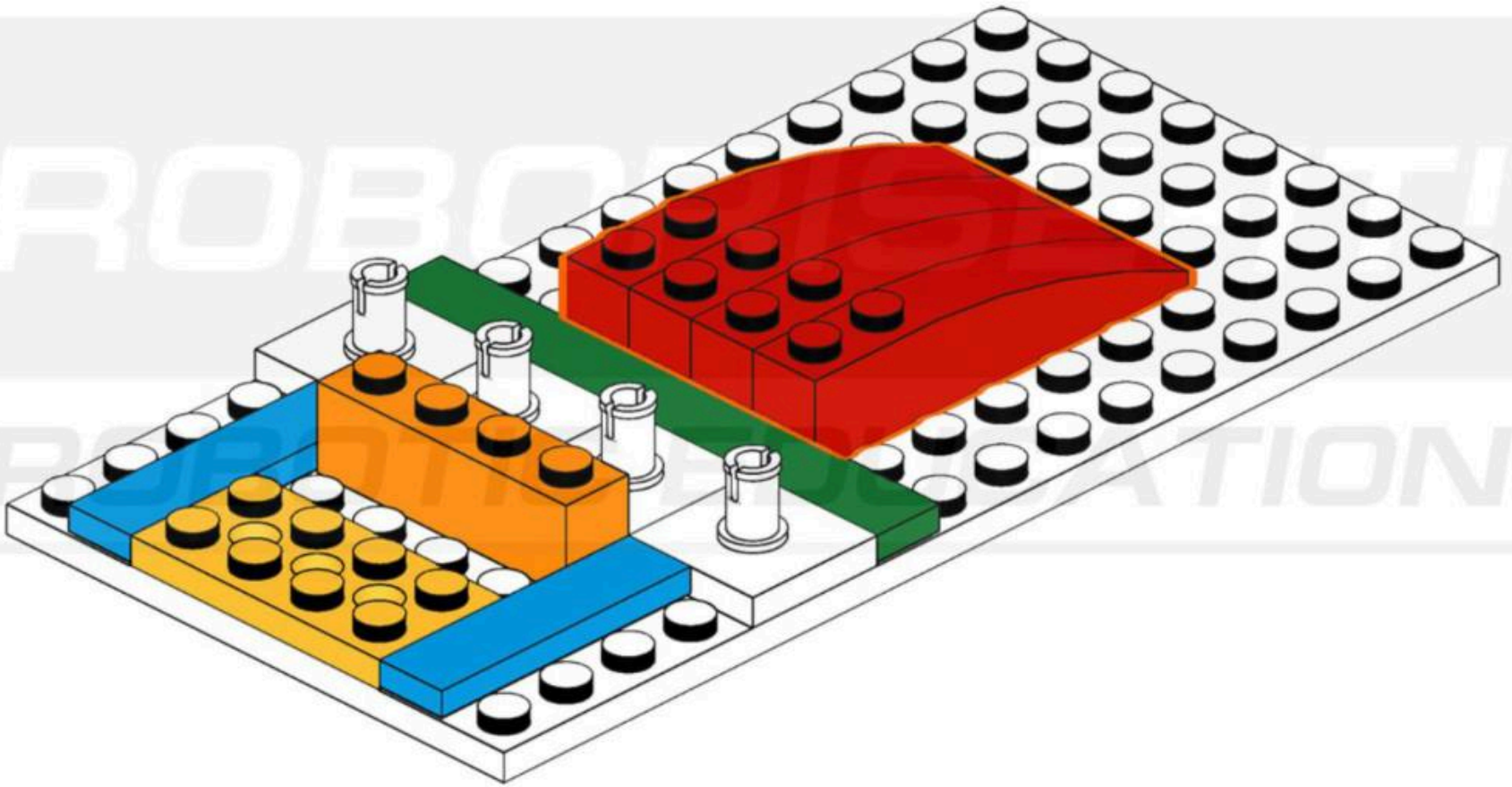
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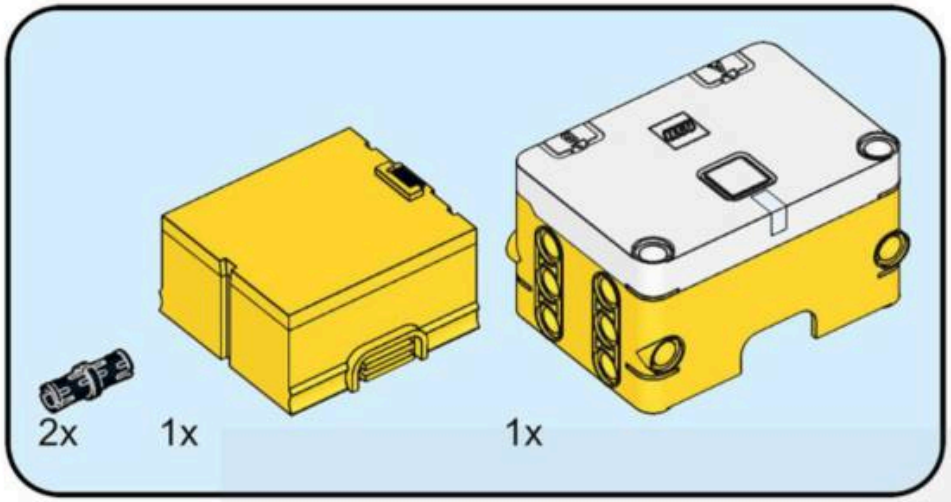


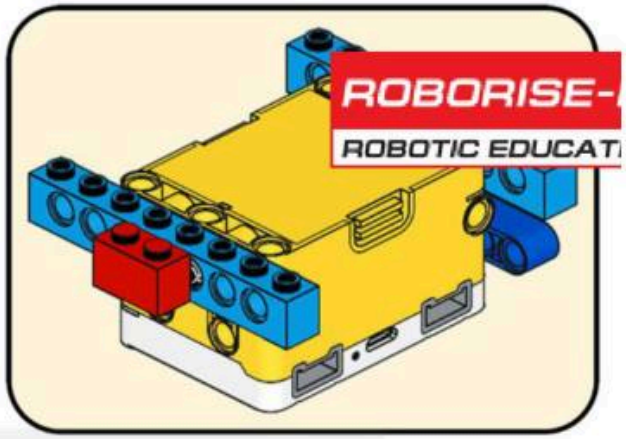
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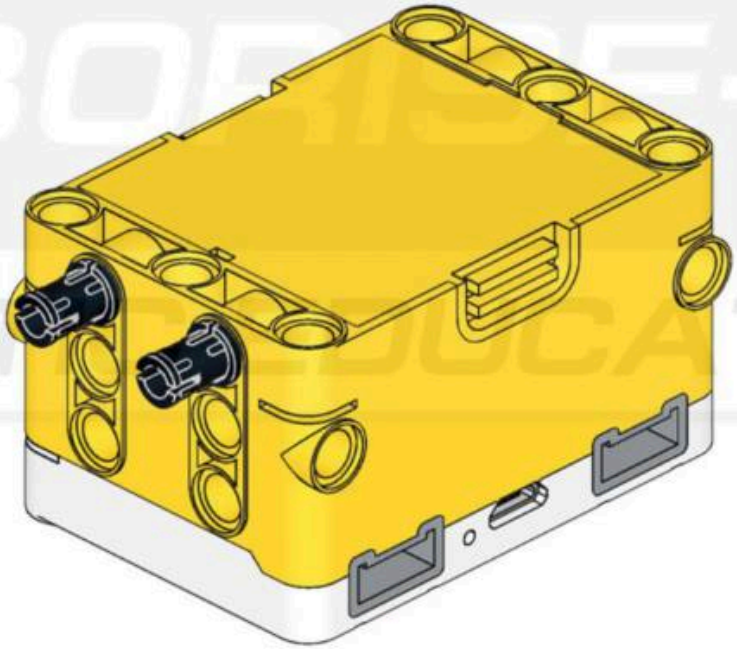
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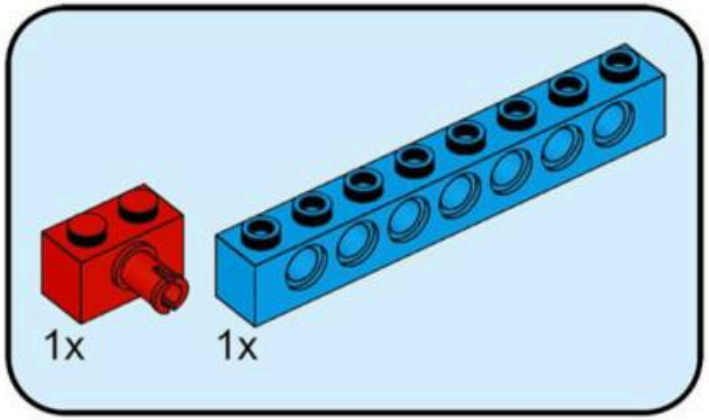




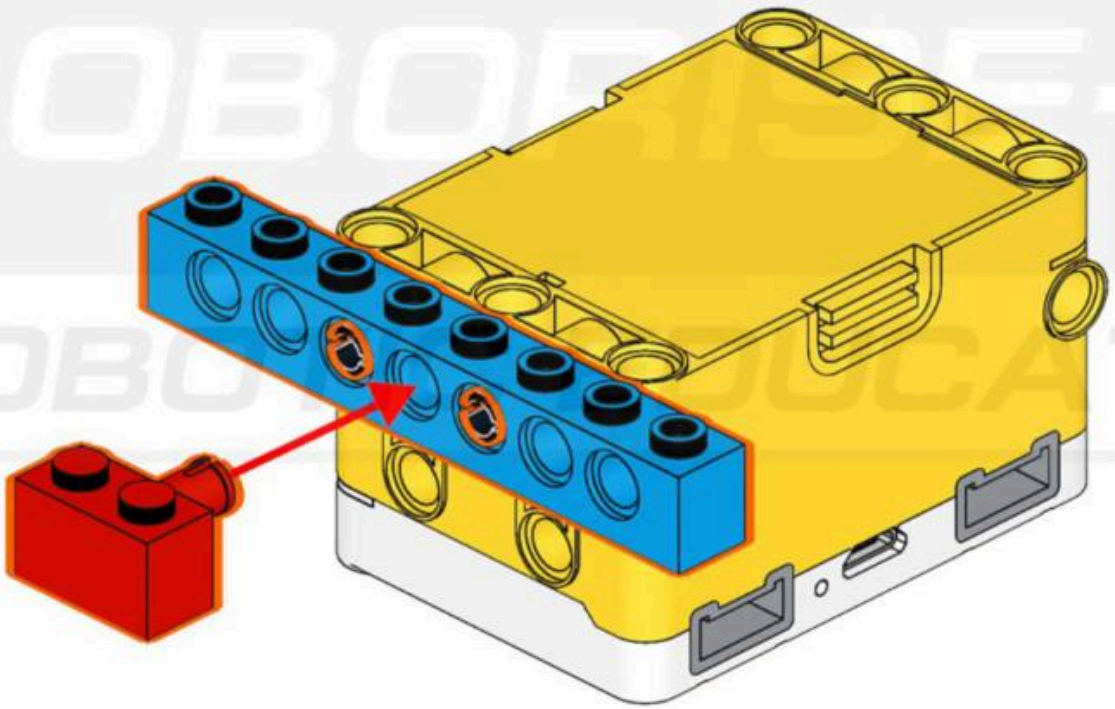
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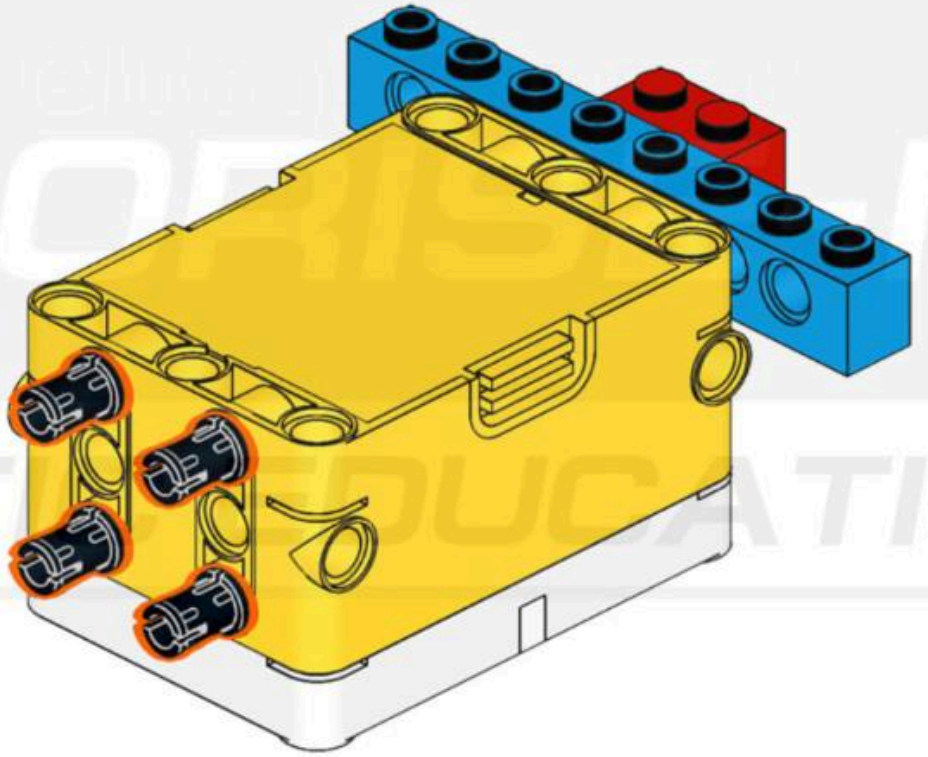
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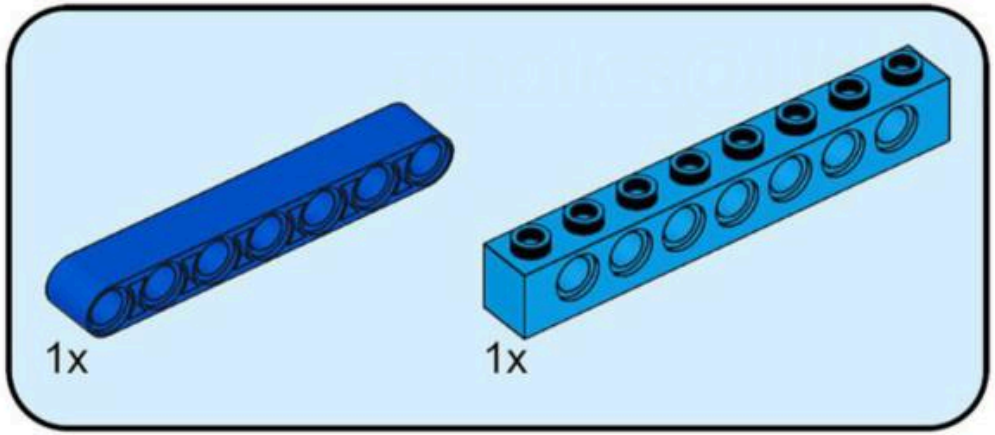
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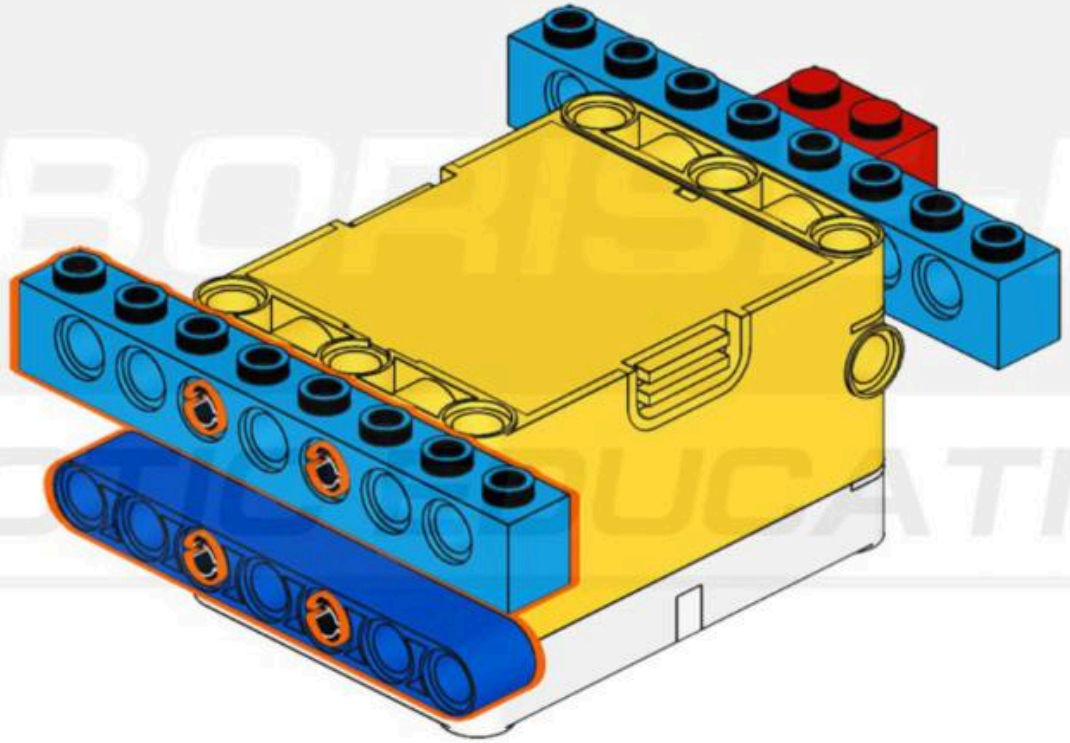




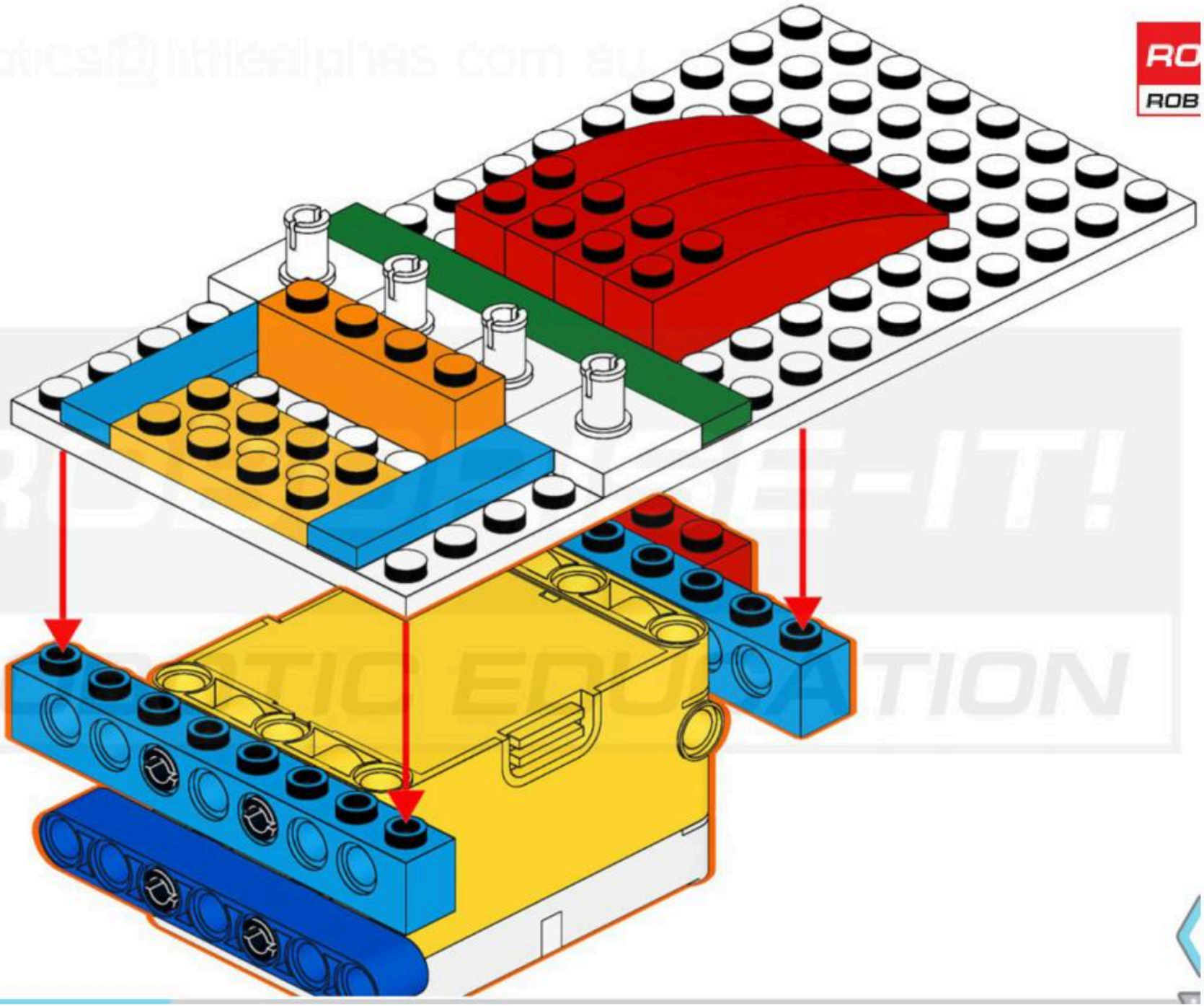




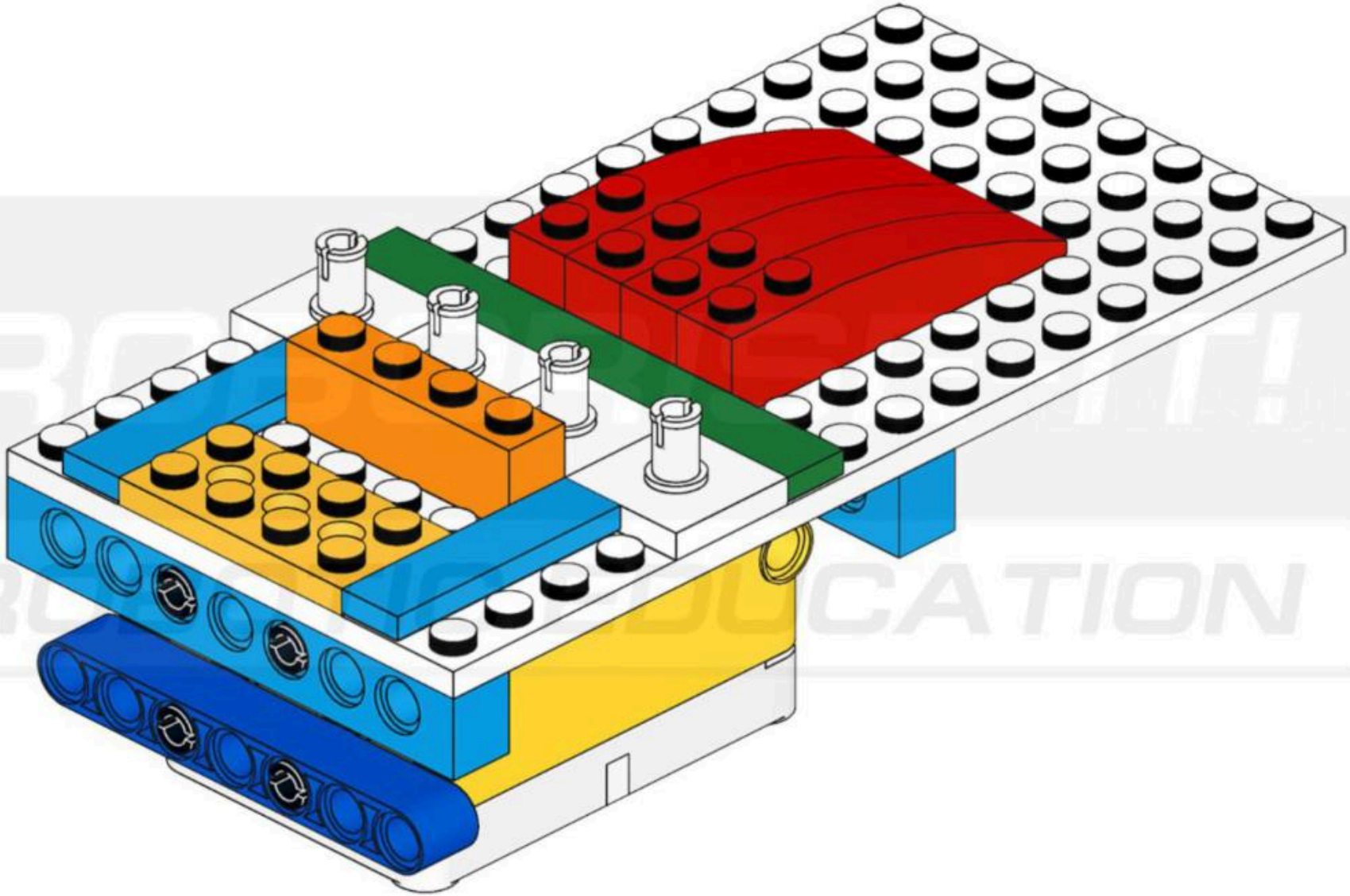
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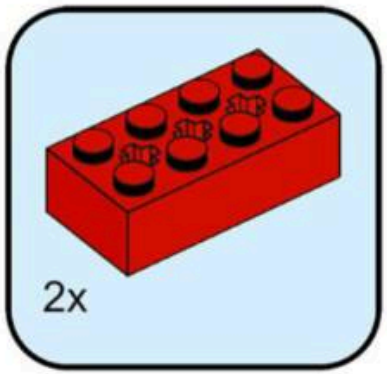


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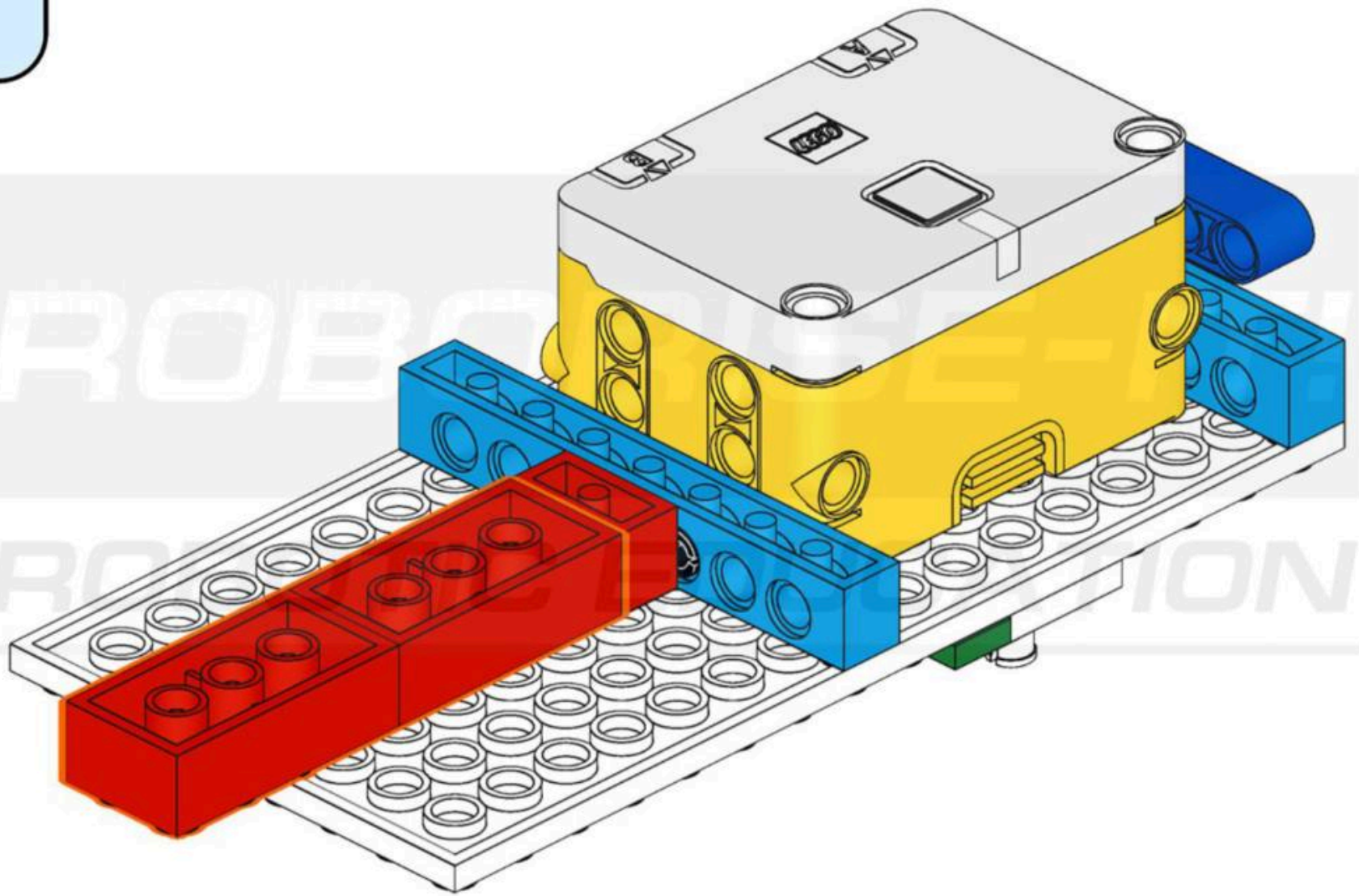


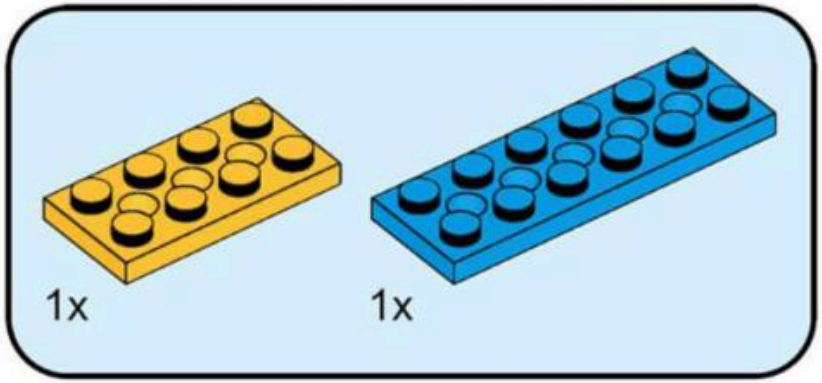
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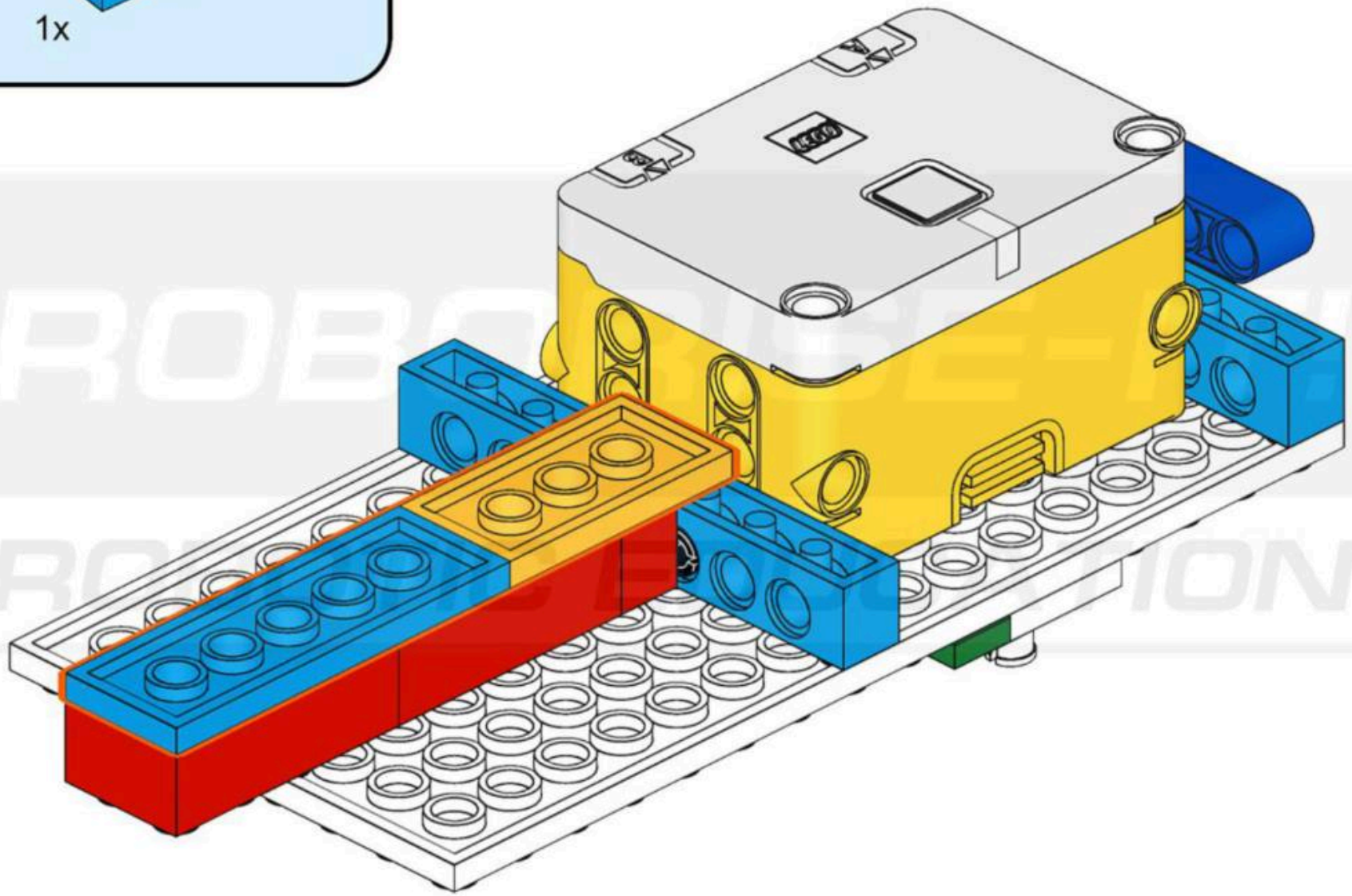


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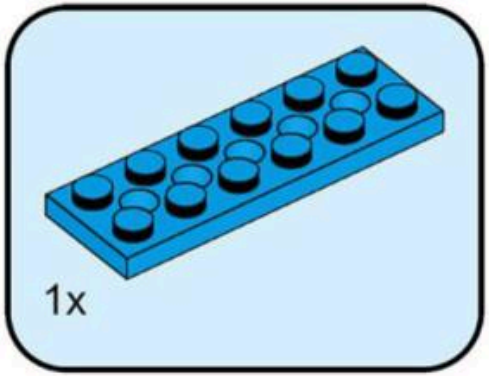




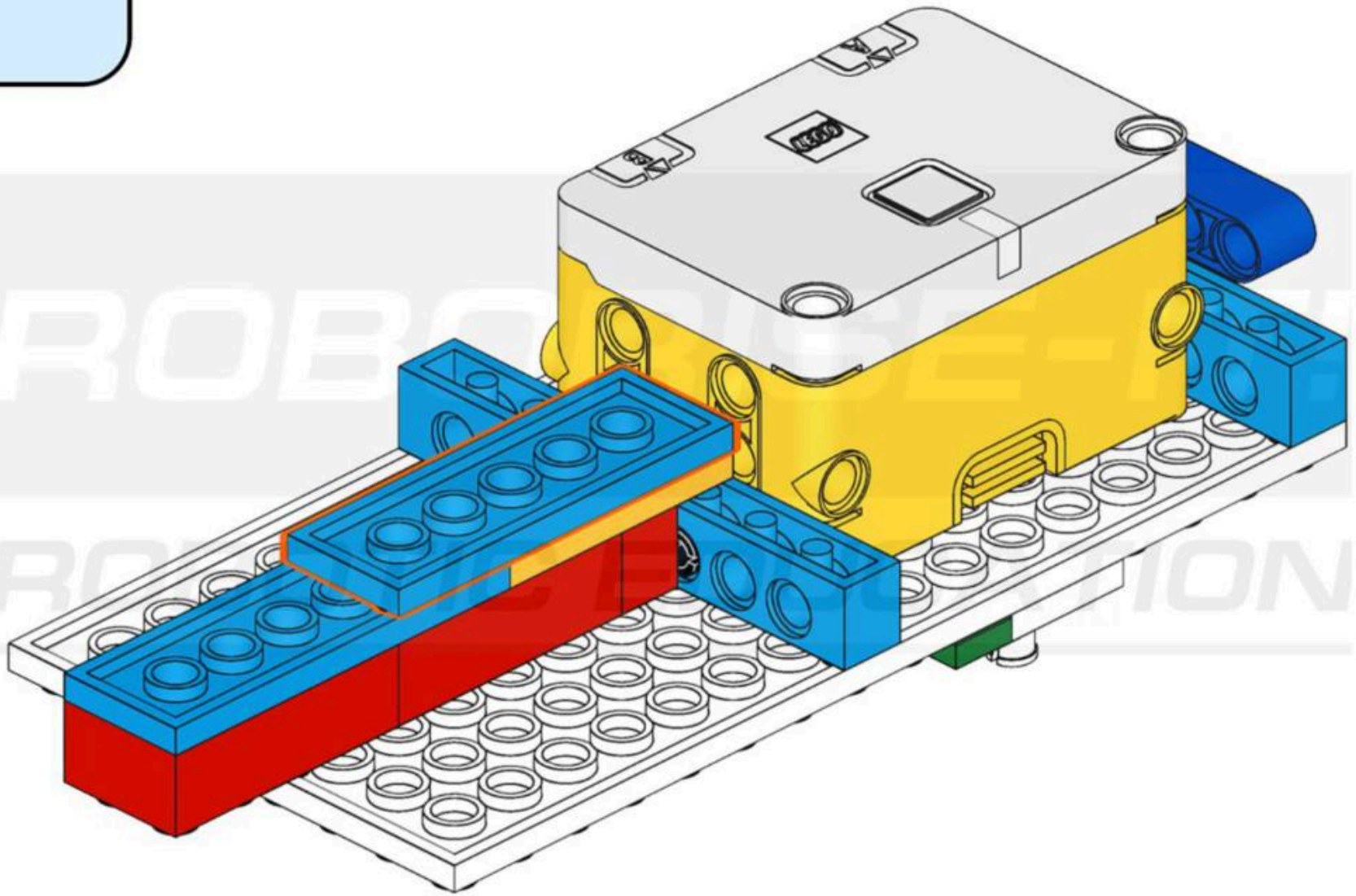
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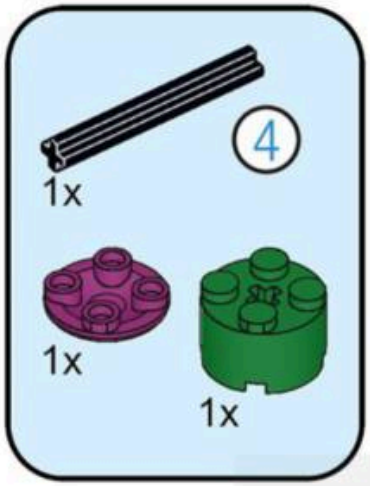


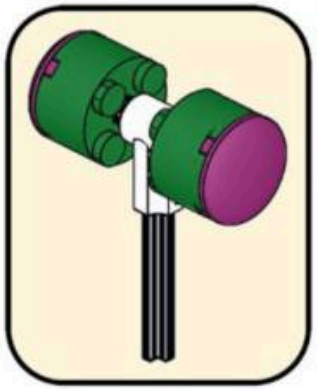
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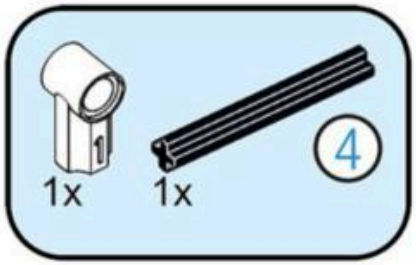




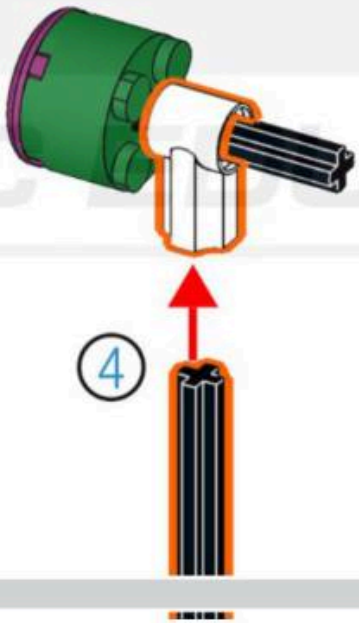


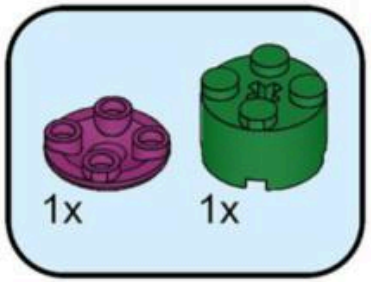
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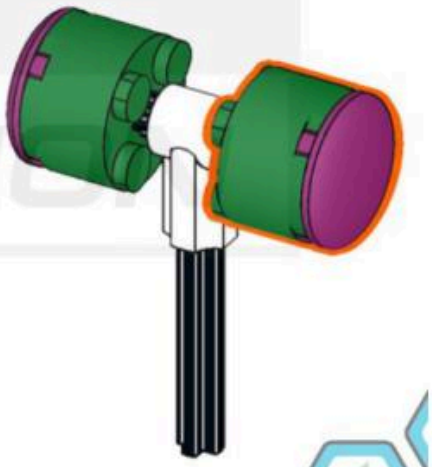


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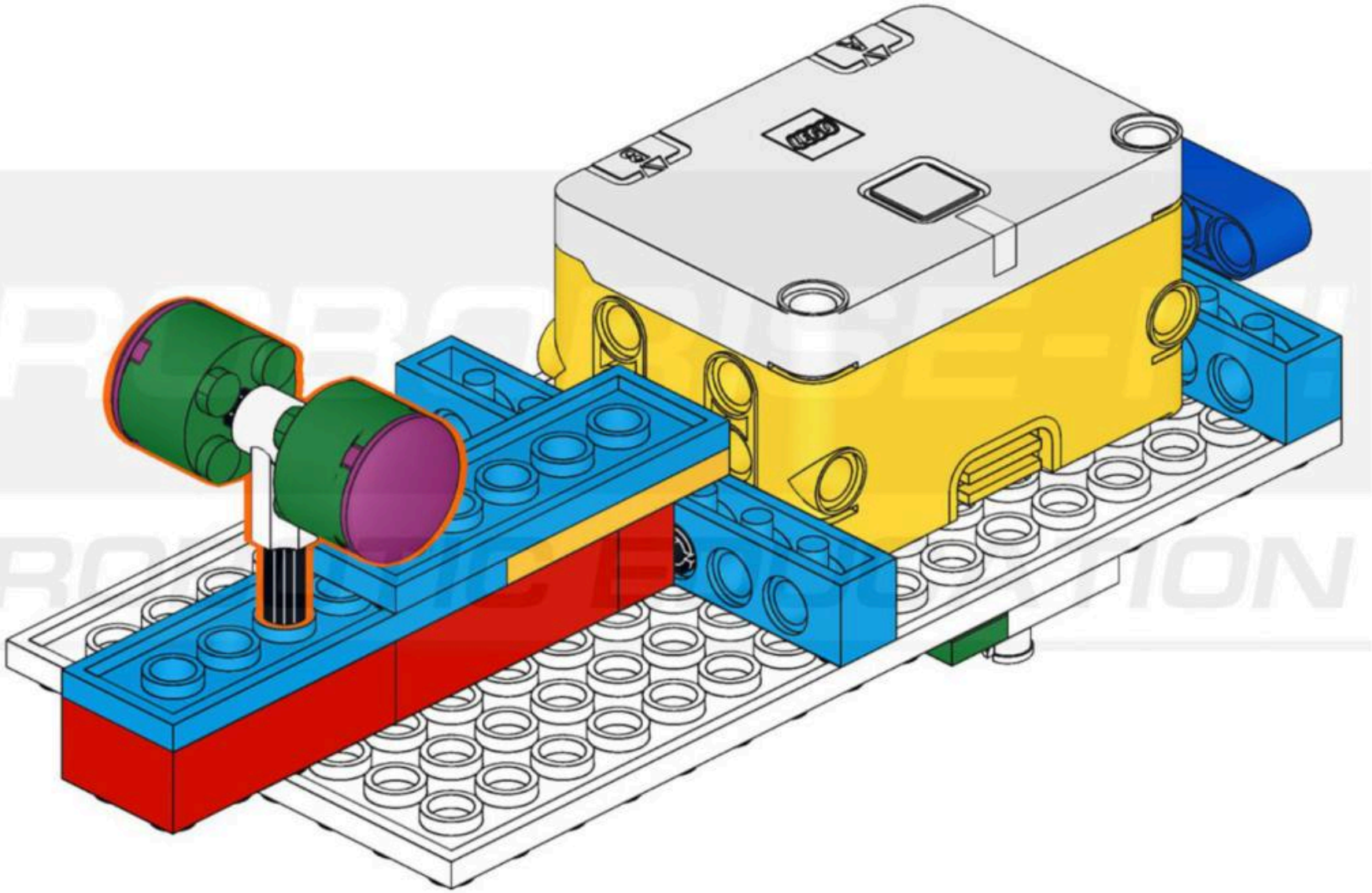


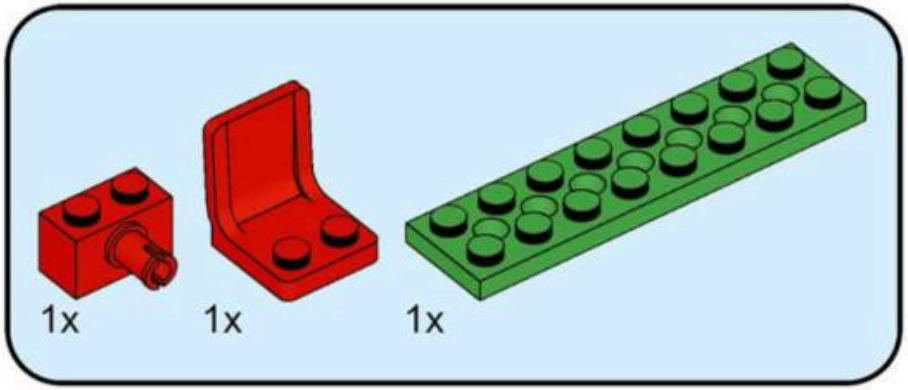
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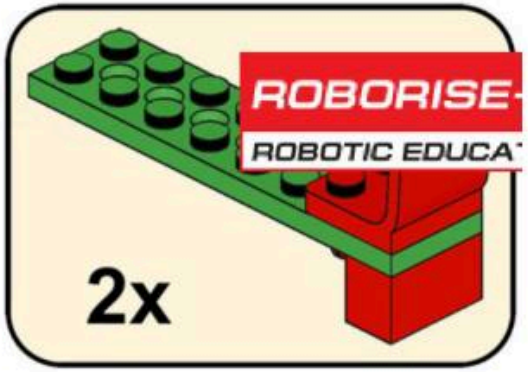


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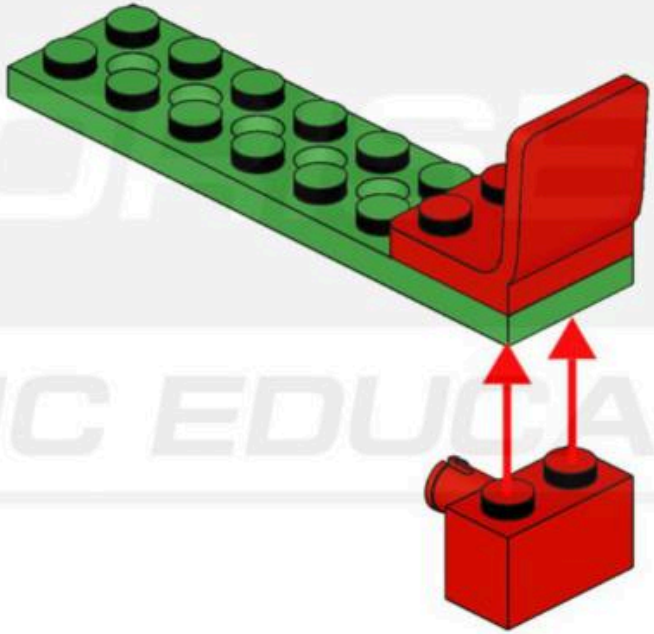
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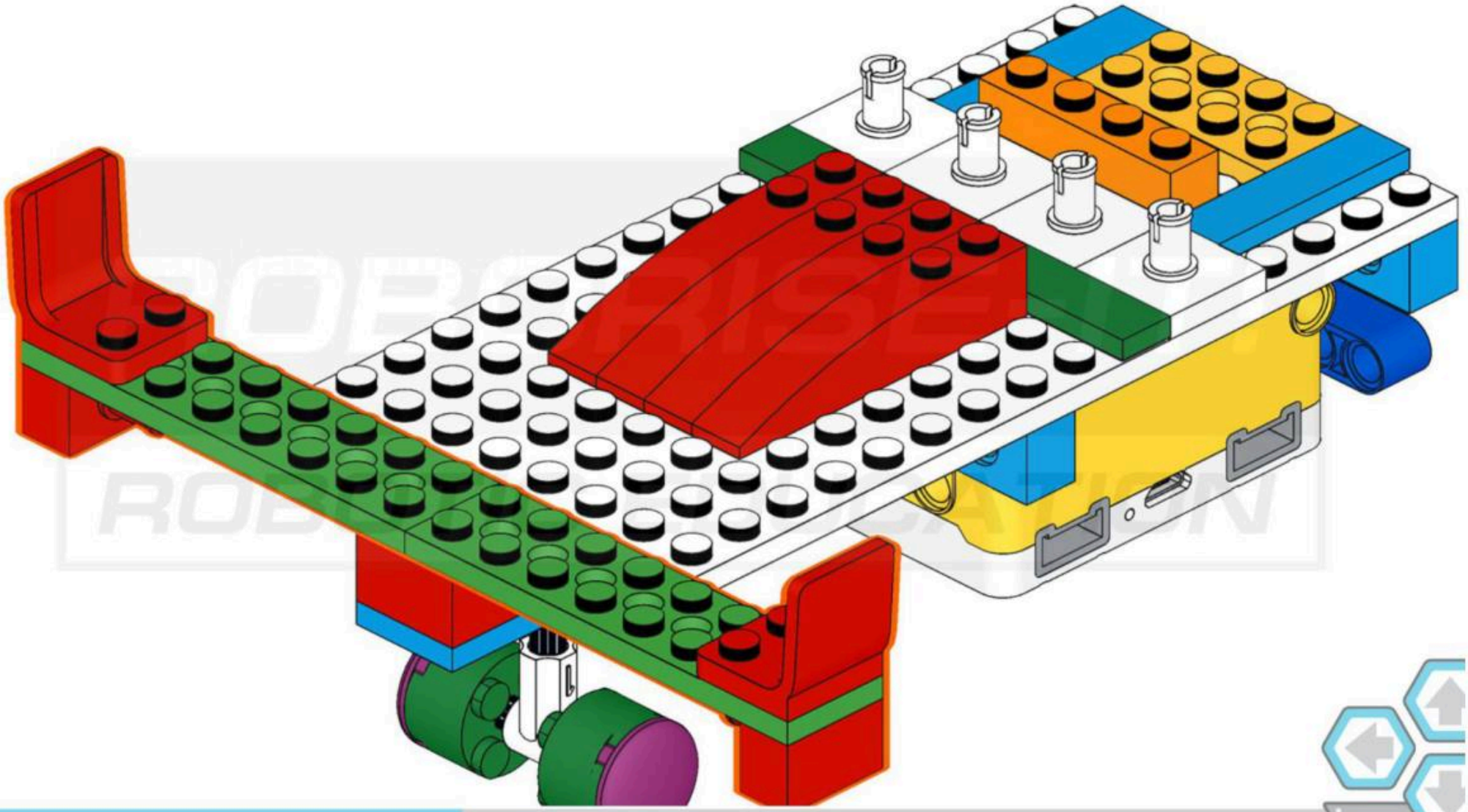


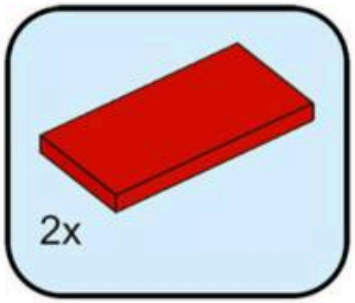
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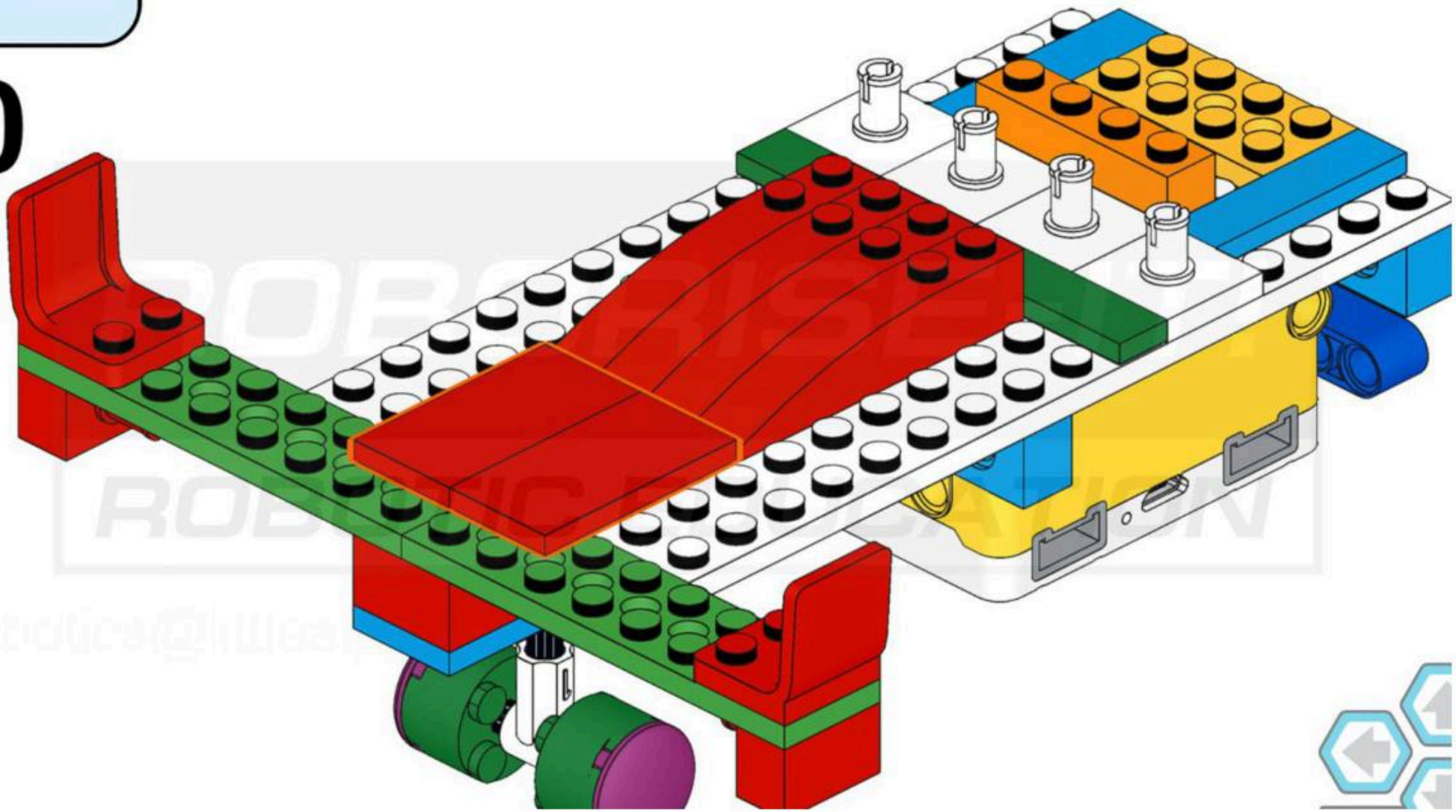
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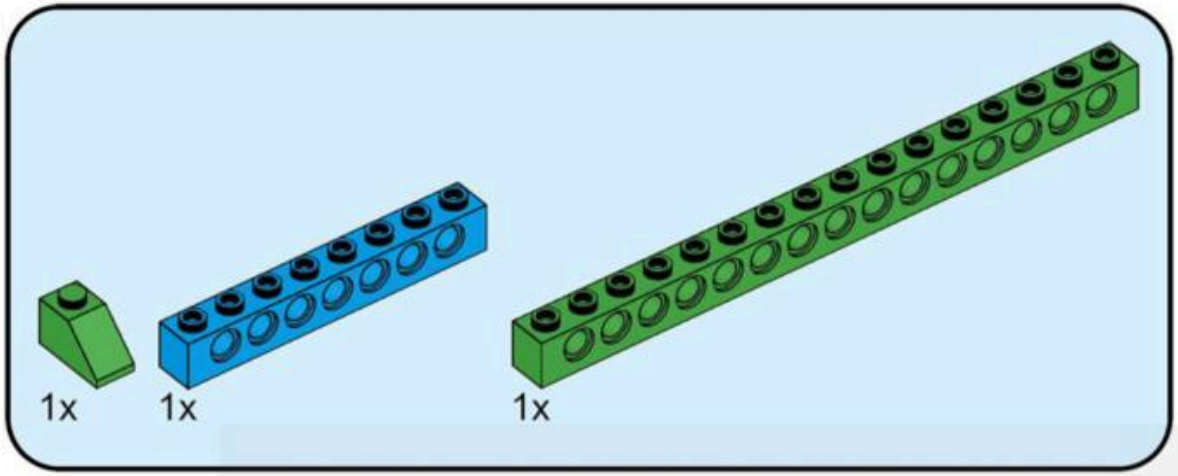
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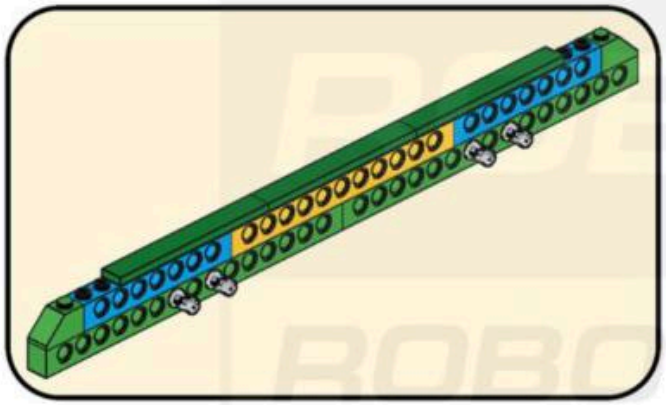


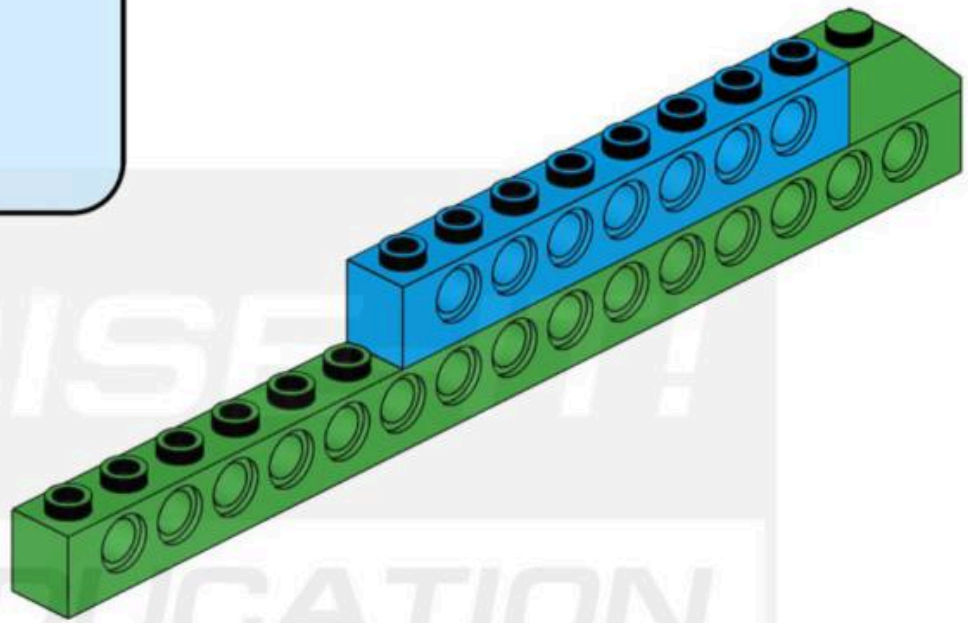


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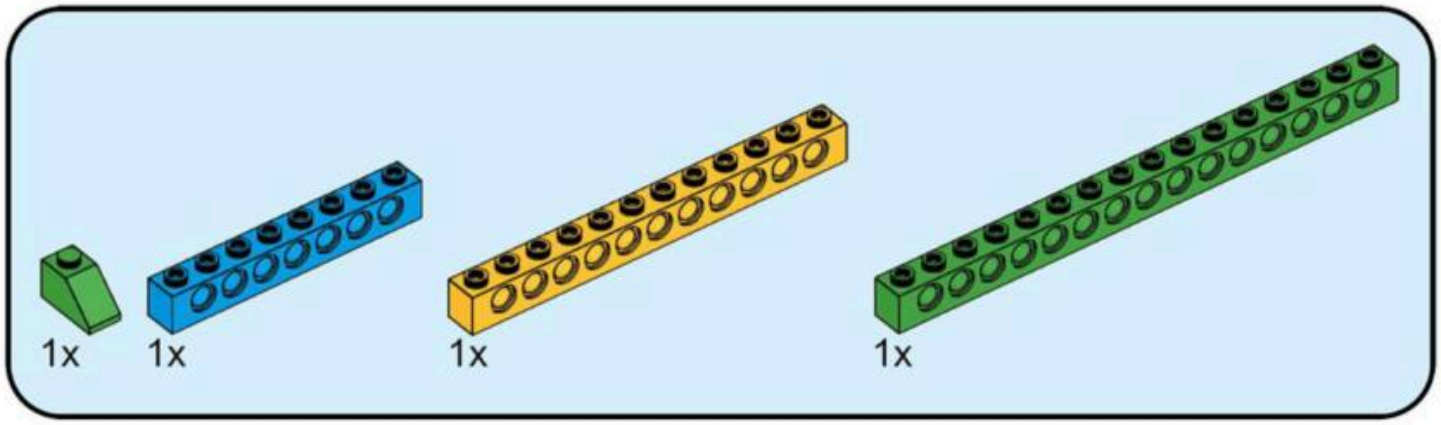




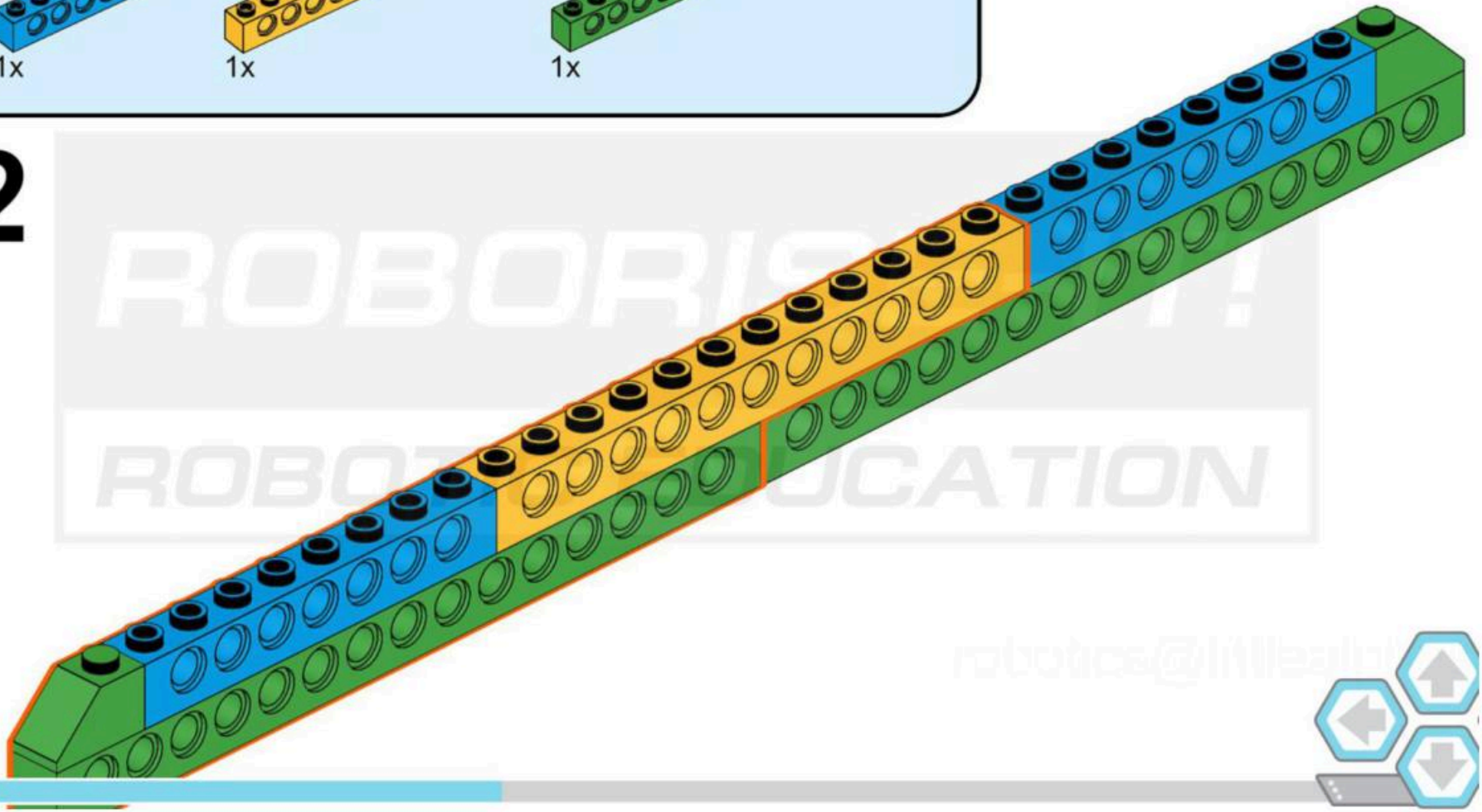




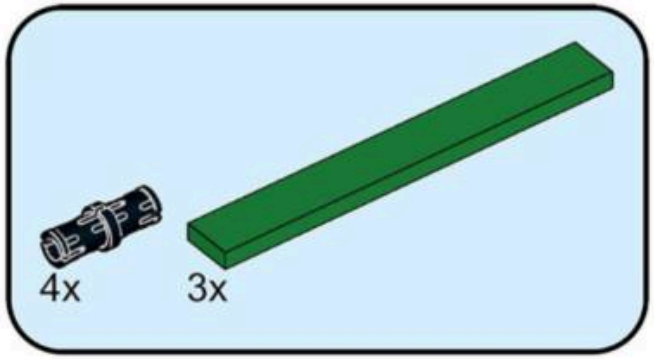
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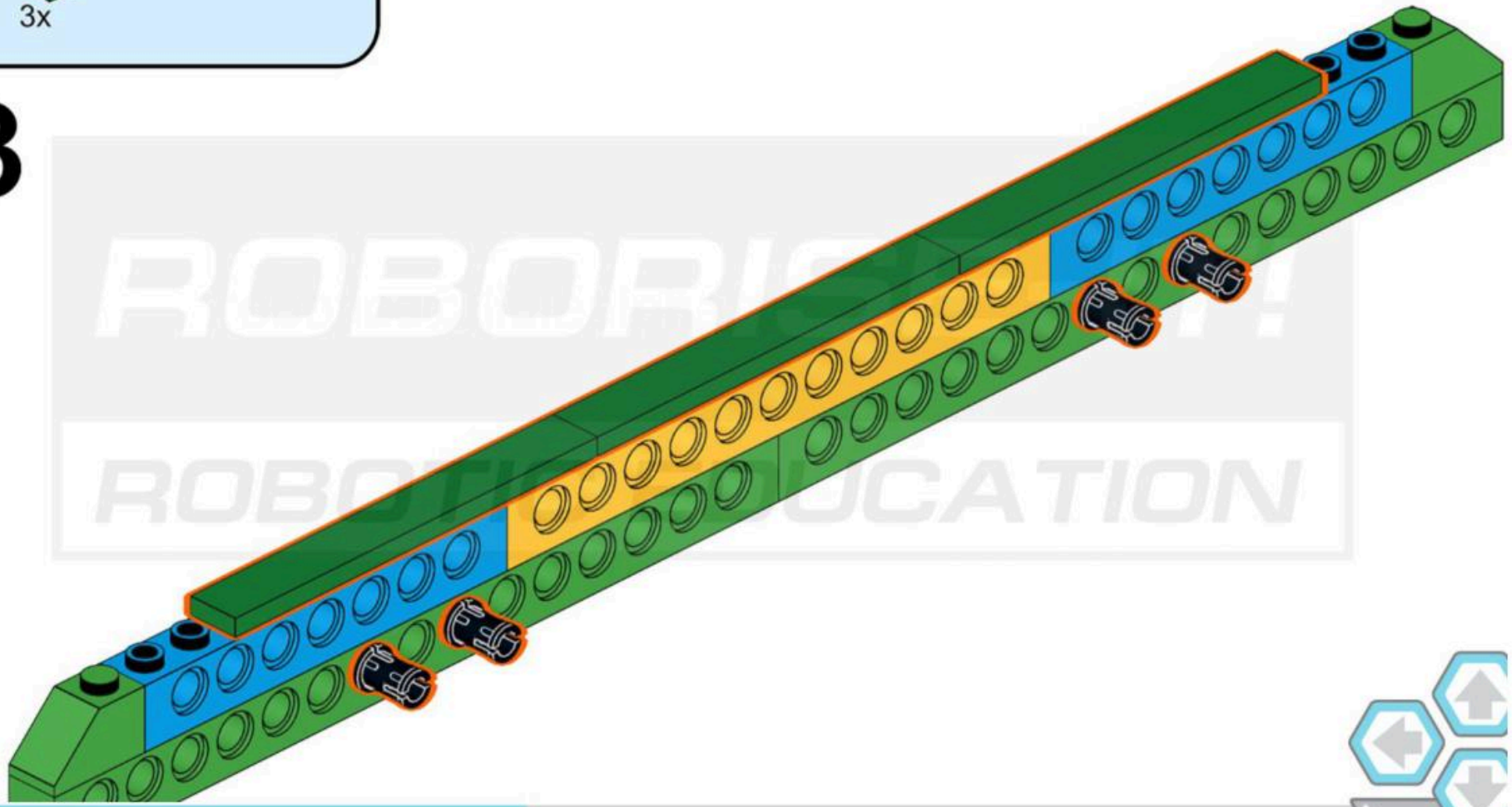
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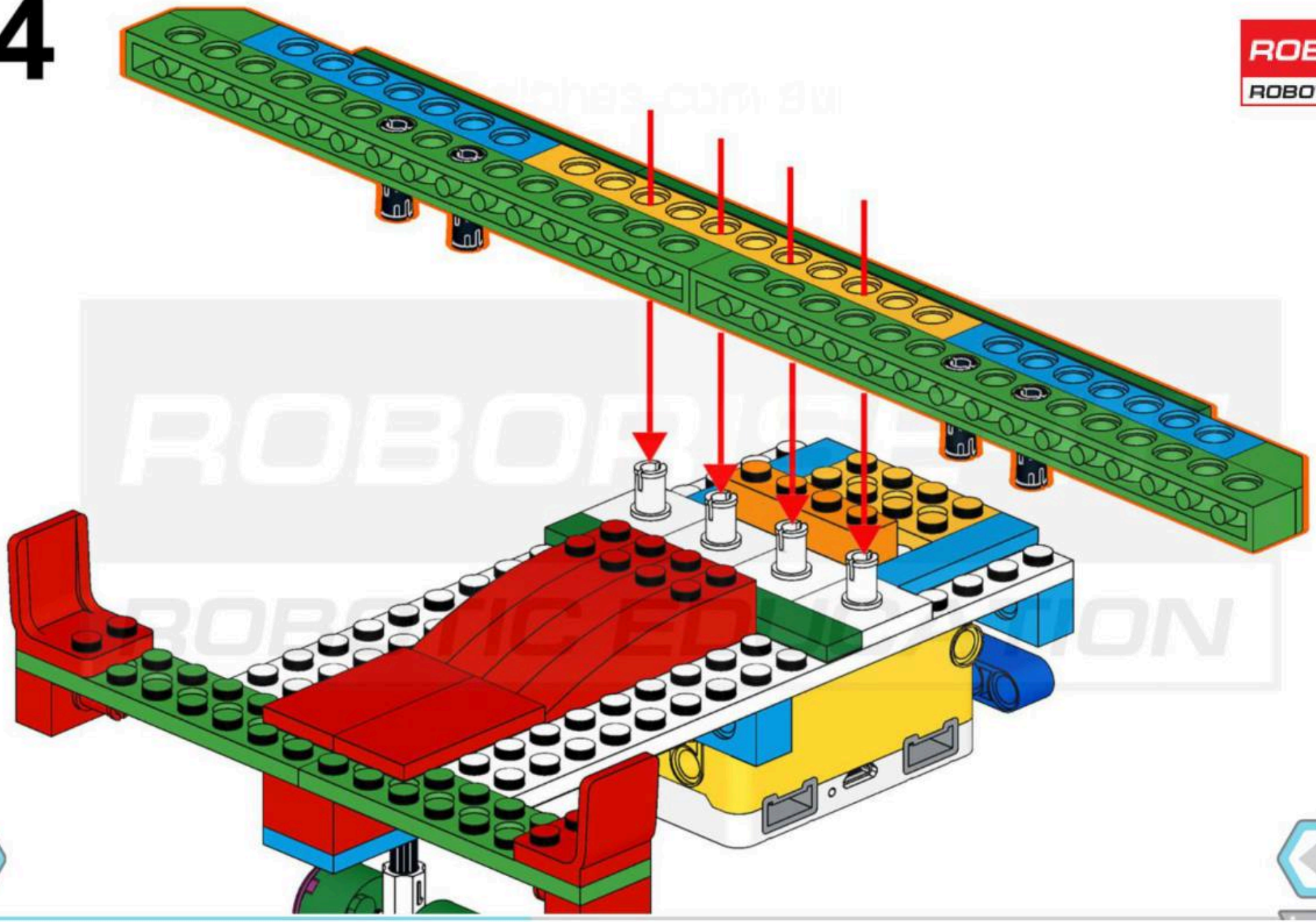


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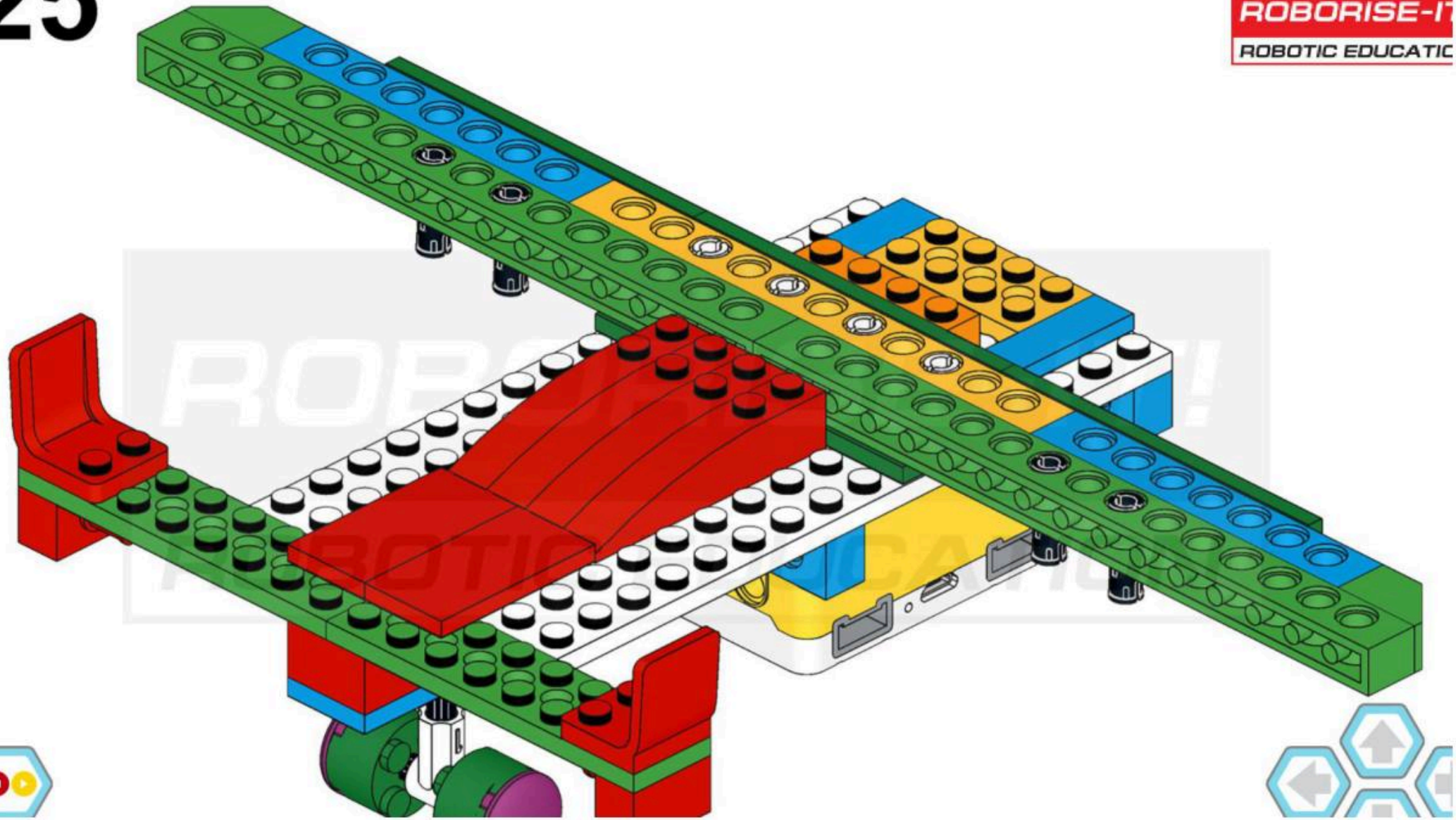


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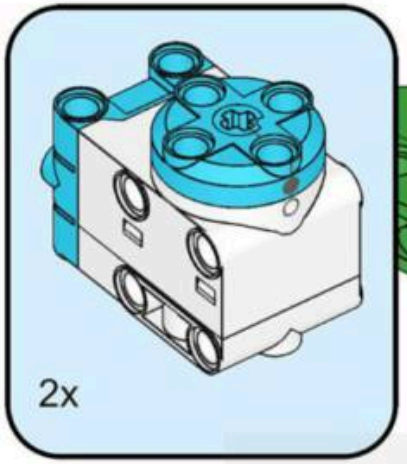
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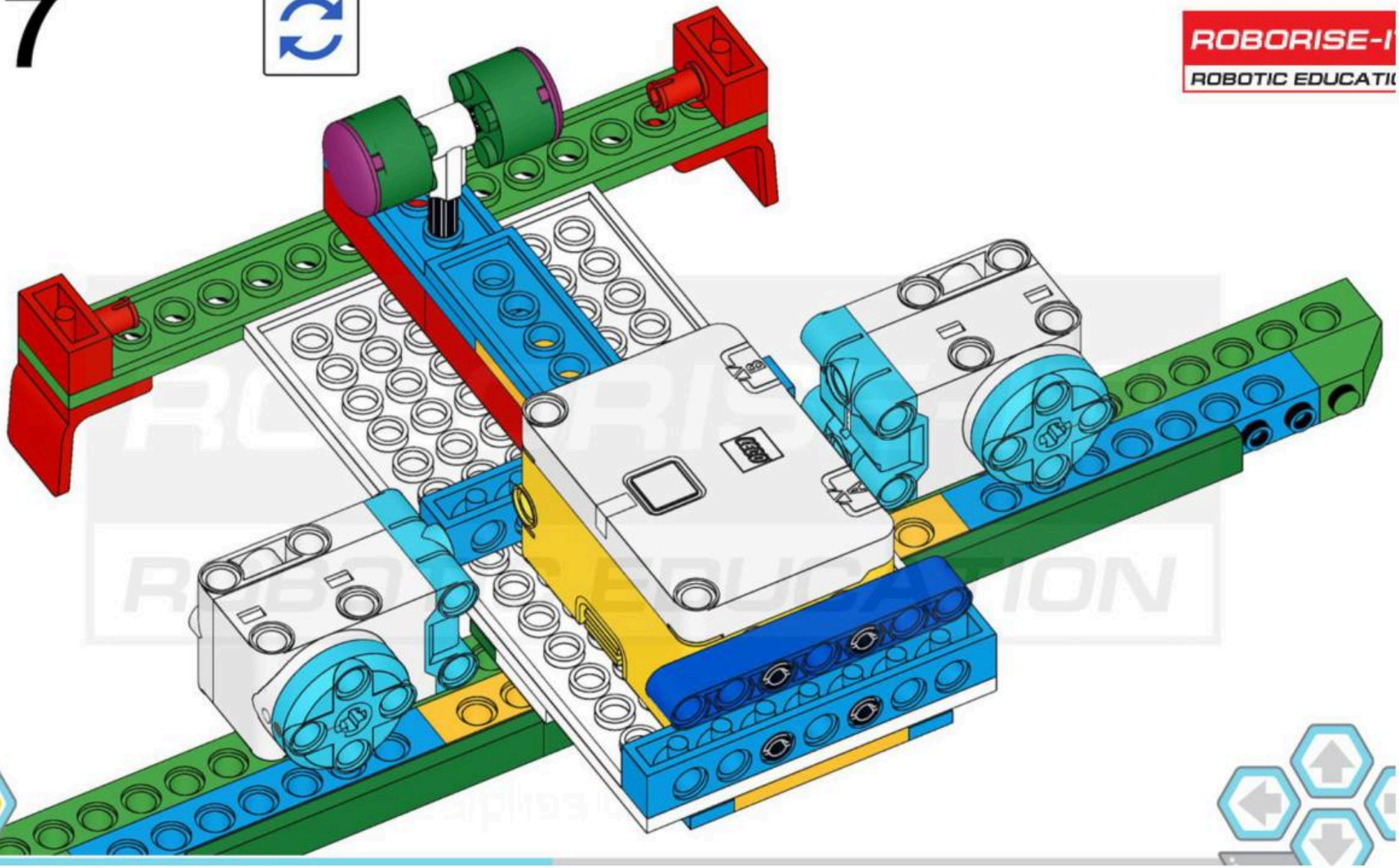
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24/43

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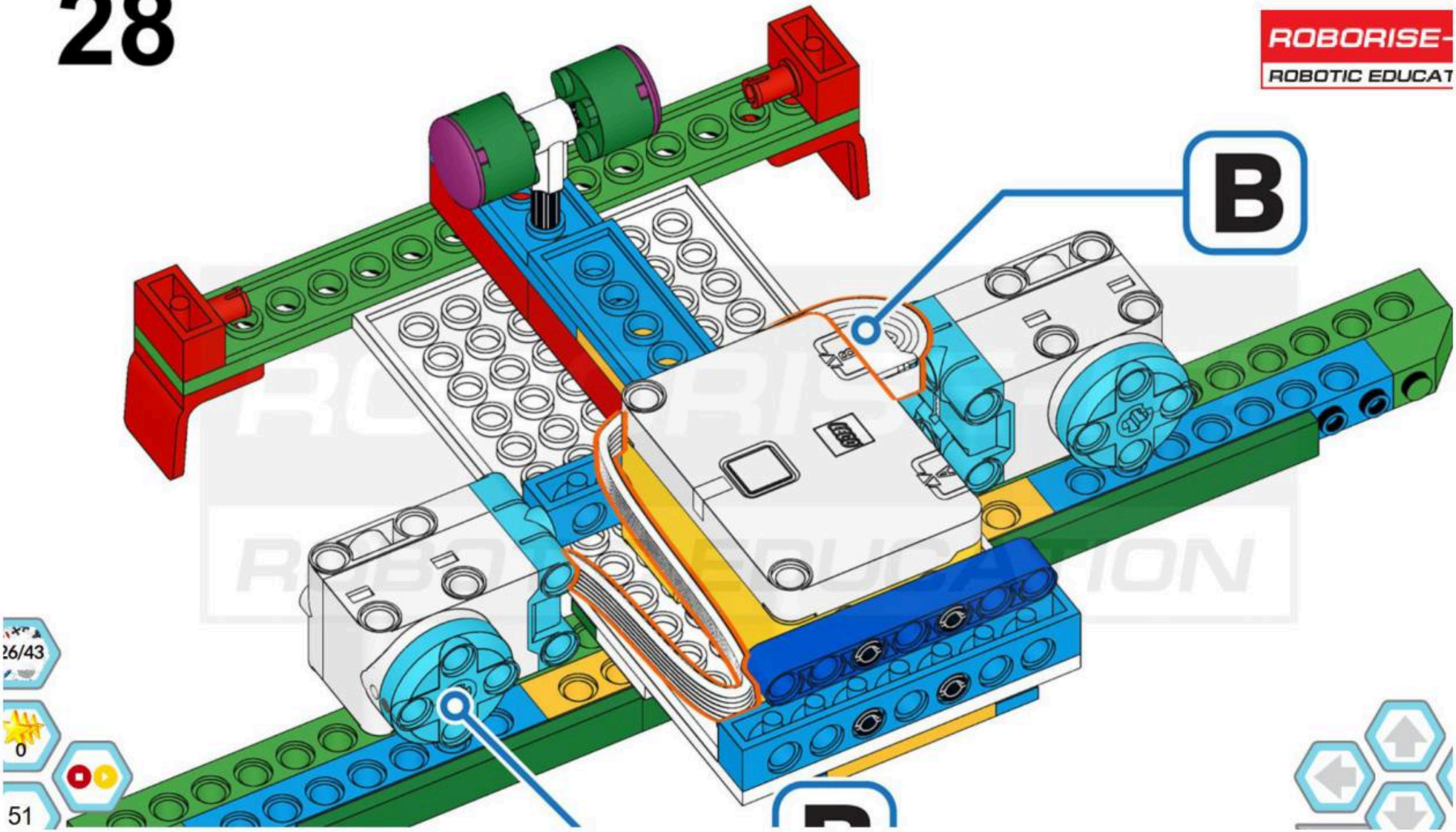
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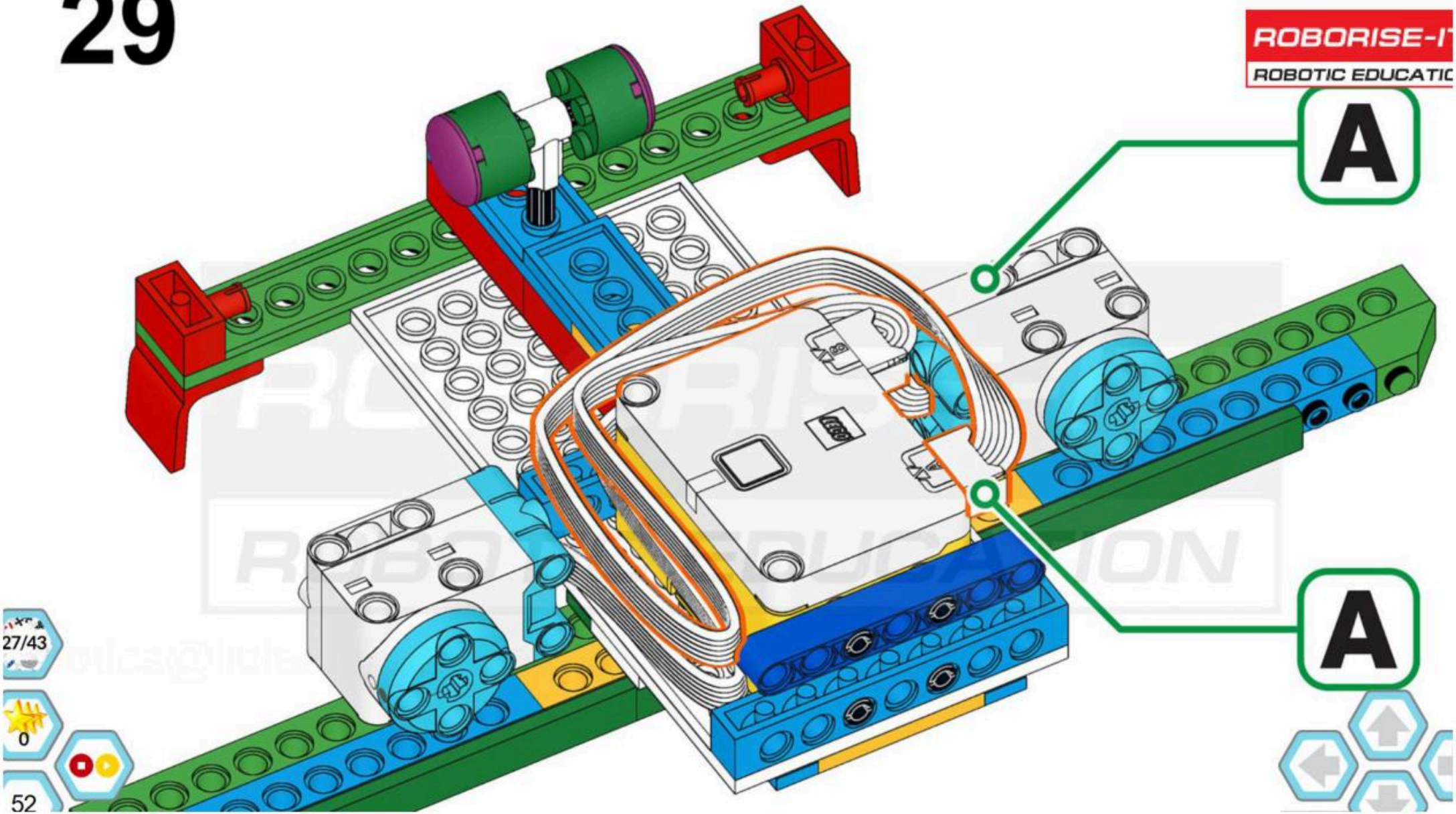
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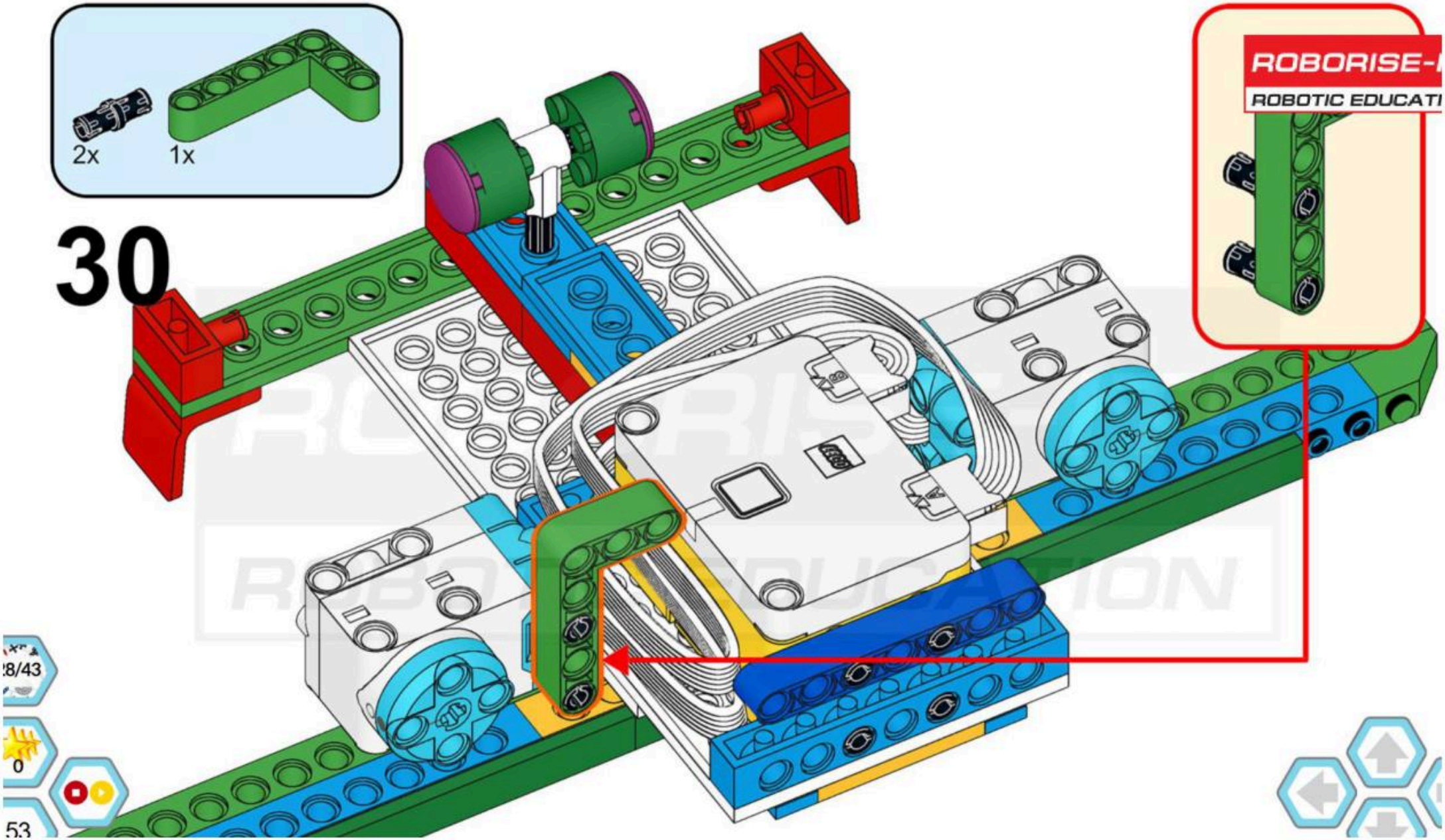
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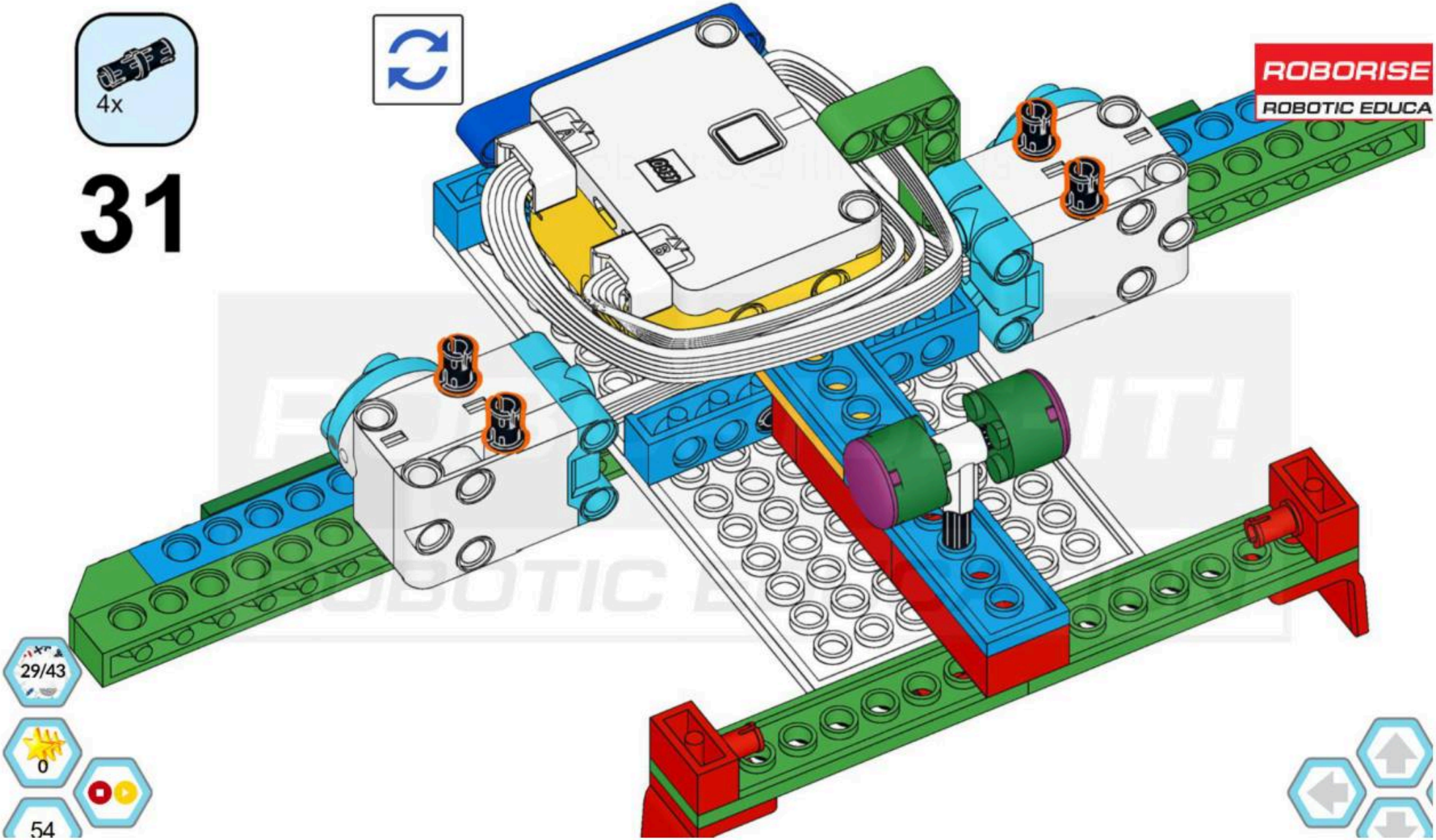


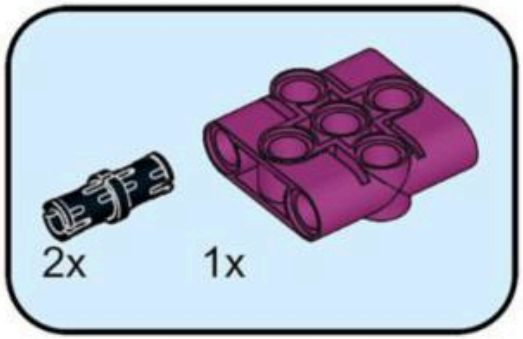
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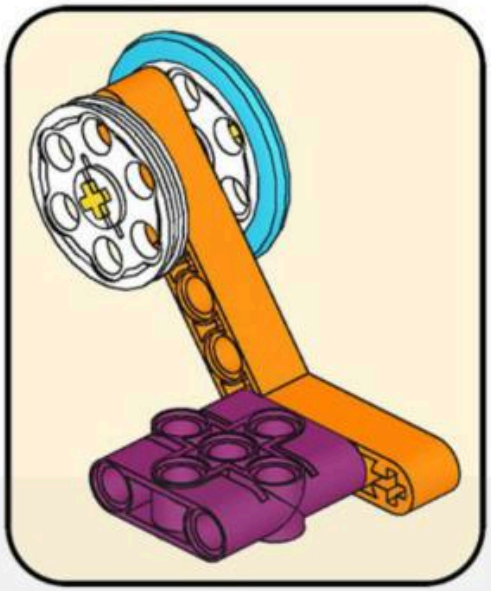


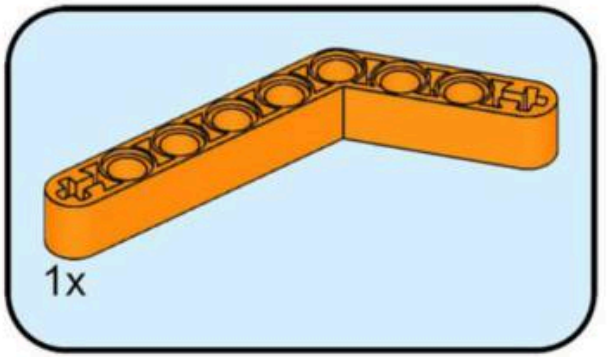

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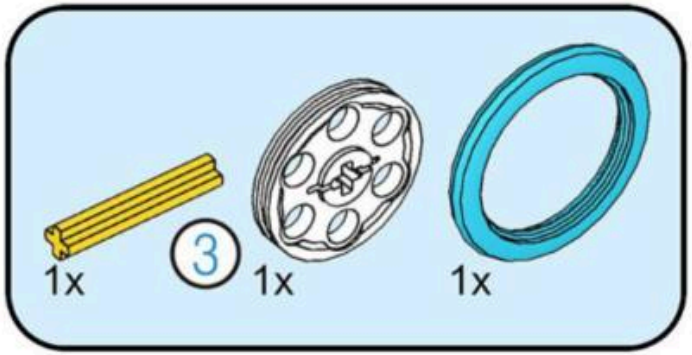


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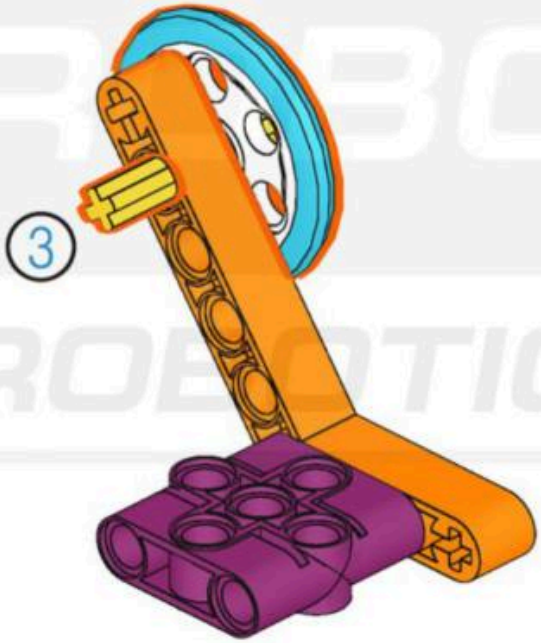


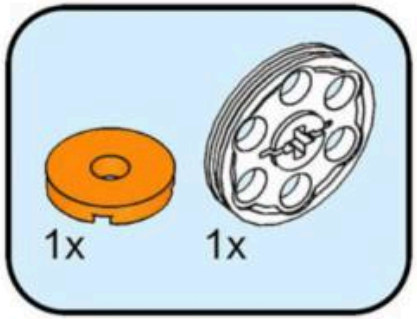


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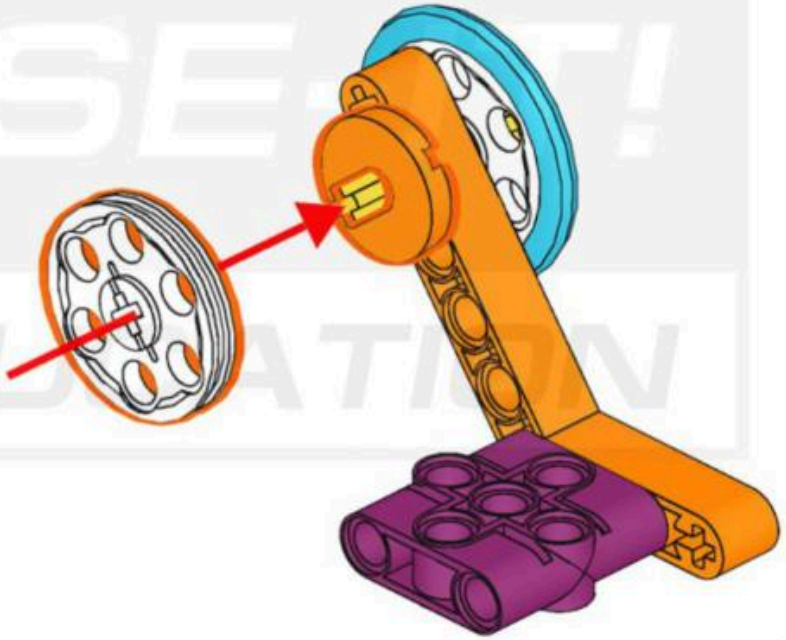


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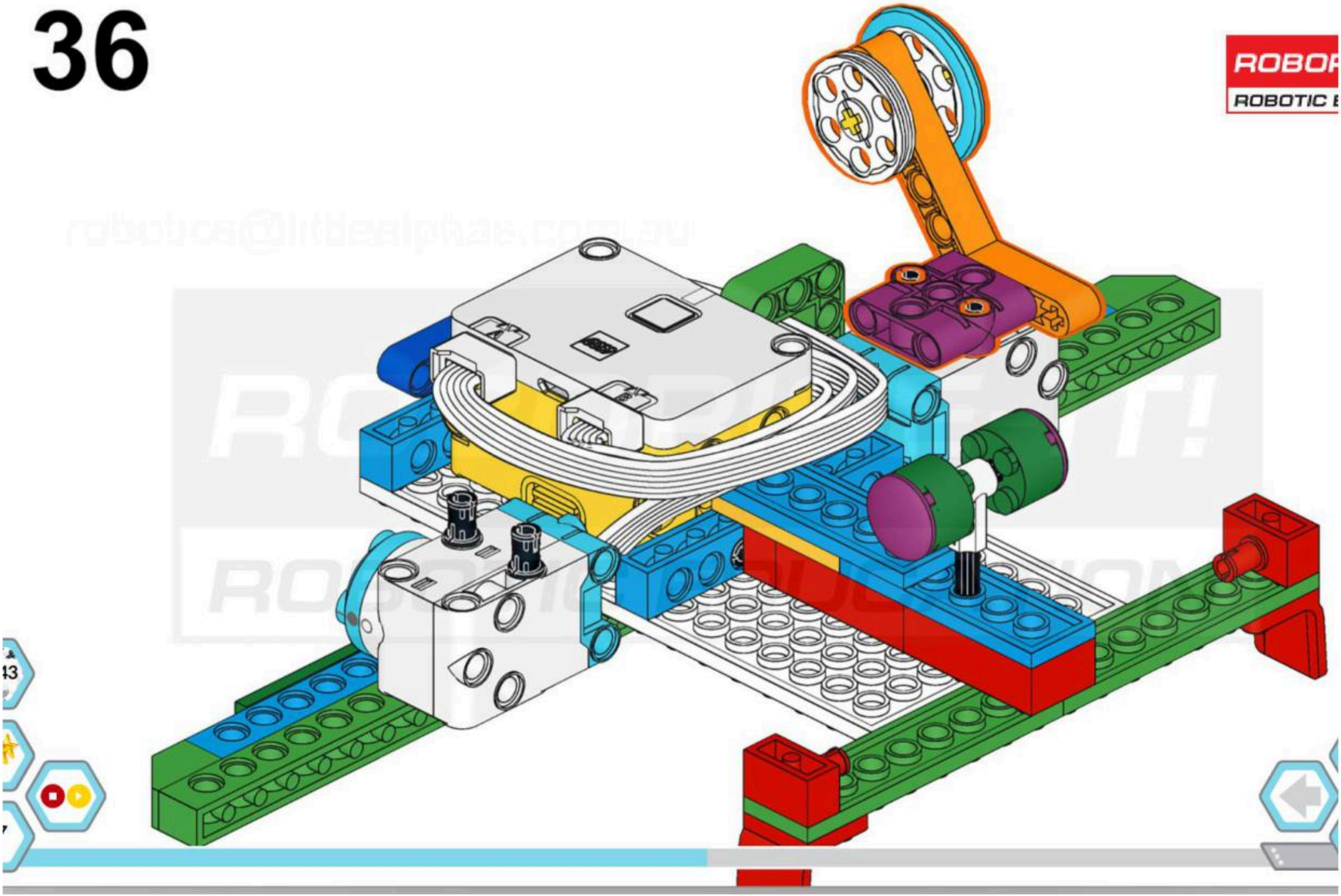


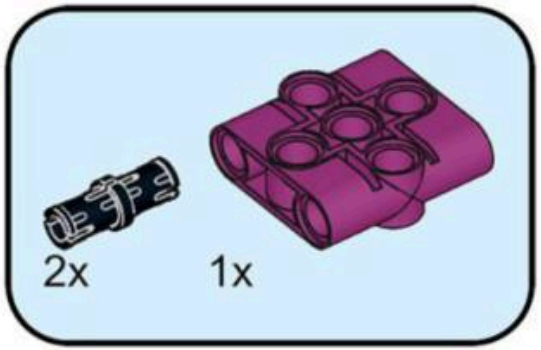
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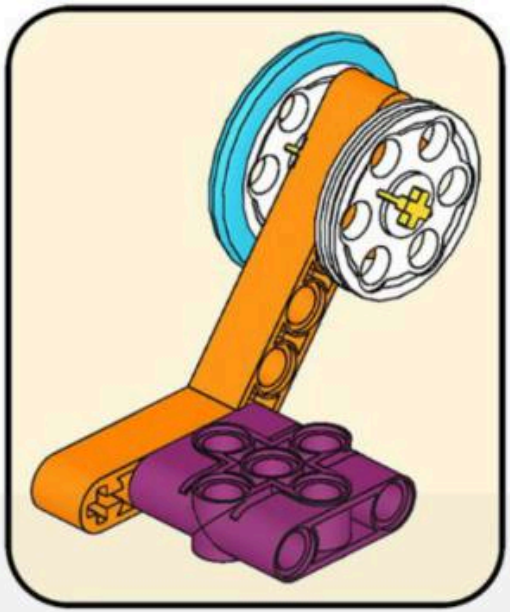
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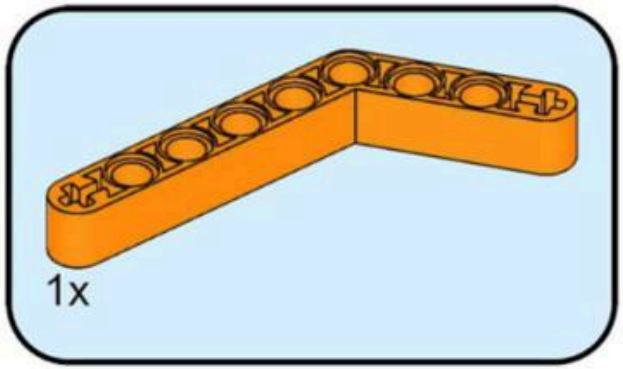
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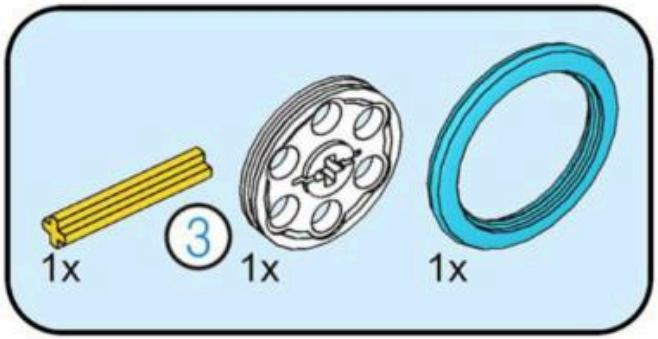


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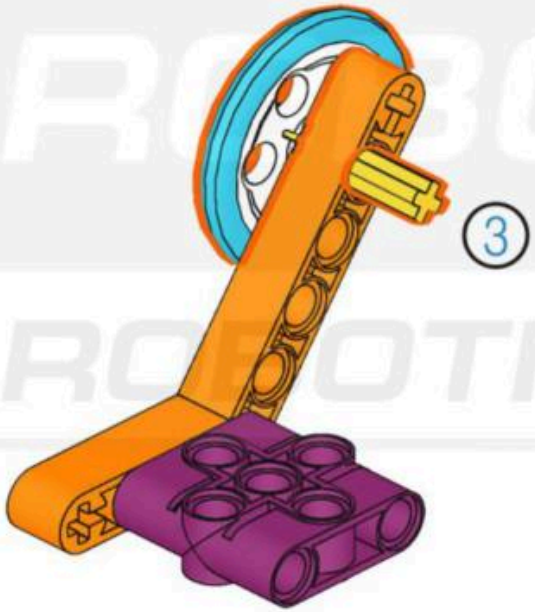


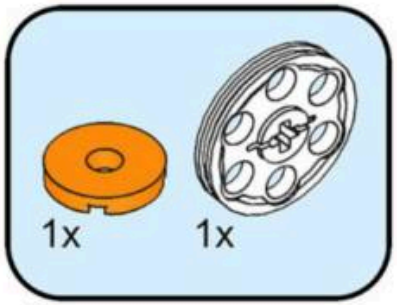


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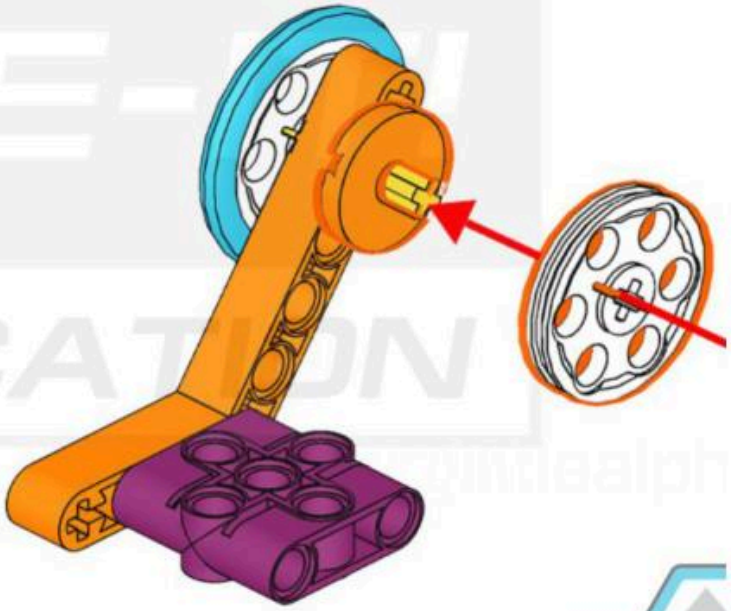


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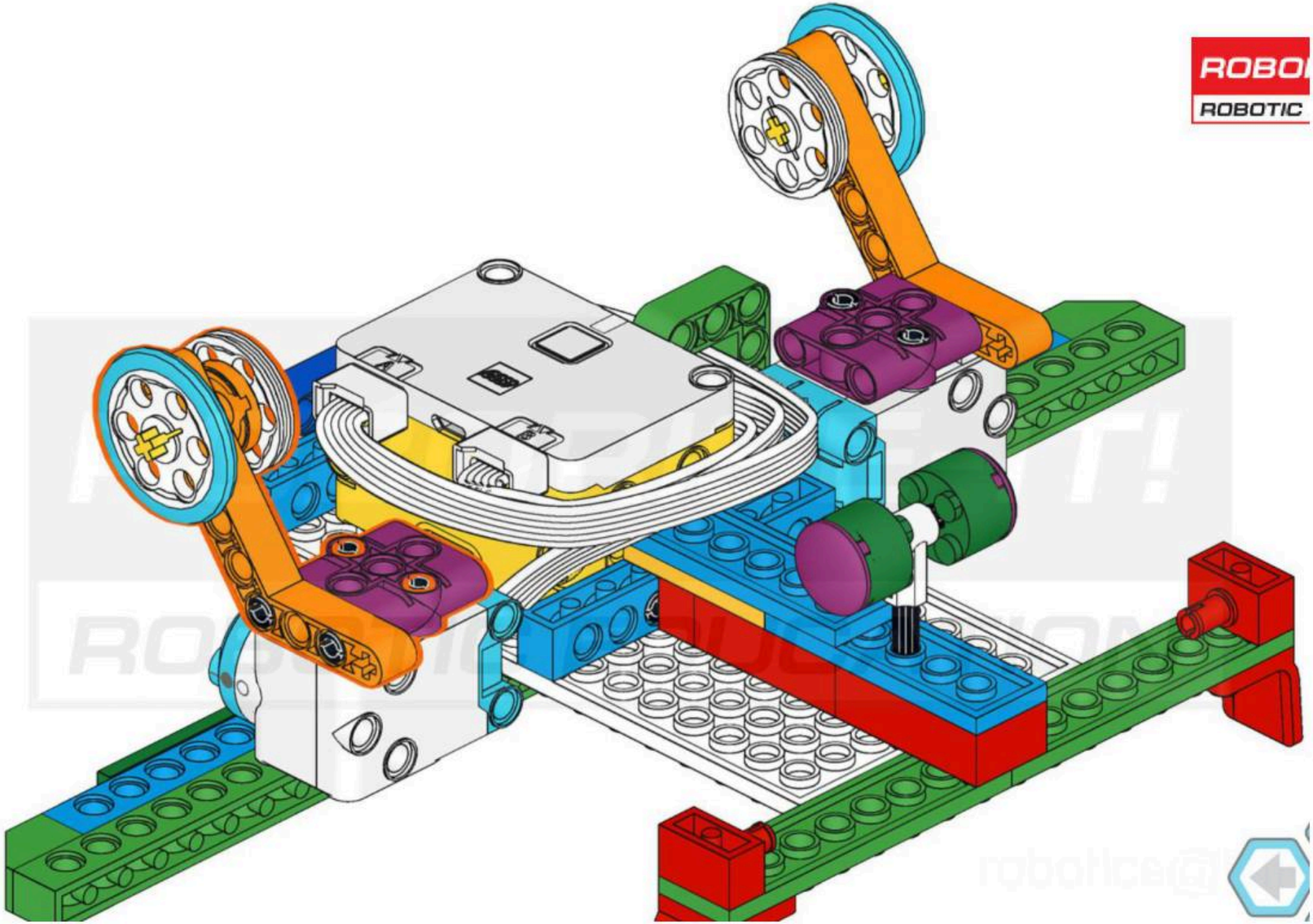
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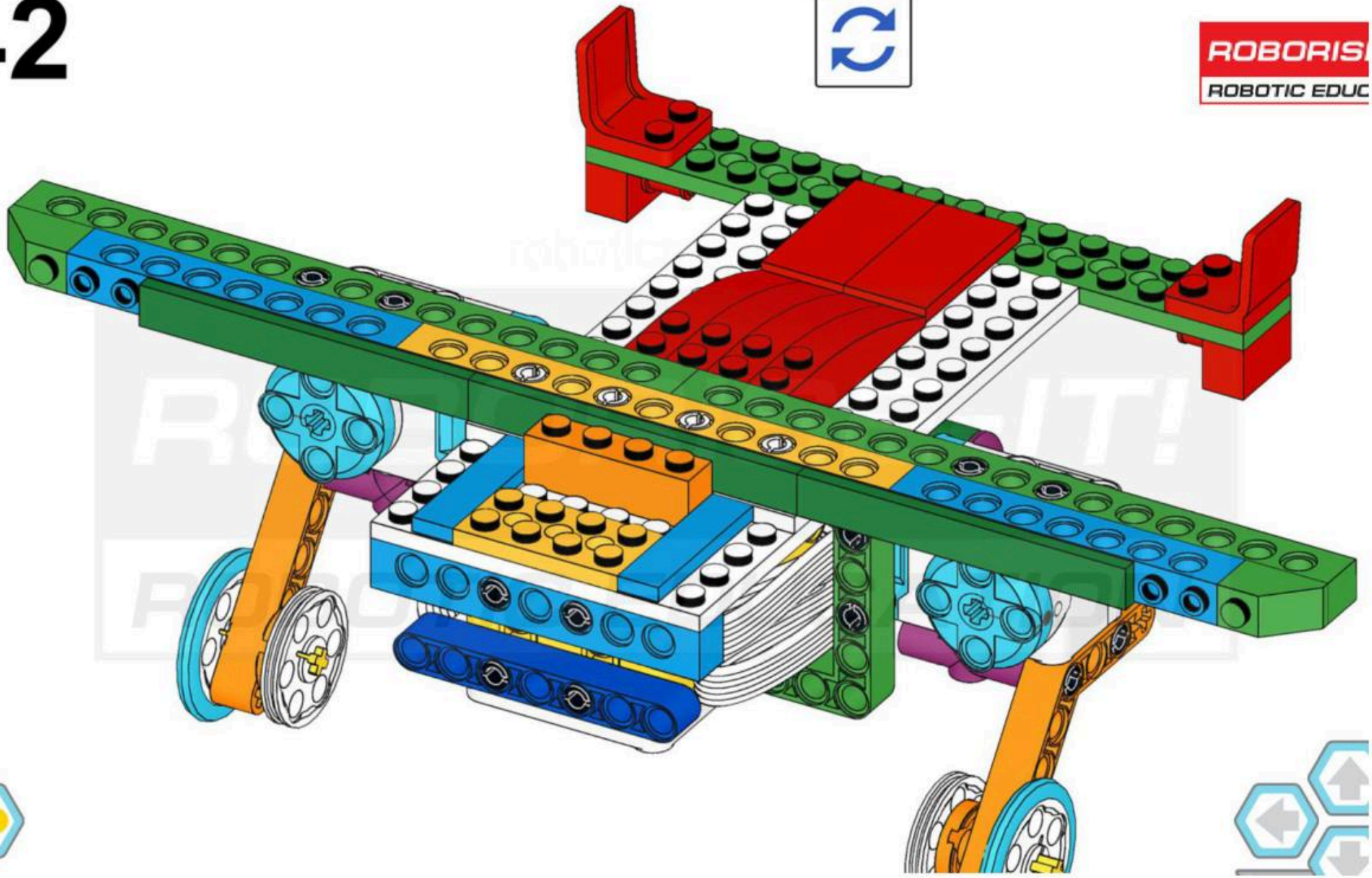
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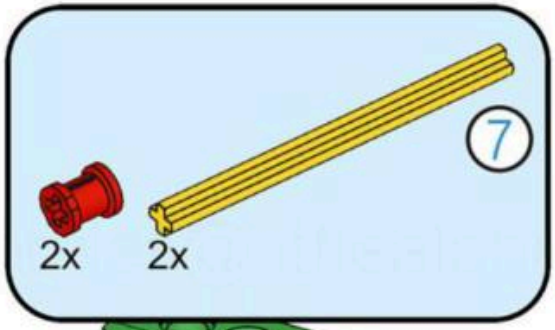
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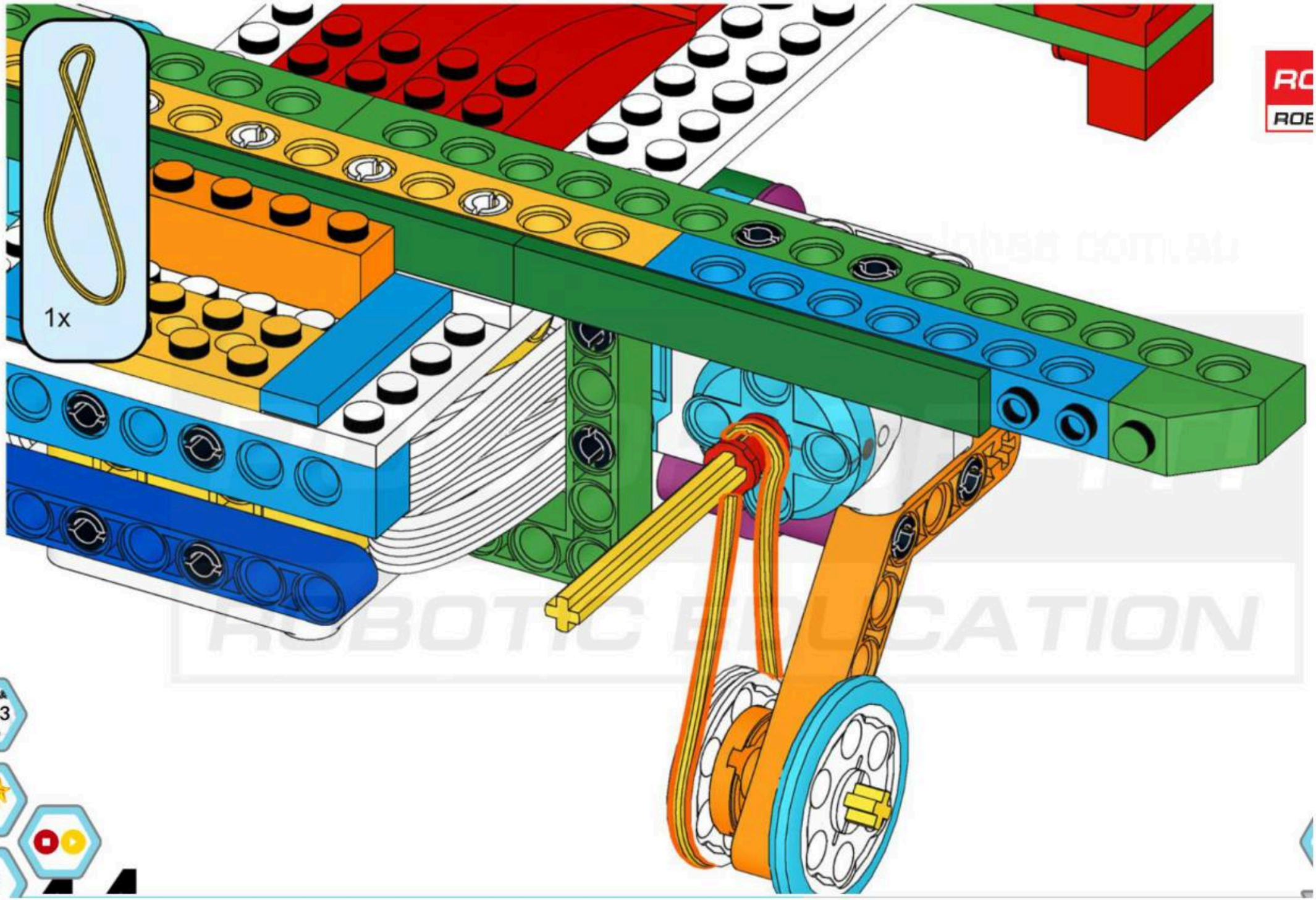
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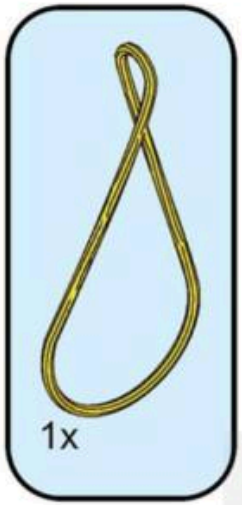
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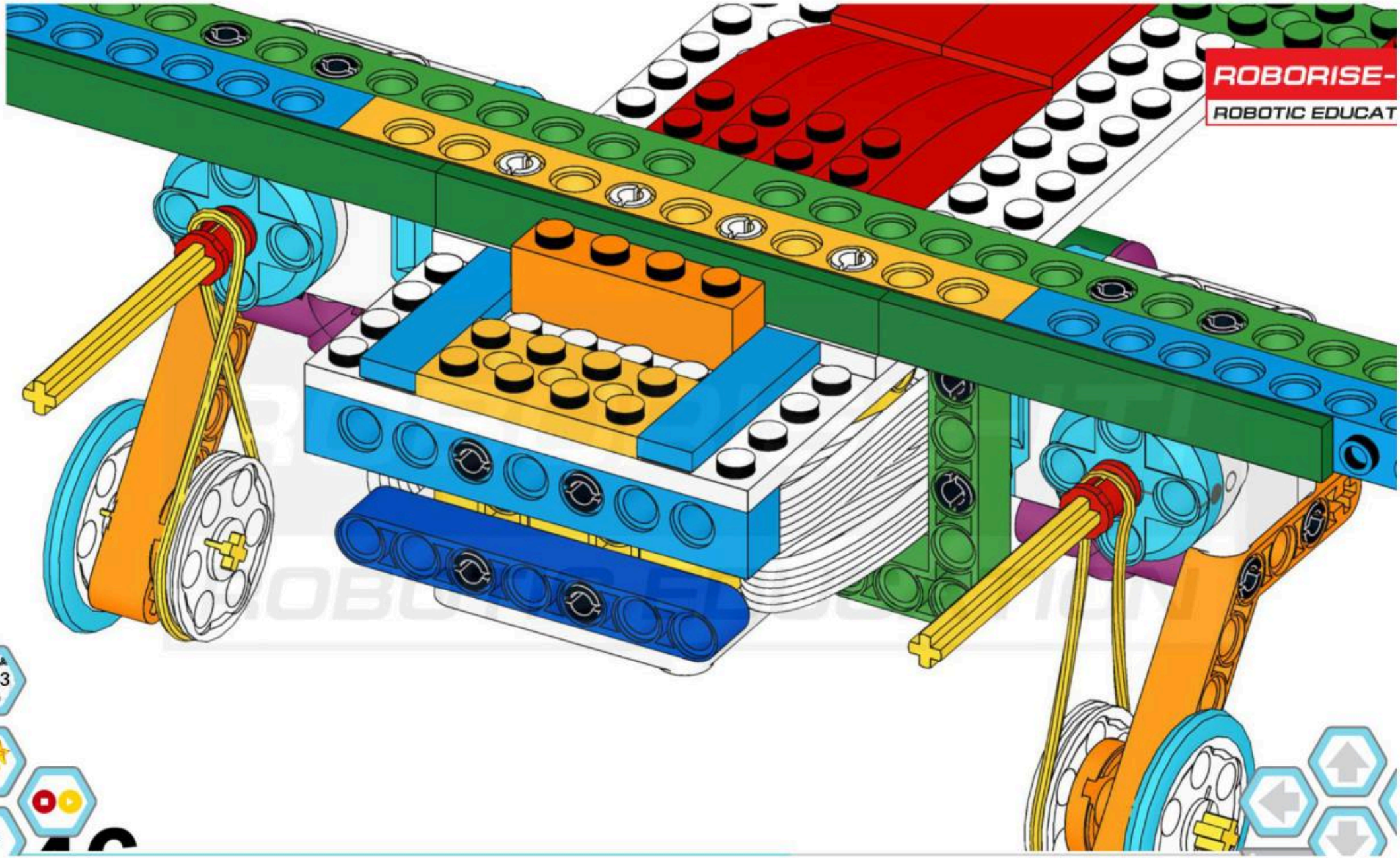
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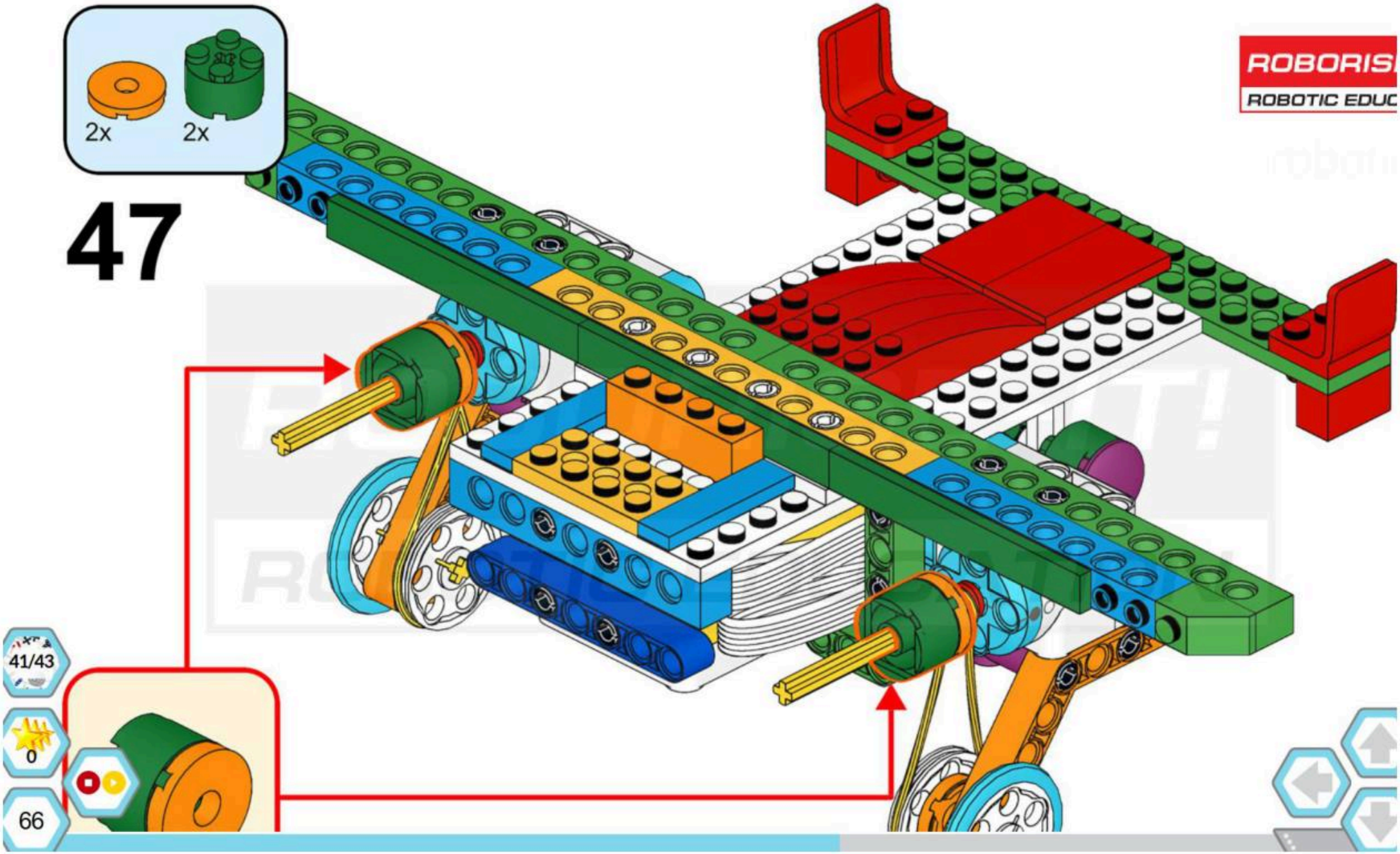

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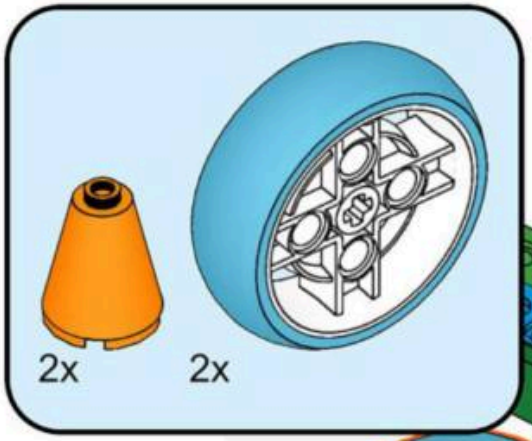


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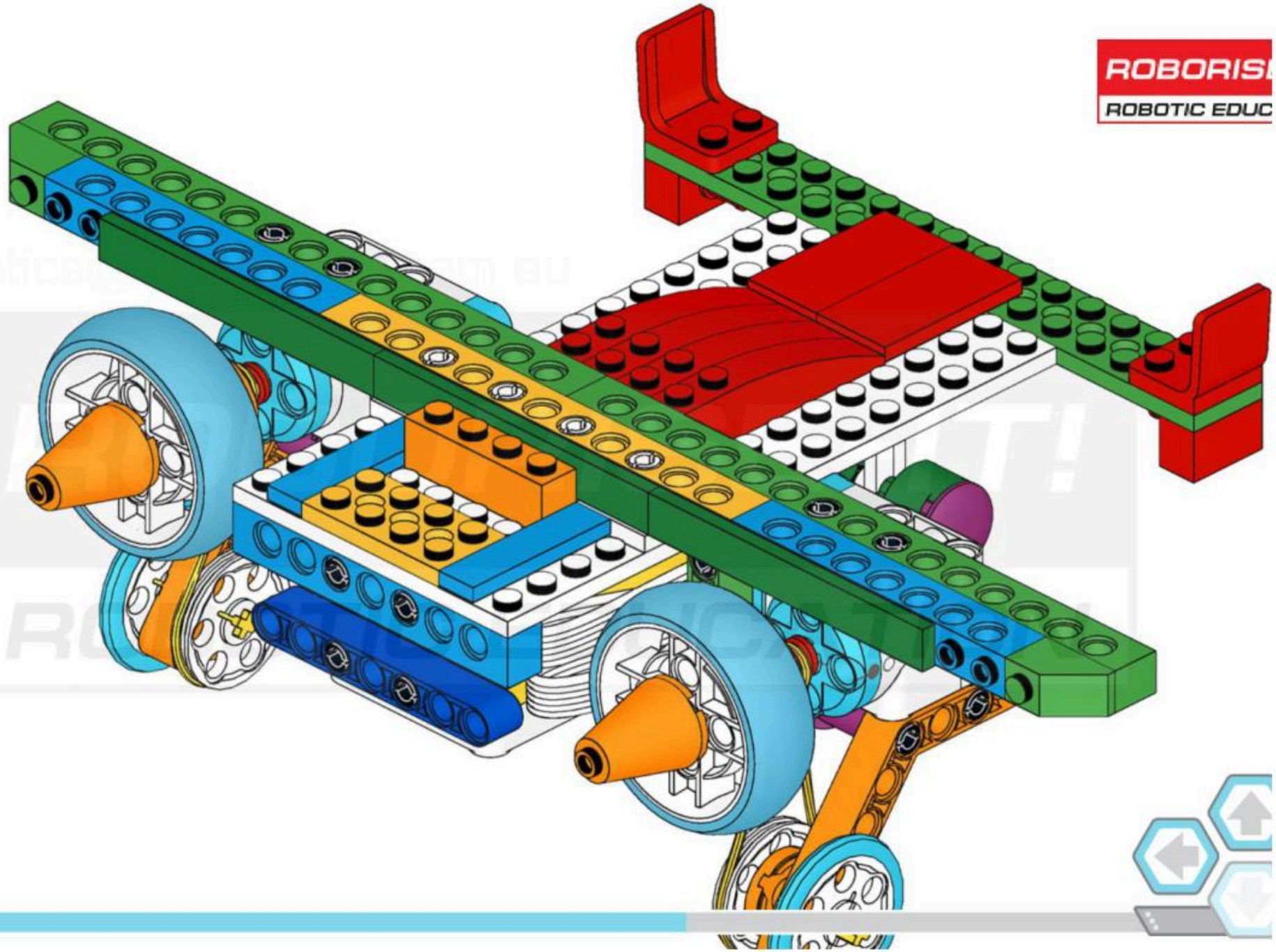

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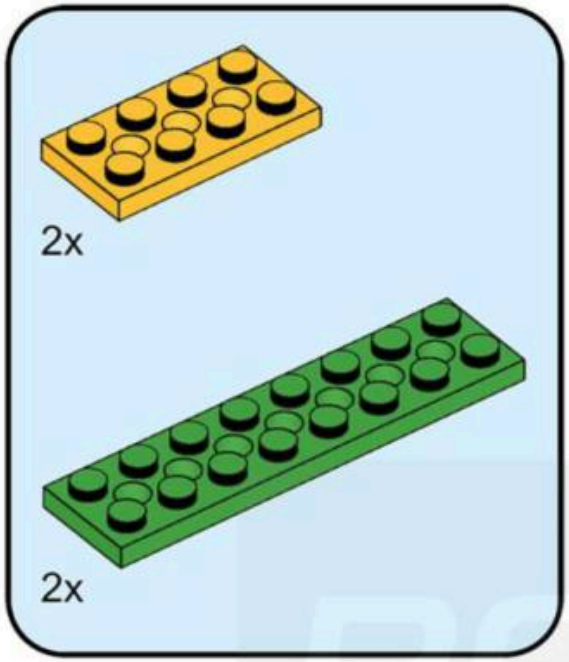
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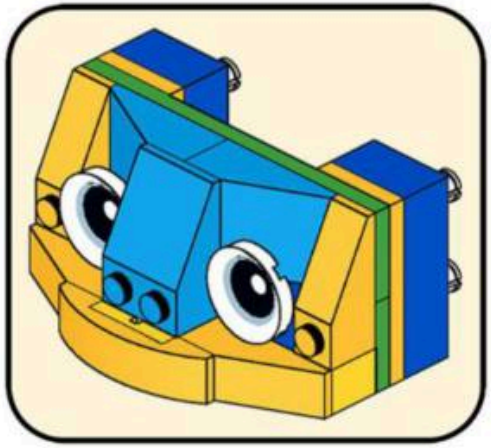
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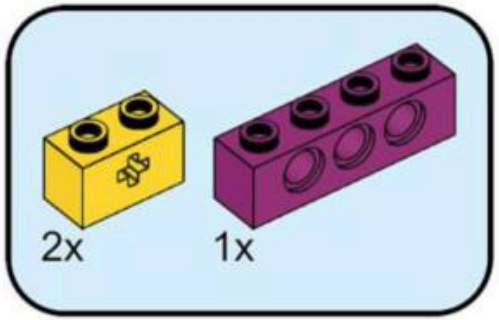
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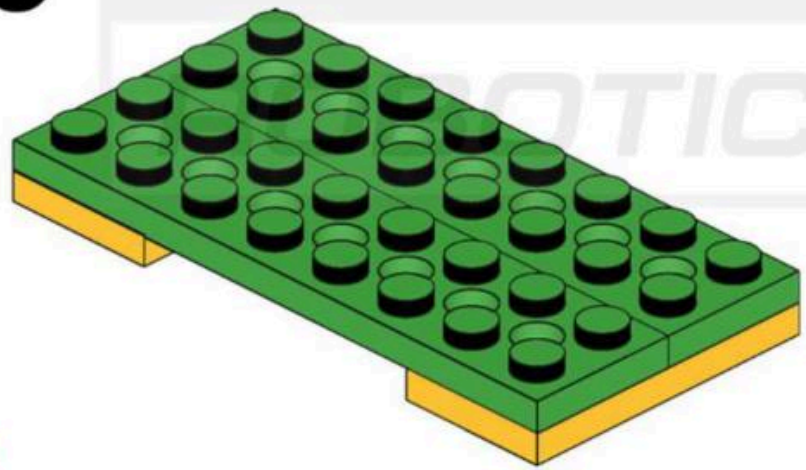


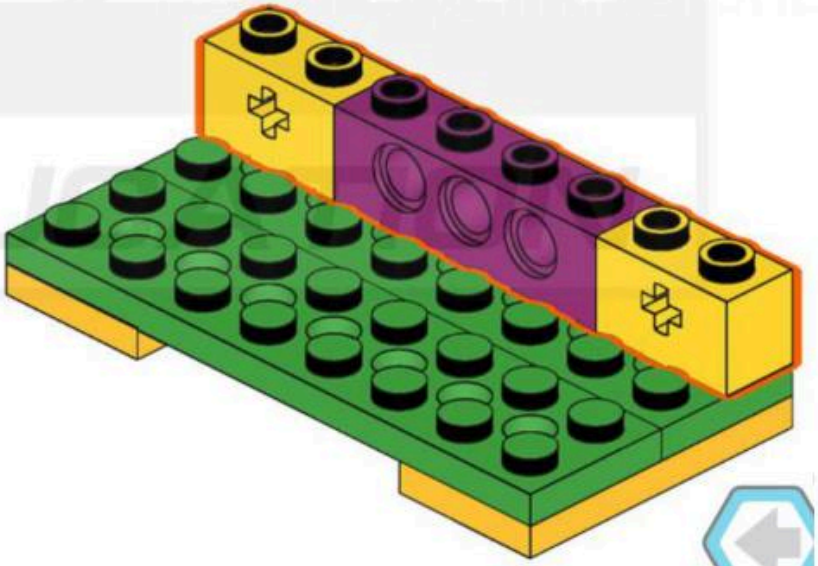


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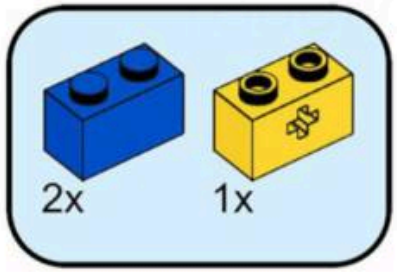
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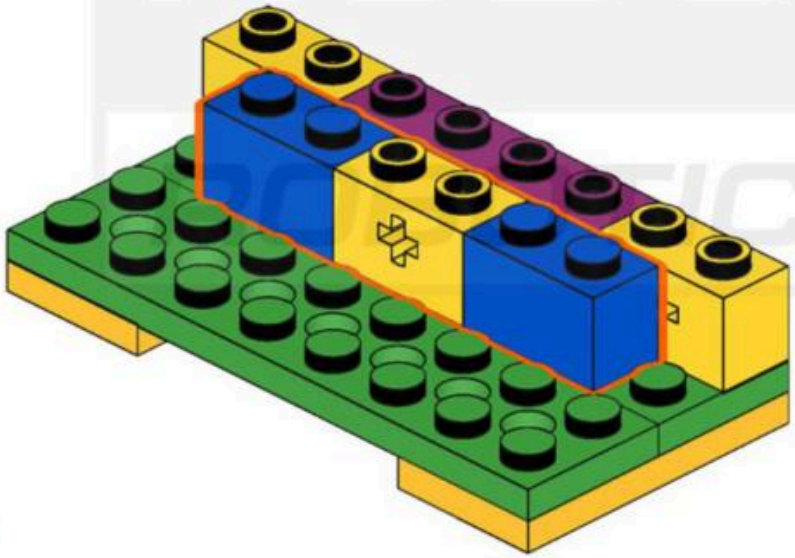


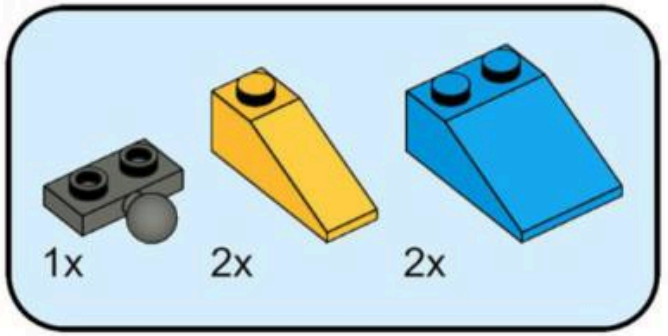


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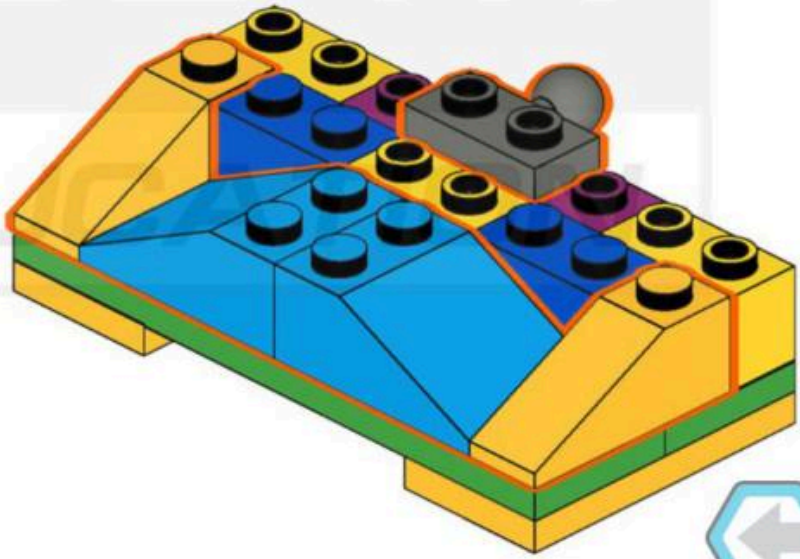


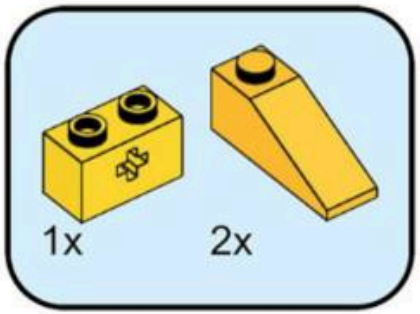
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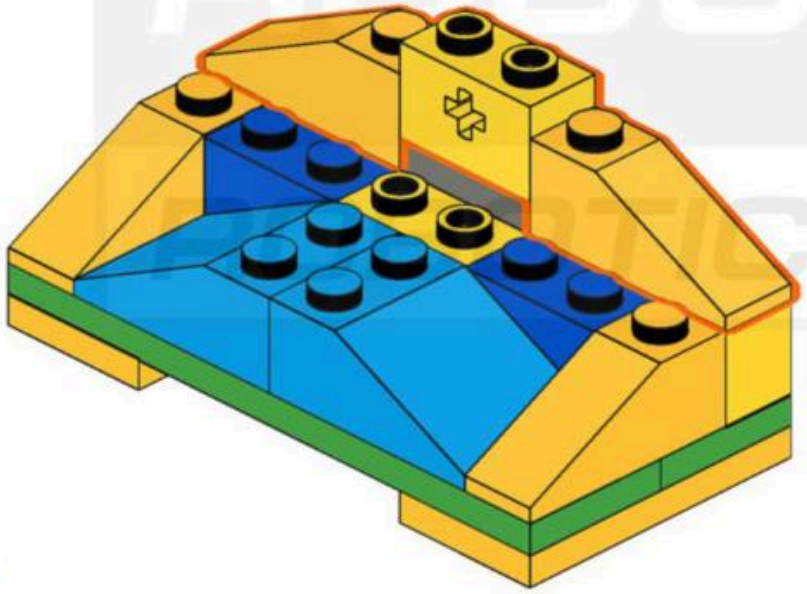


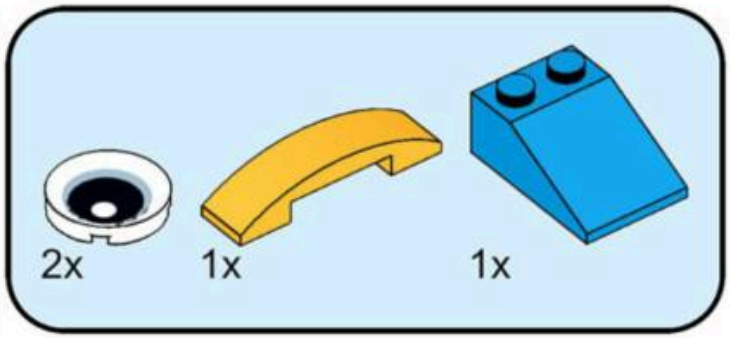
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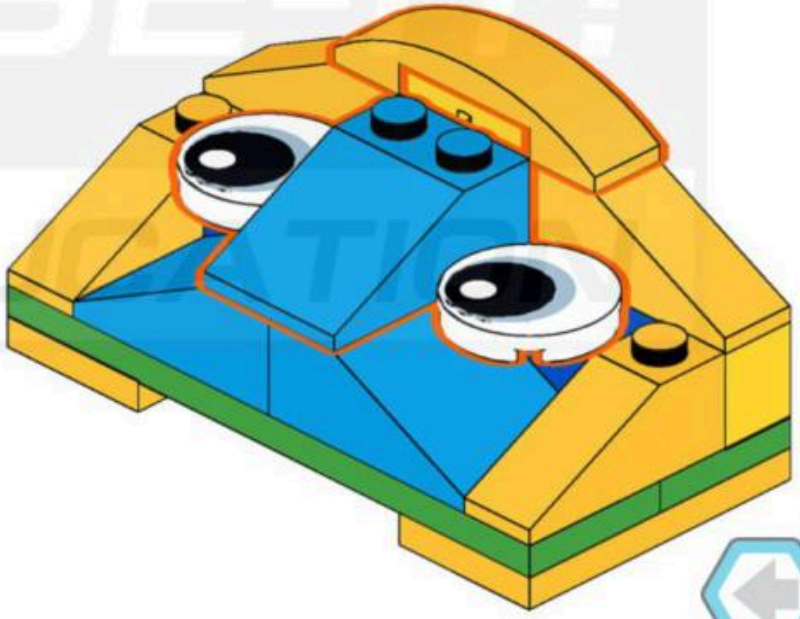


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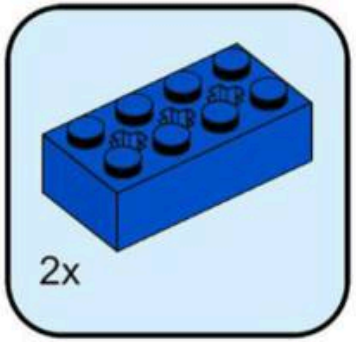




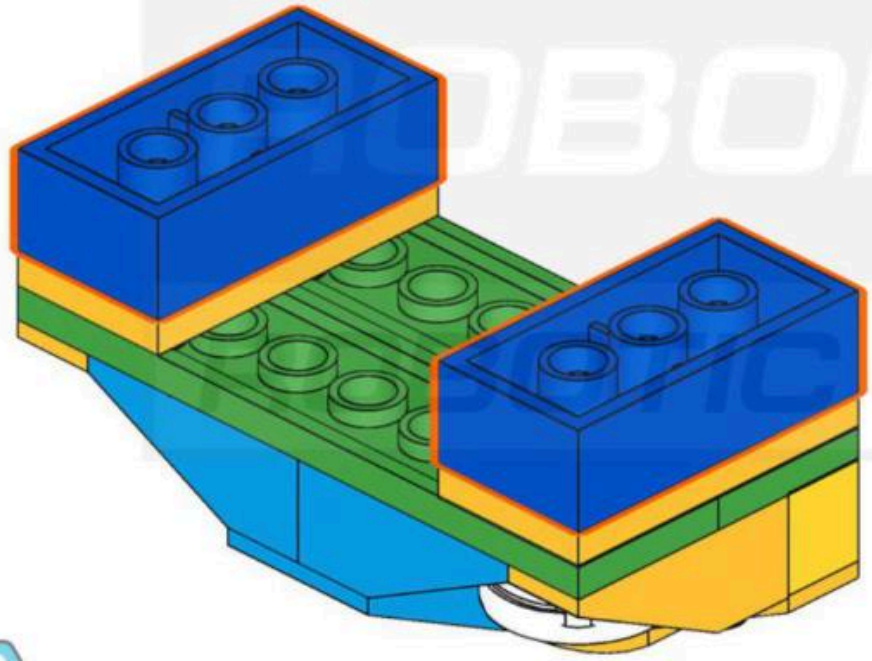
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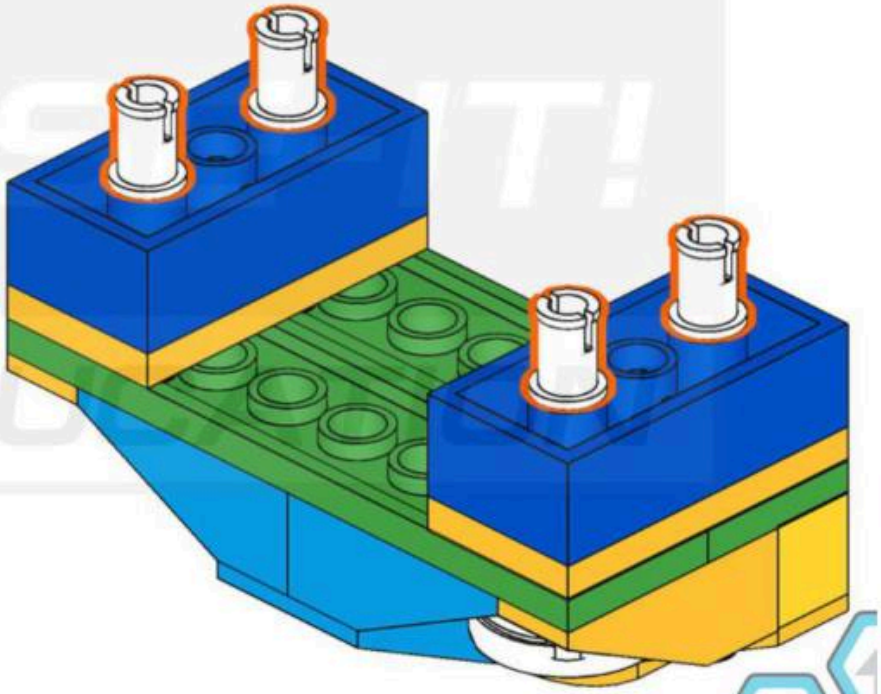


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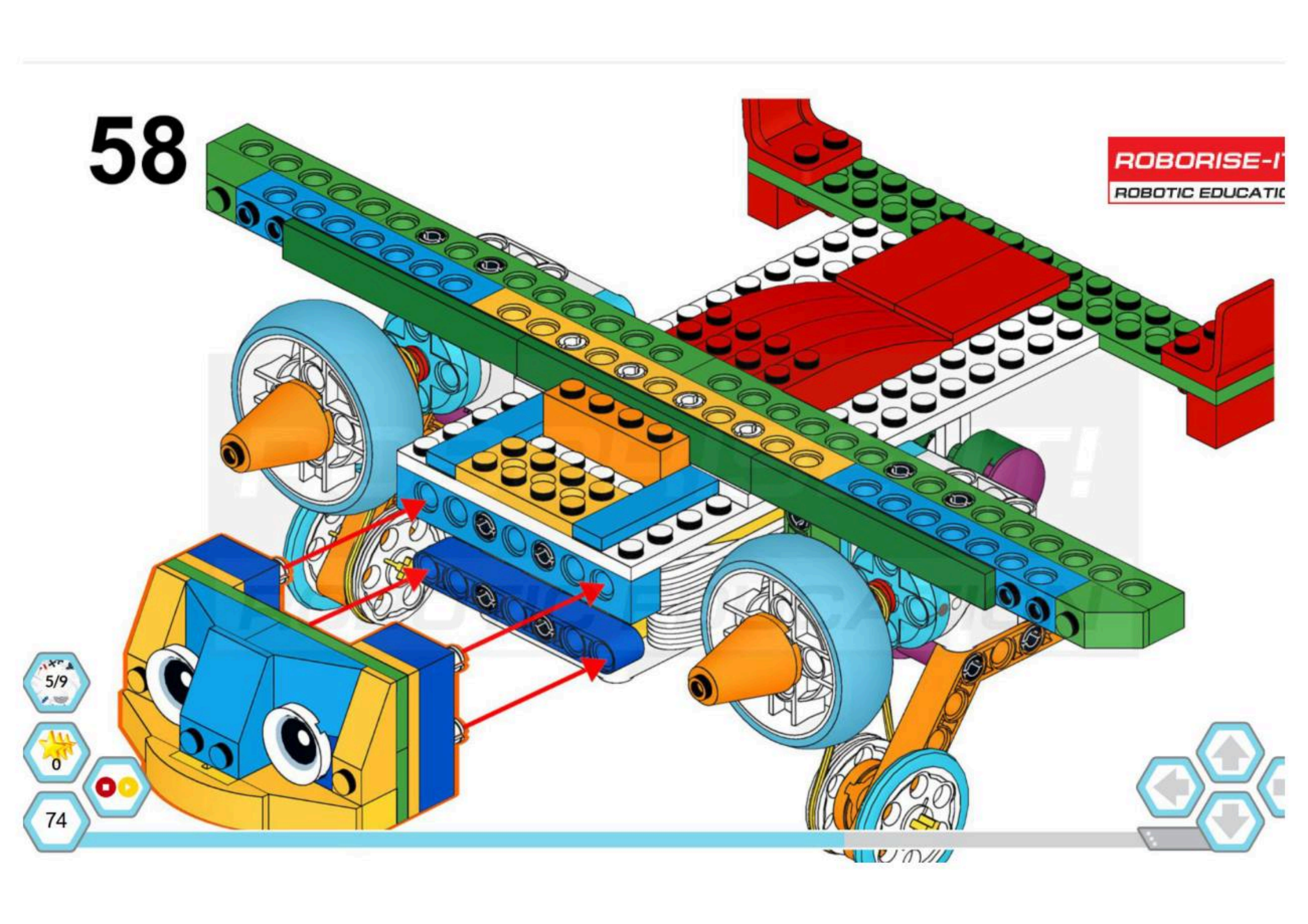




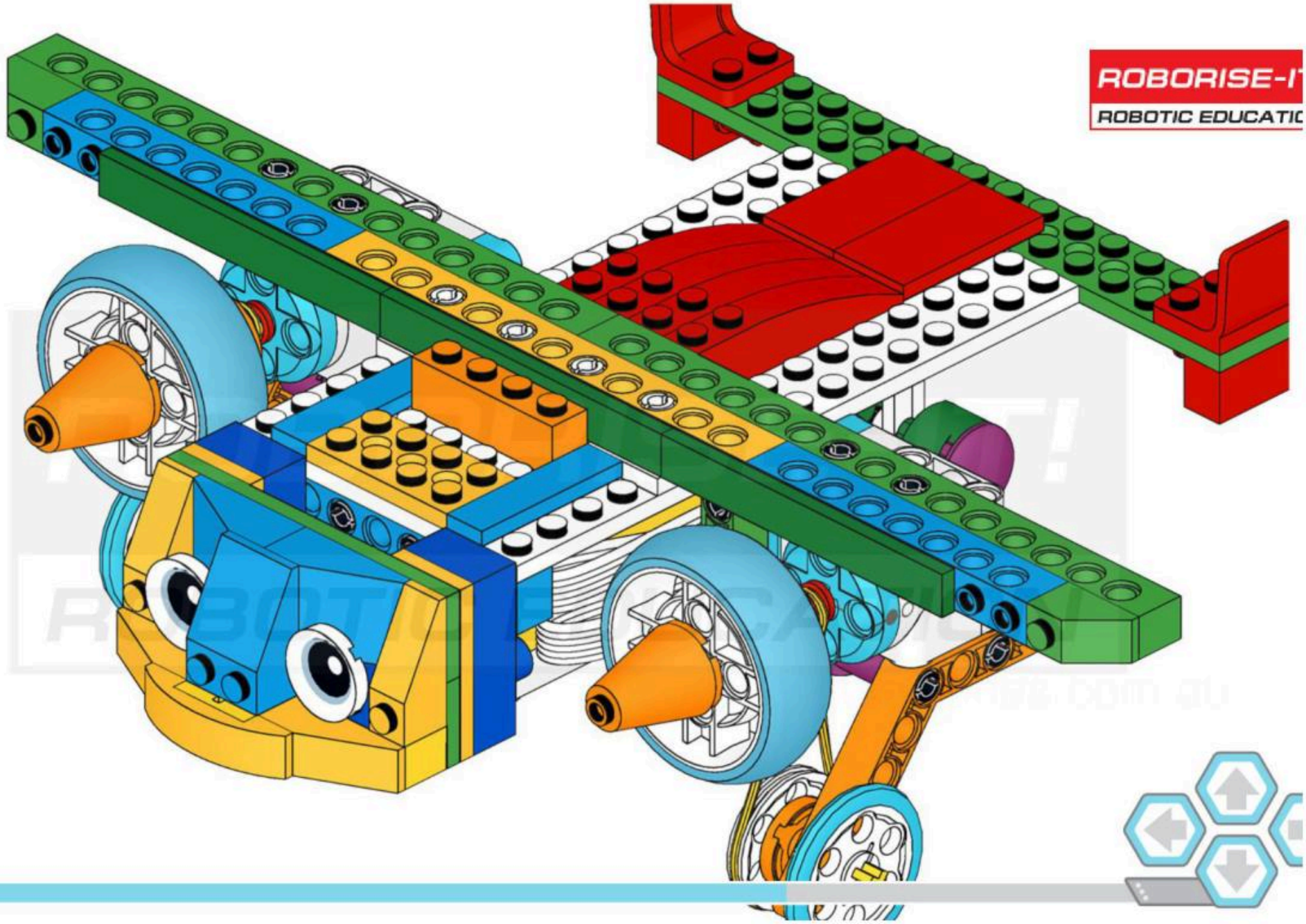
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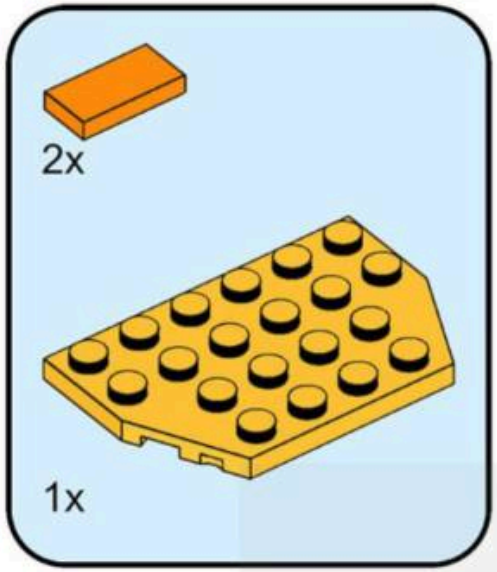


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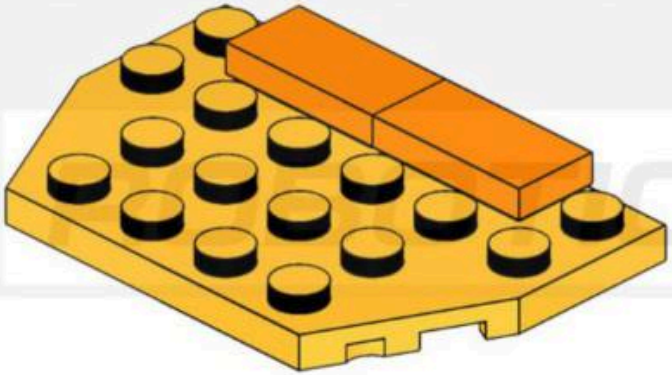

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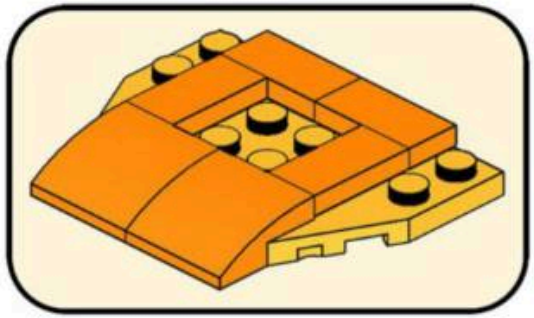
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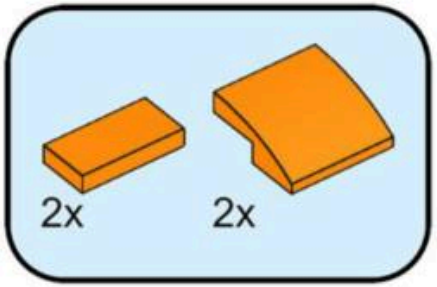




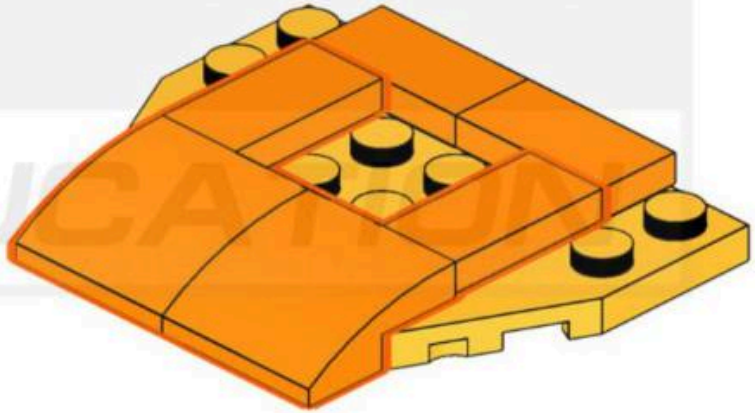
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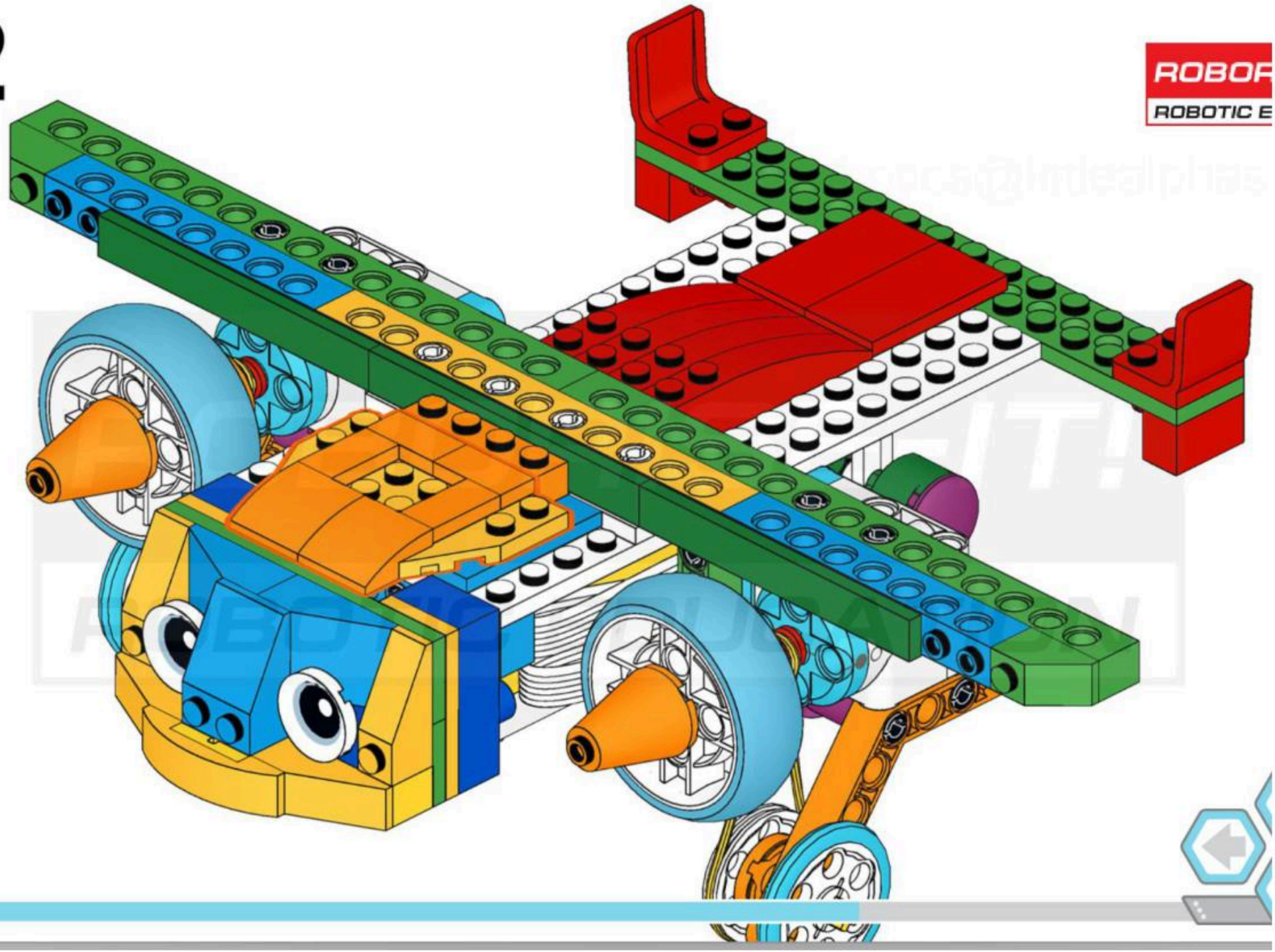




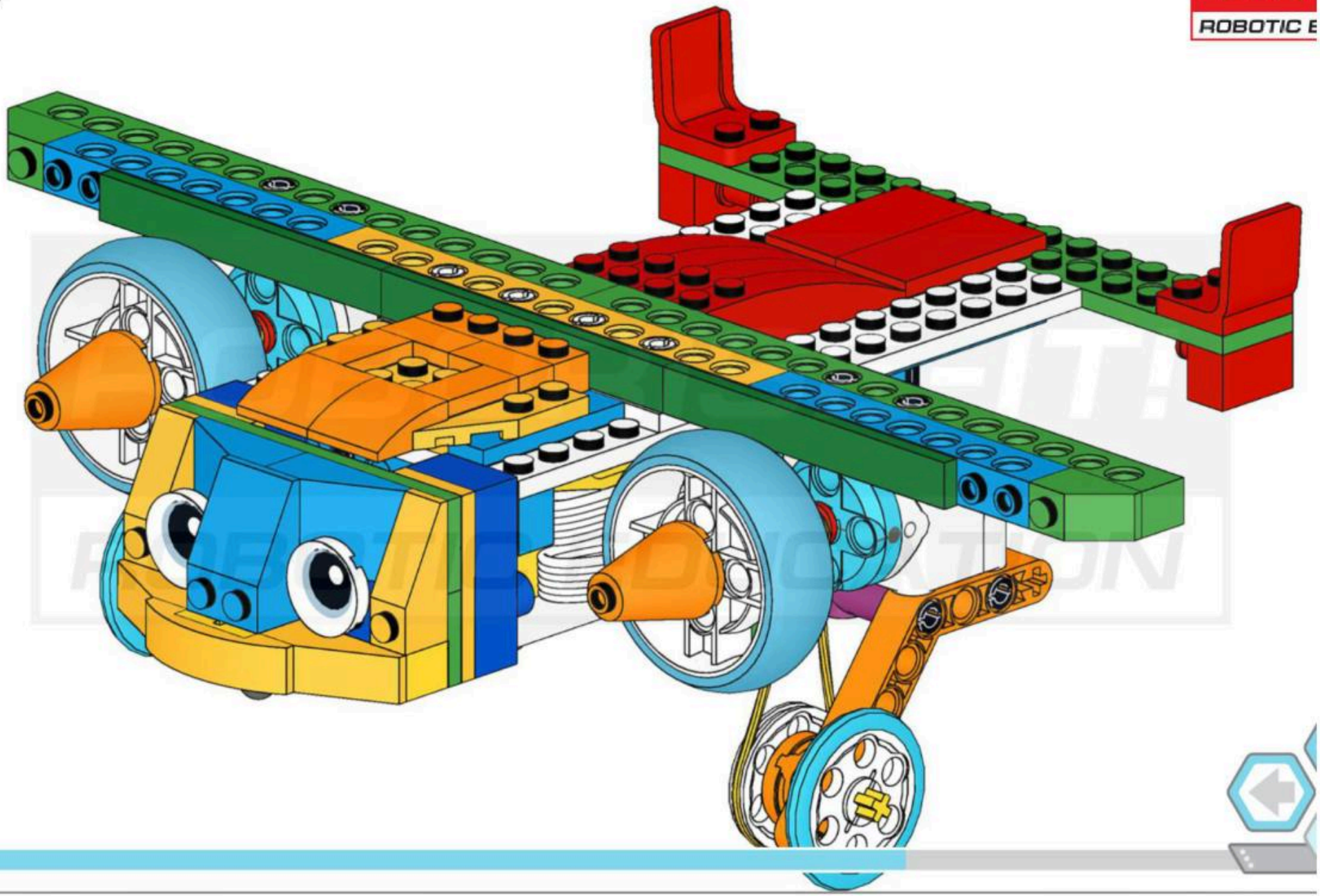
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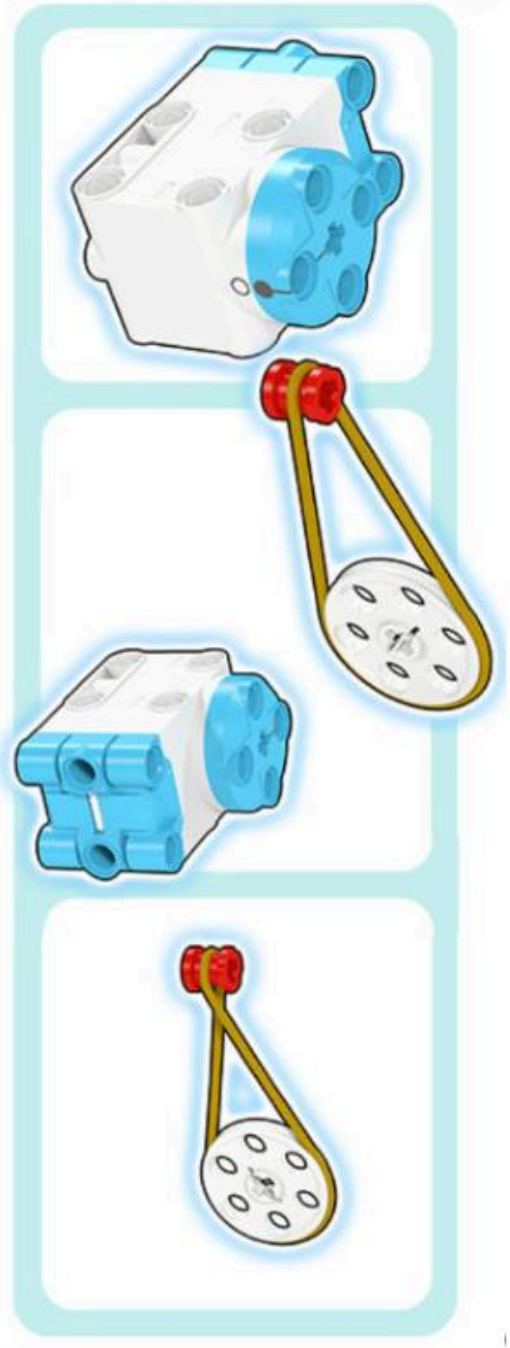


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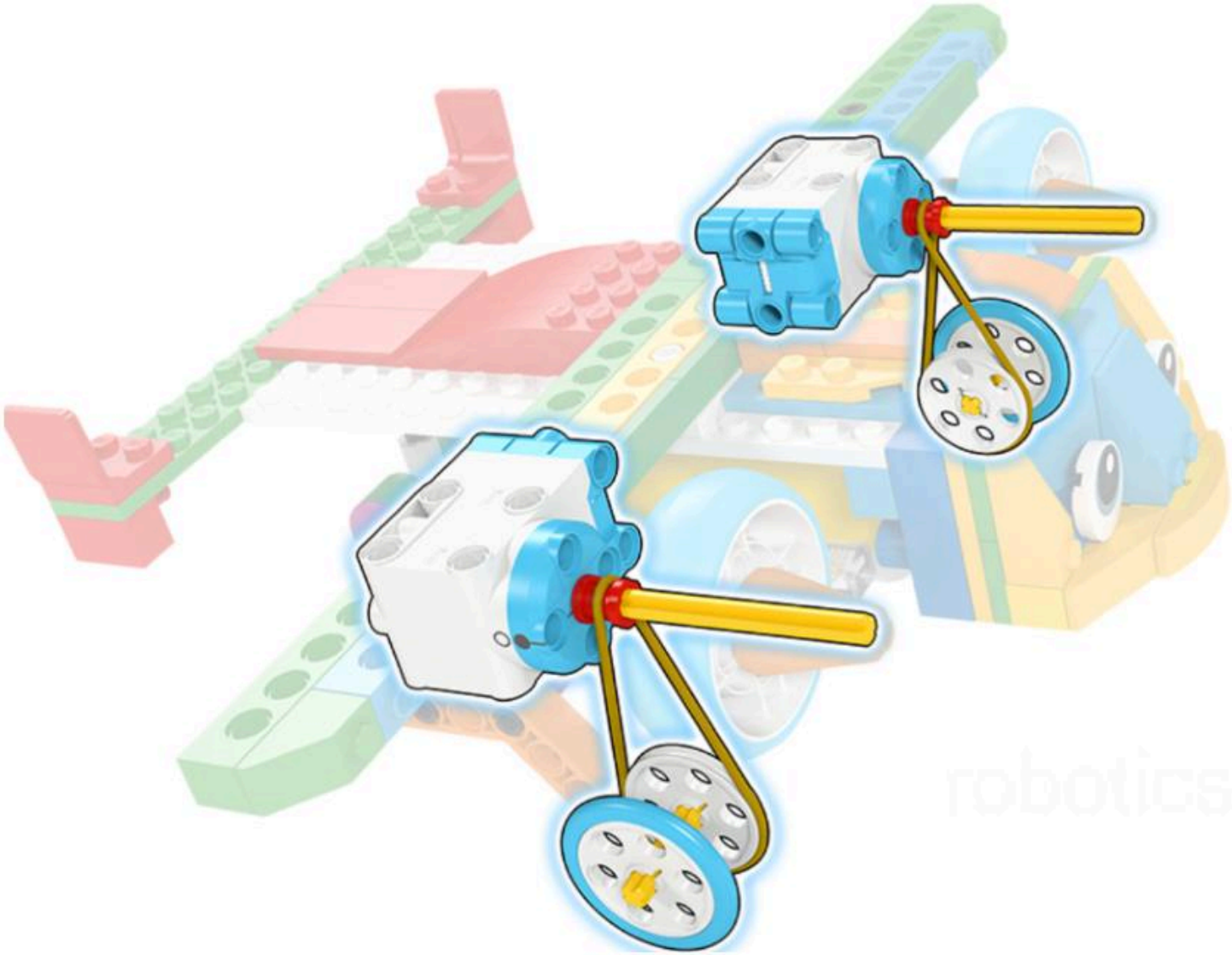


Place the parts of the robot in the correct places



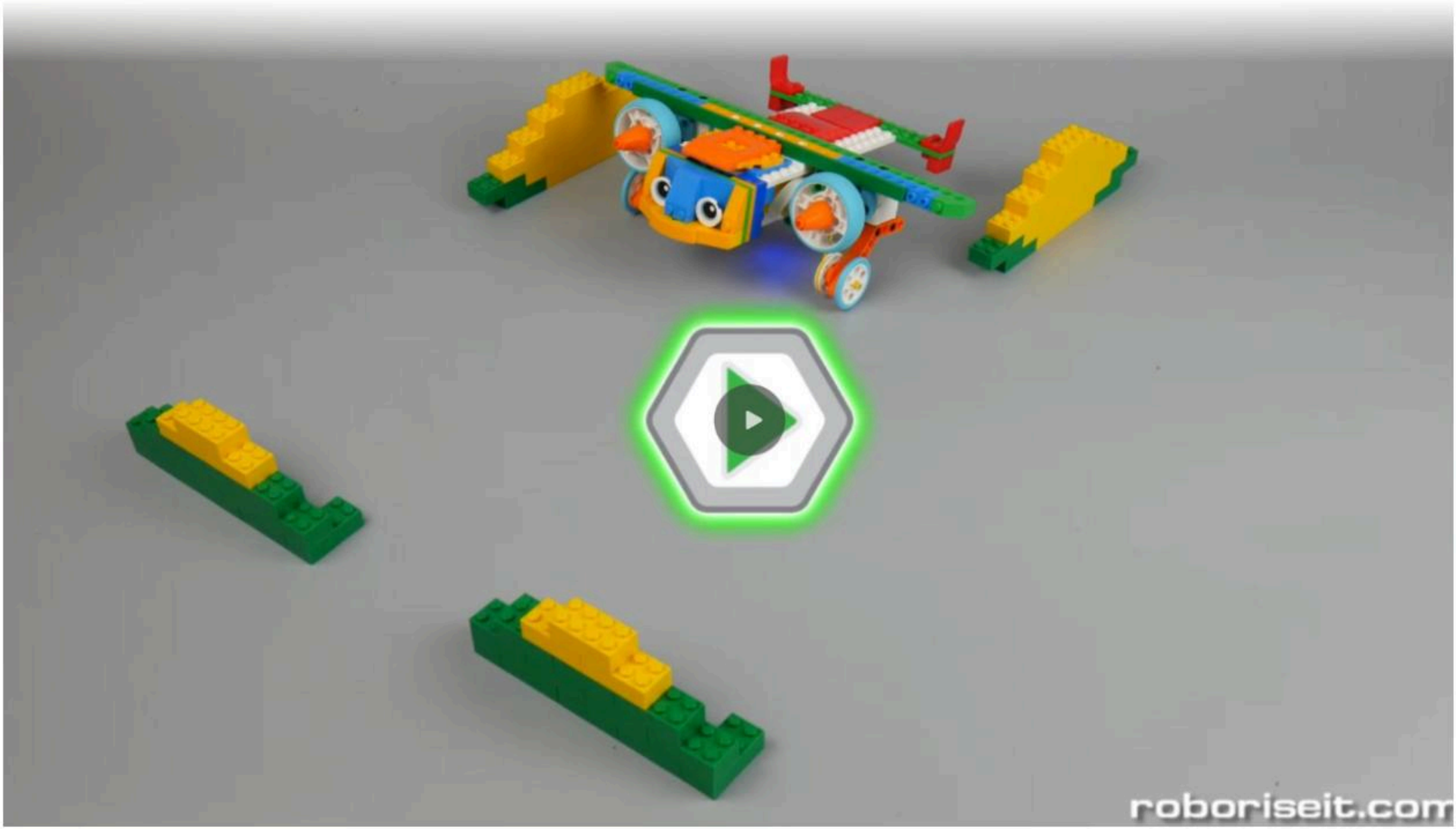


Place the parts of the robot in the correct places



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We will program the autonomous departure of the robot from the hangar and take off. This will greatly help our young pilots

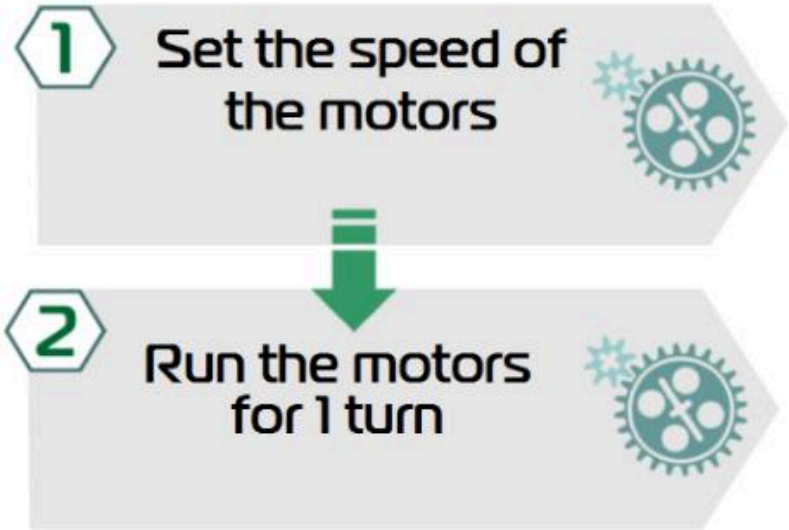




Remember which blocks you need to use in order to program the movement of a two-motor robot

Task 1. Algorithm

The program should work according to the following algorithm:



An algorithm is a sequence of actions that leads to the completion of a task. The algorithm is turned into a program, understandable work, and the program is already executed by the robot

ICON BLOCKS | WORD BLOCKS

Use the blocks to program the movement of the robot:

In ICON Blocks mode, the program looks like this:



Use the blocks to program the movement of the robot:

In Word Blocks mode, the program looks like this:



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Select the duration of the turn, at which your robot will turn left by 90 degrees.



Use the blocks to program the movement of the robot:

In ICON Blocks mode, the program looks like this:



Repeat testing the robot until it performs a sufficiently precise turn.

Use the blocks to program the movement of the robot:

In Word Blocks mode, the program looks like this:



Repeat testing the robot until it performs a sufficiently precise turn.

Task 3. Program

ICON BLOCKS | WORD BLOCKS

Use the blocks to program the movement of the robot:

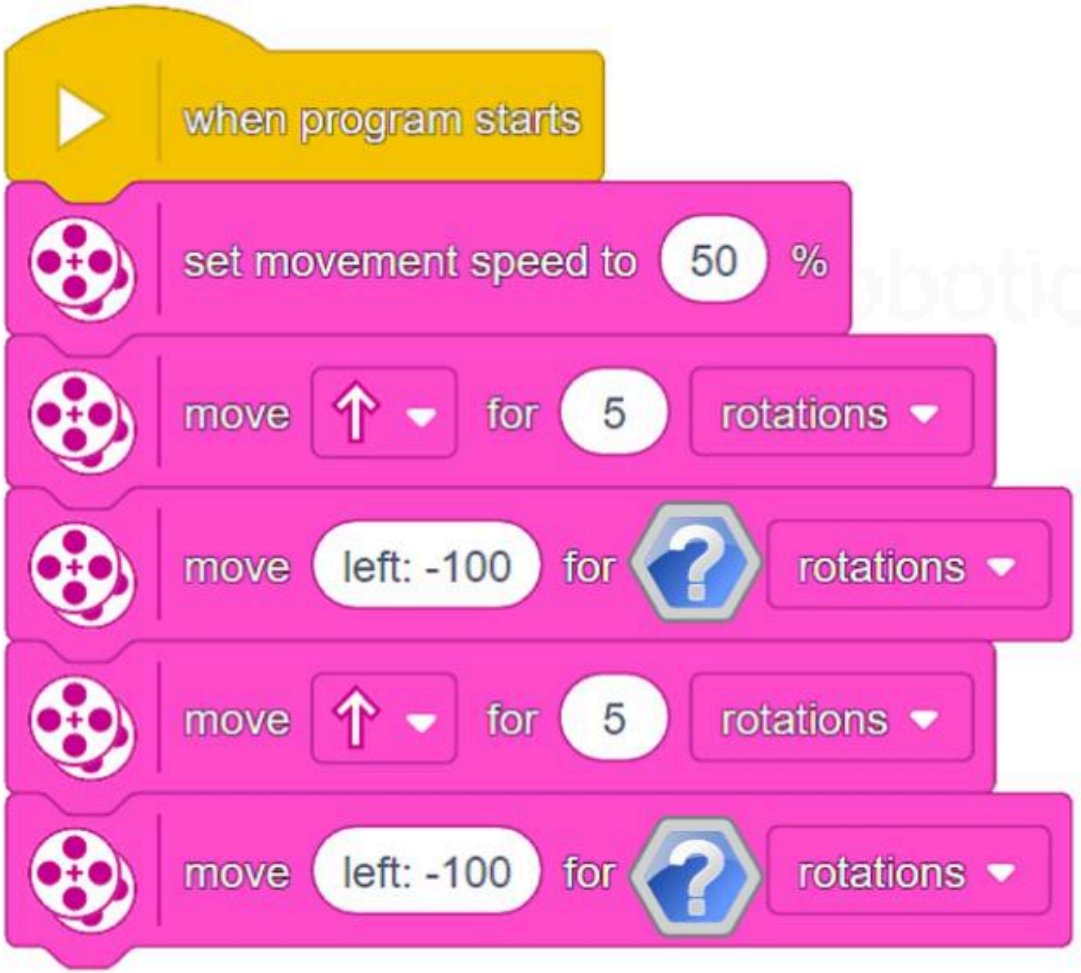
In ICON Blocks mode, the program looks like this:



ICON BLOCKS | WORD BLOCKS

Use the blocks to program the movement of the robot:

In Word Blocks mode, the program looks like this:

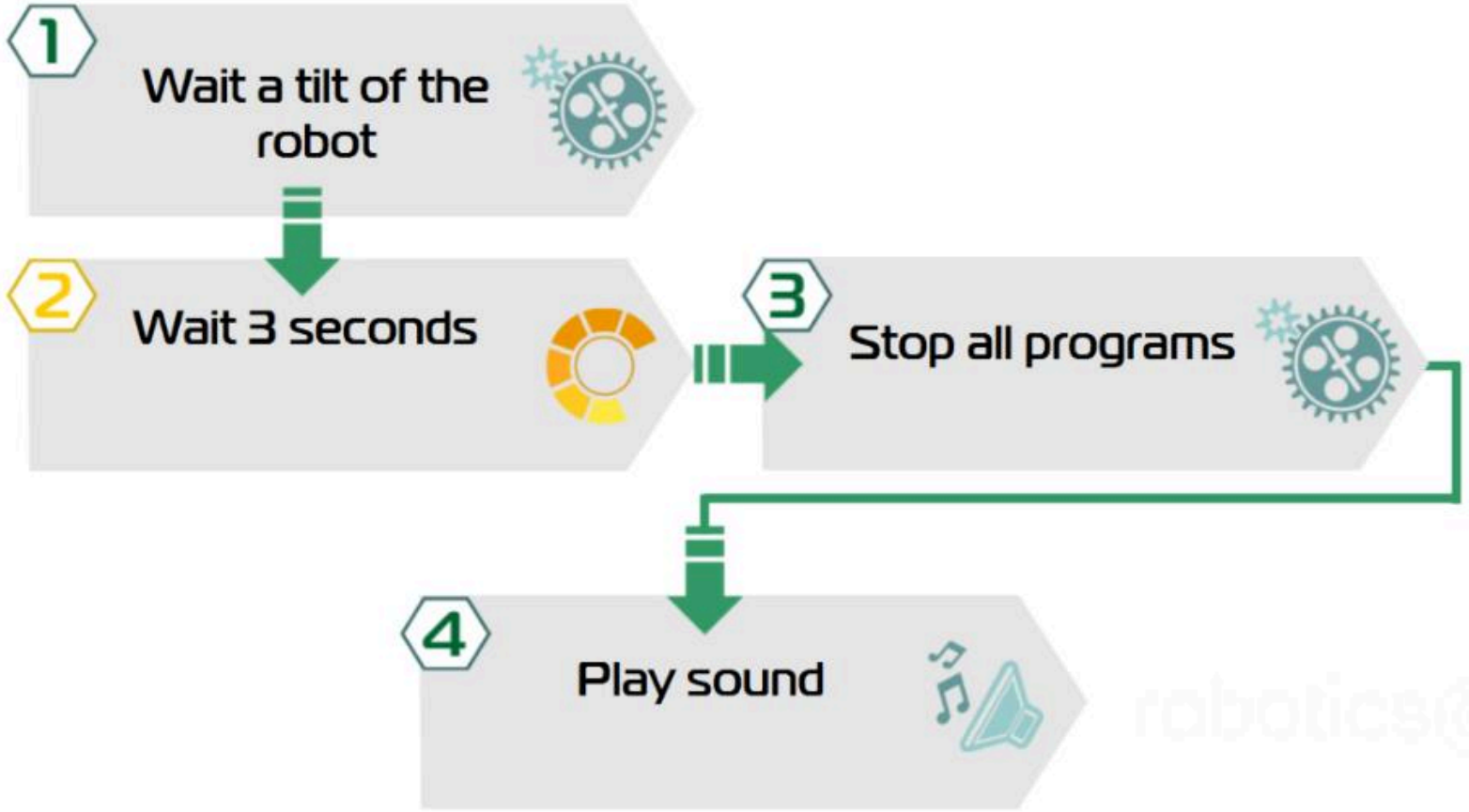


Add your robot boarding assistant to the program. It should detect the robot's descent and stop the motors three seconds after that.



Task 5. Algorithm

The program should work according to the following algorithm:





What kind of sensor do you think can help us detect the tilt of the robot?

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Robot sensors

Take a look at the kit box. In it you will find the following electronic parts that can be connected to the robot:



Color sensor



Color matrix 3x3

Thanks to the color sensor, your robot can recognize the color of objects, and you can use this data in programs. The color matrix is used as a small screen with only 9 color dots.

Programmable hub

In search of additional sensors for our robot, you need to take a closer look at the Programmable hub - the heart of your robot:

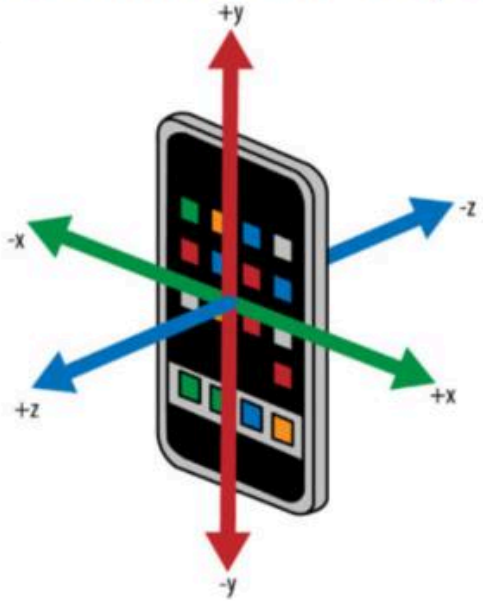


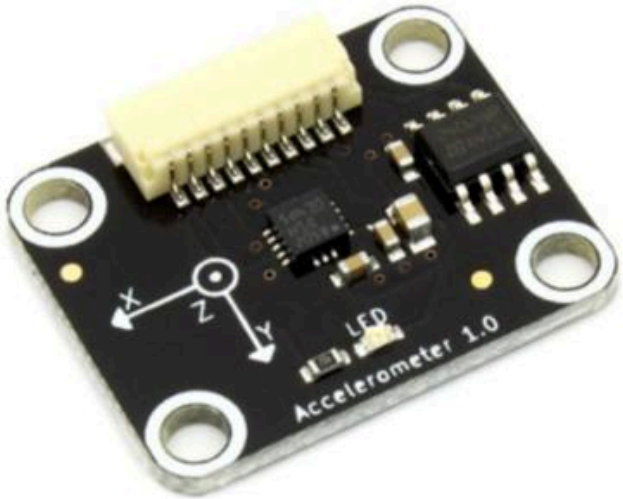
It contains an accelerometer and a gyroscope - sensors that make it possible to determine the position of the hub in space.

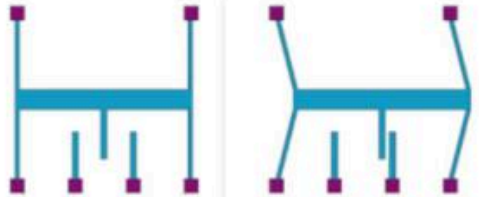
Accelerometer

This is a sensor that allows you to determine, for example, which of the sides of the hub is directed upwards.

And in a phone or smart watch, an accelerometer can help count the number of steps.



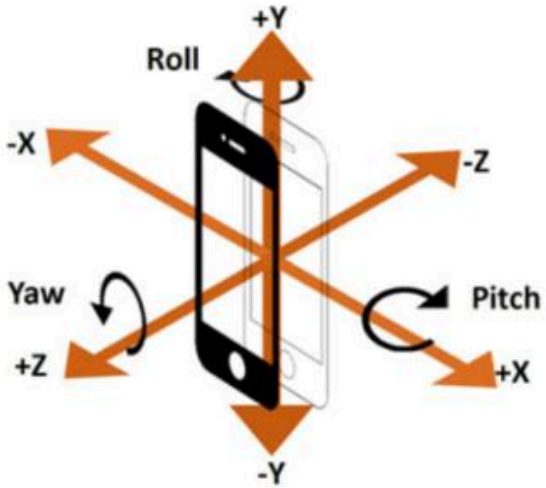


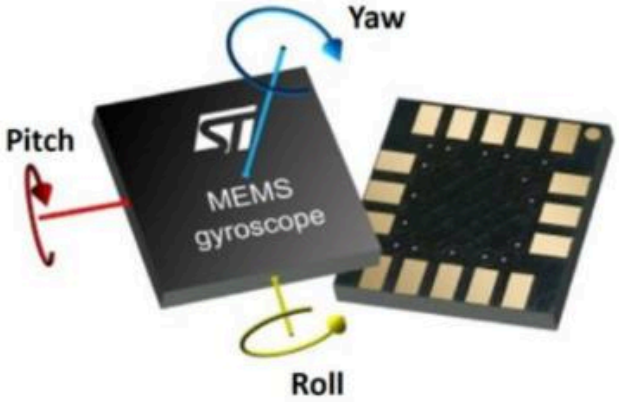


An electronic accelerometer is a microcircuit in which there are moving parts that move under the influence of gravity and various shocks. These displacements are detected

Gyroscope

Unlike an accelerometer, a gyroscope measures the speed of rotation. That is, how fast you spin the device containing it.





In mobile games, these sensors are widely used, for example, to control virtual cars. In this case, the rotation of the phone or tablet leads to turns of the virtual steering wheel.

Additional blocks

You already know from which block the program starts when you press the yellow Start button



And there are blocks that start the execution of the program when some event occurs. Such an event can be the inclination of the hub in a certain direction:



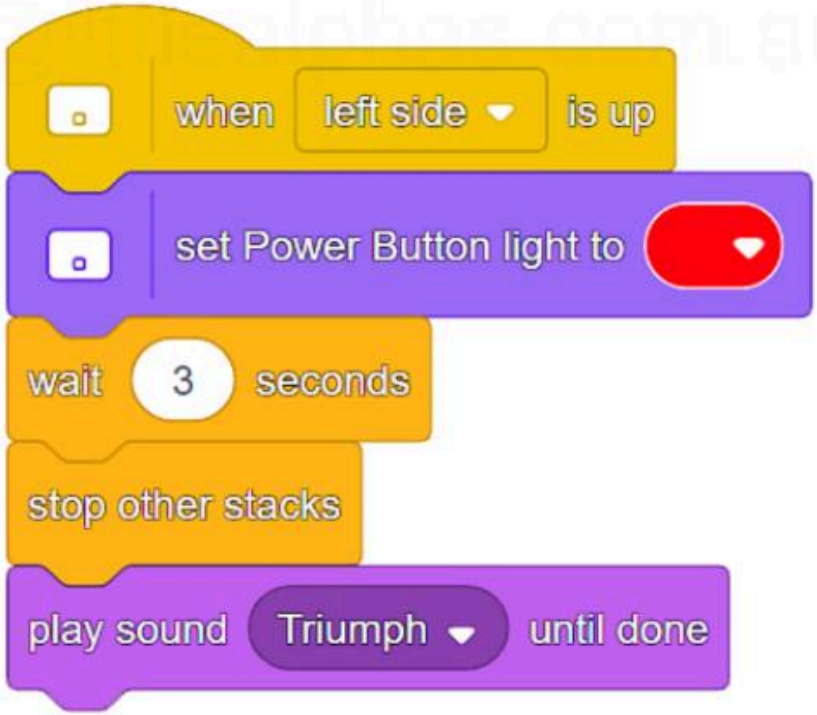
Here we will use them to launch the corresponding actions.

Task 5. Program

ICON BLOCKS | WORD BLOCKS

Program and test your robot.

In Word Blocks mode, the program looks like this. Here, in addition to the sound, you can also program the color change of the LED.

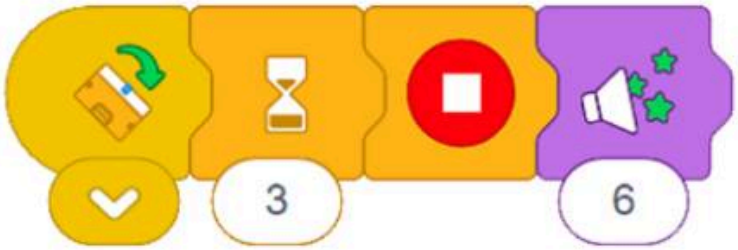


Task 5. Program

ICON BLOCKS | WORD BLOCKS

Program and test your robot.

In ICON Blocks mode, the program looks like this:



Your robot has two motors. Which parts of the robot are driven by the motors?



screws

wheels

both









Select the device that can detect the speed of rotation of your robot









Discuss!

- Why did you build a robotic plane?
- What functions of the plane help young explorers?
- Why is your plane more environmentally friendly than the vast majority of other planes?
- According to what algorithm was the departure of the plane from the hangar?
- What new software block did you use today? What is it for?
- What is the name of the robot's sensor that helped it detect the plane's descent?



Your achievements

Total: 4

1



2



3



4

